Ringraziamenti

Colgo questa pagina per fermare, brevemente, gli ultimi due anni — e le persone e i momenti che li hanno attraversati.

Sono stati anni importanti, dentro e fuori l'università. Ne conservo immagini spensierate: i pomeriggi milanesi del dopo-esame a chiacchierare al bar, le giornate australiane passate a decidere cosa avremmo visitato. E frangenti in cui ho dovuto impegnarmi davvero: le giornate lunghissime prima degli esami, passate a fare e rifare il giro del programma, e i cinque esami in trenta giorni della prima sessione, che ricordo bene. Nel complesso, di questo tempo porto con me un ricordo positivo, che mi ha arricchito, e mi sento fortunato di averlo vissuto.

Tutto ciò che faccio, però, poggia sul sostegno della mia **FAMIGLIA**. Sono sempre stato un ragazzo a cui la realtà di paese andava stretta, e ho sempre cercato di andare oltre — per curiosità, e per mettermi in gioco. La mia famiglia non mi ha mai fatto mancare né i mezzi né il supporto per farlo. Non tutti possono dirlo: io sì, e non lo do per scontato.

Un grazie a tutte le persone che hanno fatto parte di questi due anni accademici: i professori, i compagni, ma anche chi ha condiviso soltanto una conversazione nata per caso, a Milano, ad Adelaide o ovunque sia capitato. Alcune di quelle conversazioni mi sono rimaste impresse più di altre, ma tutte fanno parte del percorso, e lo hanno reso più ricco e più interessante.

Menzione speciale, infine, per la mia cagnolina **Hope**, a cui voglio molto bene: sono felice che sia stata accanto a me e alla mia famiglia in questi anni. Ha reso tutto molto più bello.

Contents

List of Figures

List of Tables

Chapter 1

Introduction

Late-life heterogeneity, welfare regimes, and the silver-economy design problem

1.1 Motivation

Roughly one in four Italians (Istat, 2024) and one in five Swedes (Statistics Sweden, 2024) are now aged 65 or older, and the European silver economy built around them has not yet solved its design problem. Europe's over-65 populations are larger, longer-lived, more economically autonomous, and internally more heterogeneous than at any earlier moment in the demographic record. Yet policy responses and product strategies still tend to treat them as a single demographic block, defined externally by age and income. This thesis argues that beneath the apparent uniformity, the same multidimensional segmentation pipeline, applied to comparable European data, returns systematically different typologies depending on the welfare regime that surrounds each population.

Italy and Sweden are at advanced stages of the same demographic transition, yet the institutions that surround the late-life condition differ structurally between them. In Italy, elderly support runs primarily through the household and the family; in Sweden, through the public sector, with the family in a complementary role. The comparative welfare-state literature has documented this contrast, Mediterranean-familistic versus Nordic-universalistic, at the aggregate level: labour-market participation, intergenerational transfers, gender equality, and healthcare access.

The hypothesis pursued here is that the regime contrast also operates at the typological level. The over-65 populations of the two countries partition into archetypes that differ systematically in structural shape, in the balance between objective and subjective dimensions, and in their positioning along age and resource gradients. These differences reflect the institutional treatment of the population. The payoff of the hypothesis is operational: a market where most of the older population is digitally autonomous and integrated into a universal-access infrastructure differs structurally from one where only a minority is reachable through direct digital channels, while the rest must be reached through social mediation, family-augmented intermediaries, and affect-aware adoption channels. The typological reading developed here is therefore relevant not only to the comparative welfare-regime literature but to the design of European silver-economy products and policies.

1.2 Research question

The thesis pursues one principal research question and three subsidiary ones.

The principal question concerns the over-65 populations of Italy and Sweden, surveyed under a comparable instrument (SHARE Wave 9) and analysed through an identical multidimensional segmentation pipeline. It asks whether these populations partition into typologies whose structural differences are systematic and large enough to be read as the institutional signature of the welfare-regime contrast between the two countries.

The first subsidiary question is whether including the subjective block of late-life experience (subjective well-being, loneliness, hope, interest, expected survival) as a constituent input to the partition delivers a typological gain over the alternative architecture that uses the subjective measures as external validators.

The second subsidiary question is whether the partition recovered in each country contains country-specific archetypes whose configuration is not replicated in the other country and that carry the bulk of the comparative content.

The third subsidiary question is whether the typological reading delivers operational implications for the structure of the silver-economy market that the alternative country-aggregate reading does not.

The principal question and its three subsidiaries are addressed empirically in Chapters 4–7; the principal findings are previewed in §1.3 below.

1.3 Contribution

Four contributions structure the analytical and operational case of the thesis.

The first is the typological identification of the Isolation Paradox in the Italian over-65 population: a 26.9% segment of the analytical sample (Moderate Isolated, the largest single cluster; Chapter 4) that is functionally and economically autonomous but socially and digitally disconnected. It under-utilises the discretionary, navigation-intensive segments of the formal healthcare system. Regression-adjusted analysis, controlling for age, chronic conditions, self-rated health, income, gender, and depressive symptoms, returns 0.67 fewer specialist contacts per year ($p < 0.001$) and a 28-percentage-point lower preventive dental-care attendance rate

($p < 0.001$) for Moderate Isolated relative to the structurally comparable Traditional Social cluster (§4.7.2). The configuration has no structural counterpart in the Swedish typology and is invisible to the existing operational categories of the Italian welfare regime.

The second contribution is the typological extension of the comparative welfare-regime literature: the documented regime contrast operates not only on aggregate outcomes but on the structural shape of the late-life partition that the same segmentation pipeline returns from each population. The contrast operates jointly on three latent dimensions of the typology: the binary versus graduated structure of fragility, the operational meaning of the digital channel (substitute or complement of the institutional infrastructure), and the late-life trajectory of the high-resource pole. The thesis reads this coherent pattern as the typological signature of the regime.

The third contribution is methodological: the empirical case for treating the subjective dimensions of late-life experience as constituent inputs to the segmentation rather than as external validation outcomes. Two findings support the case. First, the 4D specification that drops the subjective block returns a materially lower external discriminative power on life satisfaction in both countries (§3.8). Second, the qualitative contrast between Fragile Resigned and Fragile Depressed collapses under the same specification, eliminating the affective distinction that drives the differential policy responses developed in §4.7.

The fourth contribution is operational: framing the silver-economy design problem as a regime-conditioned typological problem. The three unmatched profiles, Moderate Isolated in Italy and Asset Rich and Wealthy Digital in Sweden, carry the country-specific signal in the standardised analytical space; they identify the empirical anchors of the design space, the segments on which product strategies cannot be transferred across regimes. The cross-country dental-care gap on the four matched pairs delivers, in turn, an order-of-magnitude indication of the preventive-care under-engagement that the Swedish reference reveals.

1.4 Structure of the thesis

The thesis is organised in nine chapters. Chapter 2 reviews the five strands of literature that anchor the strategy: comparative welfare regimes, gerontological subjective dimensions, segmentation methodology, silver economy and gerontechnology, and European typological studies. Chapter 3 sets out the pipeline: SHARE Wave 9 sample preparation, factor-analytic diagnostics, K-means clustering ($k = 5$ Italy, $k = 6$ Sweden), Latent Class Analysis triangulation, and the empirical justification of the subjective block as input.

Chapters 4 and 5 report the parallel typologies: Chapter 4 develops the Italian typology of five archetypes and the Isolation Paradox, Chapter 5 the Swedish typology of six archetypes and the three structural contrasts that distinguish it. Chapter 6 integrates the two through the matched cross-country reading and its implications for silver-economy market structure. Chapter 7 reports the robustness battery (imputation, internal validity, external validation, cross-method convergence, cross-wave stability). Chapter 8 presents the operational deliverable, the Italian Silver Atlas platform and its cohort-scoring machinery; Chapter 9 synthesises the findings, contributions, limitations, and future-research directions. Figure 1.1 maps this organisation.

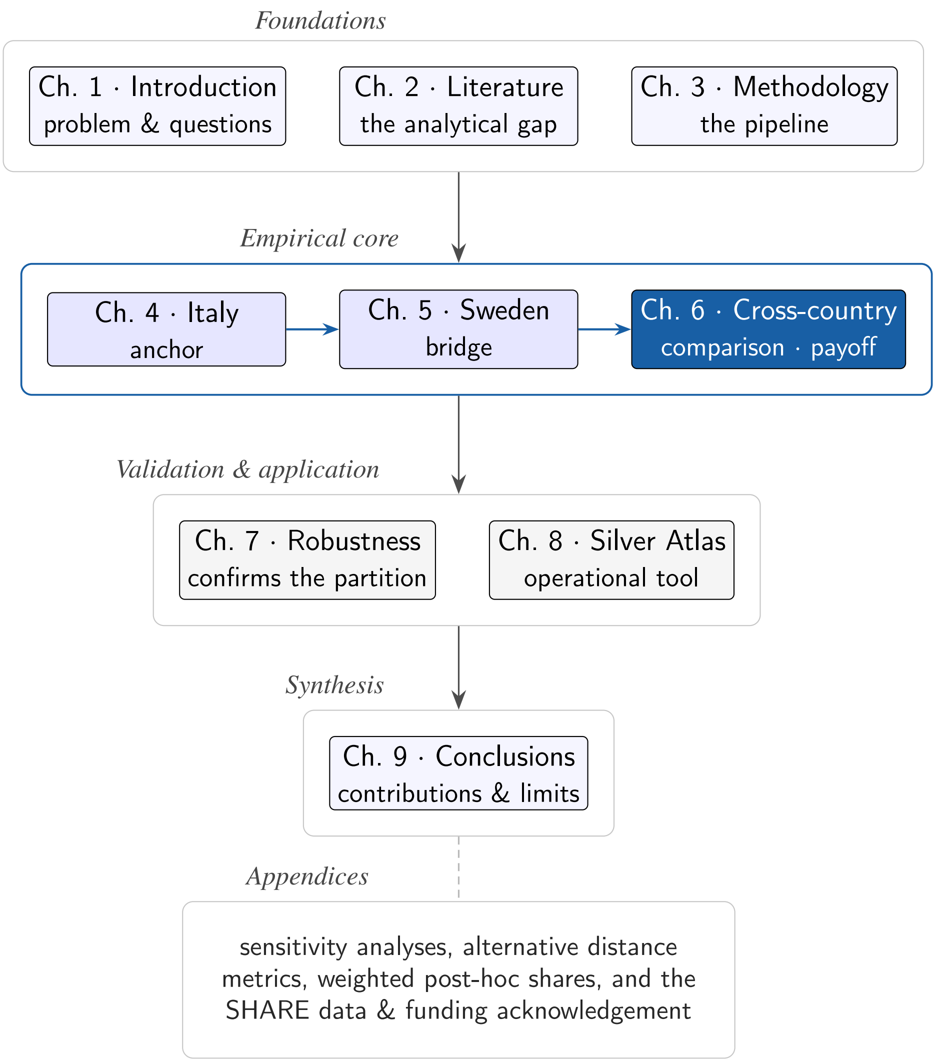


Figure 1.1: Structure of the thesis. The nine chapters move from the foundations through the empirical core—where the Italian (anchor) and Swedish (bridge) typologies feed the cross-country comparison—to validation, operationalisation, and synthesis, with the appendices collecting the supporting analyses and data documentation. The through-line is the welfare regime as the structuring institution of the late-life partition, with the Italian Isolation Paradox as its headline.

Chapter 2

Literature Review and Theoretical Framework

Welfare regimes, subjective late-life experience, and the segmentation tradition

2.1 Overview

The analytical strategy of this thesis rests on a single proposition: the over-65 population is not a demographic block but a structured partition of substantively distinct configurations, and the institution that structures it is the welfare regime. This chapter builds the proposition from the existing literature and motivates each design choice that the pipeline of Chapter 3 implements. The argument moves from the heterogeneity of the late-life population to the welfare-regime framework that structures it, then to the case for subjective experience as a constituent, rather than derivative, dimension of that heterogeneity. It then turns to two further strands: the segmentation tradition that recovers the partition, and the silver-economy domain in which the partition acquires operational meaning. It closes by locating the analytical gap the empirical chapters address.

2.2 Demographic context: population ageing in Europe

The European demographic transition has produced an over-65 population that is both materially larger and, on the dimensions of late-life experience, substantially more heterogeneous than the cohort of two decades earlier. In Italy the over-65 share stands at approximately 24%, corresponding to a national over-65 population of 14.18 million in 2024 (Istat, 2024); in Sweden it is approximately 21%, or 2.06 million (Statistics Sweden, 2024). Both countries sit at comparably advanced stages of the transition but under structurally different welfare-state architectures—the comparative property the cross-country analysis exploits.

That this heterogeneity is structured, rather than dispersion around a representative trajectory, is the premise on which the segmentation rests. Dannefer (2003) formalises the cumulative-disadvantage mechanism, under which working-age inequalities compound across the life course into a more dispersed late-life distribution on resources, health, and subjective dimensions. Netuveli and Blane (2008) show that quality-of-life trajectories in the over-50 European population follow distinct latent classes that age-cohort aggregation obscures; Lestari et al. (2022) show that the typology recovered from such data reflects the multidimensional input admitted into the analysis. The heterogeneity is therefore not noise to be averaged away but a partition to be recovered. What structures that partition, the next section argues, is the welfare regime.

2.3 Comparative welfare regimes and late-life institutions

The comparative welfare-state literature initiated by Esping-Andersen (1990) establishes that the institutional treatment of social risk sorts into a small number of regime types, each producing systematically different outcomes on labour-market participation, family structure, intergenerational transfers, and old-age income. The original three-regime typology (liberal, conservative-corporatist, and social-democratic) rests on two properties: de-commodification, the degree to which the welfare state insulates the individual from market dependency, and stratification, its production of class and status differentials.

Two refinements of that typology bear directly on the present analysis. The first is a fourth regime type, the Mediterranean-familistic regime articulated by Ferrera (1996) and developed by Saraceno (2010) and Naldini and Saraceno (2008), in which the household and the family are the primary loci of social provision in domains the Nordic regime delegates to the public sector. This regime couples a fragmented public-service infrastructure with a residual elderly-care branch and high intra-family responsibility for late-life support. In the cohorts carrying elderly-care obligations, it produces systematically lower female labour-force participation than the Nordic one (Bettio & Plantenga, 2004). The second refinement, drawn by Anttonen and Sipilä (1996), separates the social-care branch of the welfare state from its income-transfer branch. The Nordic regime's universal social-care infrastructure, comprising extensive home and residential care delivered by municipal authorities, is the channel through which late-life provision is de-familised. It is qualitatively distinct from the income transfers on which the Esping-Andersen typology centred.

The empirical signature of the typology is visible across late-life outcomes: old-age poverty risk is more compressed under the Nordic regime than the Mediterranean one (Ferrera, 1996); healthcare access is more uniformly distributed under universal-access institutions than under fragmented ones (Bambra, 2007); and intergenerational co-residence and informal elderly care are far more prevalent in the Mediterranean regime (Saraceno, 2010). The contrast has a direct operational expression in the formal long-term-care workforce: Sweden employs roughly twelve workers per 100 people aged 65 or older, against an OECD average of five and an Italian figure near 1.5. The Italian figure reflects the residual care branch of the Mediterranean regime, where late-life care is discharged predominantly within the family rather than through a municipal workforce (OECD, 2023).

This thesis extends the welfare-regime literature at the typological level: the regime contrast operates not only on aggregate outcomes but on the structural shape of the late-life partition that an identical segmentation pipeline returns from each population. The reading developed in Chapter 6 is that the welfare regime is the structuring institution of that partition, and that the structurally different cluster configurations recovered from Italy and Sweden are its empirical signature.

2.4 Subjective dimensions of late-life experience

If the welfare regime structures the partition, what dimensions must the partition be drawn on to register that structure? The gerontological literature gives a clear answer: the subjective dimensions of late-life experience carry empirical content that objective state alone does not predict, and they are themselves shaped by the institutional environment. Both propositions are constitutive of the analytical choice made here.

The independence of subjective evaluations is anchored in Idler and Benyamini (1997), whose meta-analysis of twenty-seven cohort studies establishes that self-rated health predicts mortality independently of clinically observed indicators. Reviewing two decades of subsequent work, Bowling (2006) concludes that subjective evaluations should be treated as primary rather than derivative late-life indicators. The implication is that objective-only segmentations systematically discard substantive information. The framing is shared across the active-ageing paradigm of Walker (2005), the life-course sociology of Lalive d'Epinay et al. (2003), and the successful-ageing tradition of Rowe and Kahn (1997), each of which treats subjective dimensions as constituent components of the late-life condition rather than as outcomes of objective state.

The measurement instruments carried in SHARE Wave 9 and adopted in the analysis are the operational counterparts of this position. CASP-12 (Control, Autonomy, Self-realisation and Pleasure) (Hyde et al., 2003) measures quality of life; EURO-D, a harmonised twelve-item depression scale (Prince et al., 1999), provides a multi-country comparable measure of depressive affect; UCLA-3 (Hughes et al., 2004) measures perceived loneliness on its three-item short form, grounded in the cognitive-discrepancy theory of loneliness. Each instrument was constructed to capture a substantive component of the late-life condition rather than an outcome of objective state, with the reliability coefficients and operational detail reported in §3.8.

That these dimensions are institutionally shaped is supported by a convergent literature. Wahrendorf and Siegrist (2010) identify welfare-regime variables as the principal cross-country mediators of work-family-late-life trajectories, and Courtin and Knapp (2017) document cross-country variation in late-life social isolation that aligns with the welfare-regime typology. The combined evidence licenses the analytical strategy adopted here: the subjective block is a constituent component of the late-life partition, and the cross-country variation observed in the empirical chapters is informative about the regime that surrounds each population.

2.5 Multidimensional segmentation of older adults

Recovering a partition on a jointly objective-and-subjective input matrix is a segmentation problem, and the methodological literature supplies two traditions. The geometric tradition, initiated by MacQueen (1967), partitions a standardised input matrix by minimising the within-cluster sum of squares (K-means); the parametric tradition, developed by Vermunt and Magidson (2002), recovers the same structure probabilistically under a finite-mixture assumption (Latent Class Analysis). Steinley (2006) reviews the properties of K-means, the selection of k, and the diagnostics for partition quality; Piccarreta and Trentini (2025) provide the applied treatment of dimensionality reduction and clustering—the joint simplification of variables and cases—on which the analytical design of this thesis builds.

The two traditions are complementary rather than competing: the convergence between a geometric partition and its probabilistic counterpart on the same respondents is itself a validation criterion. High convergence supports the substantive reading of a partition; low convergence signals thin cluster boundaries. Upstream of the clustering, the factor-analytic diagnostic step draws on the classical exploratory-factor-analysis literature: Kaiser (1974) for the sampling-adequacy diagnostics, Horn (1965) for the parallel-analysis dimensionality decision.

The decisive design question the literature leaves open is whether the subjective block enters that pipeline as a constituent input or only as external validation. The dominant practice clusters on objective indicators and validates against subjective outcomes (Wahrendorf & Siegrist, 2010); this thesis takes the opposite position, admitting the subjective block as a constituent input.

2.6 Silver economy and gerontechnology

The segmentation is not an end in itself: it intervenes in the silver economy, the aggregate of products, services, and institutional arrangements targeted at the over-50 population. This is the principal commercial and policy domain in which the demographic transition of §2.2 becomes an operational problem. The European Commission frames the silver economy as a structural opportunity of the next two decades: the over-50 silver economy was valued at EUR 3.7 trillion in 2015 and was projected to reach EUR 5.7 trillion by 2025, when its contribution to EU GDP is forecast at roughly 32% (Technopolis Group & Oxford Economics, 2018). Two features of that market bear on the present analysis. First, the addressable population is heterogeneous: commercial segmentation that has not internalised the multidimensional structure of late-life experience overstates the addressability of the target cohort. Second, the market depends structurally on digital adoption: many silver-economy products that are viable in the Nordic regimes assume an internet-mediated channel that the Mediterranean over-65 cohort does not uniformly support (OECD, 2023).

The gerontechnology literature documents this dependence directly. Hargittai and Dobransky (2017) show that the second-level digital divide, the gap in the depth and breadth of online activity conditional on access, is wider among older adults and tracks the same socio-economic gradients that structure late-life resources (Campens et al., 2024). The digital channel is therefore an empirical variable of the typological partition, not a uniform substrate against which the silver economy can be designed.

The thesis contributes to these literatures on two counts. First, the typologies developed in the empirical chapters translate the multidimensional structure of the over-65 population into addressable segments at a granularity the prevailing commercial reports do not match. Second, the regime-conditioned reading of the digital channel positions the silver-economy intervention in a regime-specific design space rather than a country-aggregate market. The design implications for the under-served Italian segment follow from the operational meaning of the digital channel under a familistic regime, a meaning the gerontechnology literature documents indirectly but does not typologise.

2.7 European typological studies and analytical positioning

The four preceding strands converge on an analytical position that the existing European typological literature leaves open. Multi-country typological analyses of older Europeans (Lestari et al., 2022) recover latent classes or segments but pool or compare them descriptively across countries, without estimating within-country typologies and aligning them through an explicit structural matching procedure. Country-specific segmentation studies on Italian or Swedish older adults are typically built on objective health and resource indicators alone—the objective-only specification questioned in §2.4—and do not extend to the welfare-regime contrast. The protocol adopted here is defined relative to that gap by three explicit choices.

First, the input matrix admits the subjective block on equal footing with the objective indicators, against the more common objective-only clustering with subjective validation; the case rests on the gerontological tradition reviewed above and is made empirically in Chapter 3. Second, the analysis is cross-country comparative on a single regime axis rather than multi-country descriptive: the two-country design trades breadth for depth of interpretation, which the typological extension requires. Third, the comparison is anchored in the welfare-regime literature of §2.3 rather than a country-as-control design, so the cross-country contrasts are read as institutional signatures rather than residual differences.

These three choices, together with the structural matching of the two within-country typologies, define an analytical instrument that no existing European typological study assembles in full: each component appears in isolation in the literature, and their assembly is the contribution operationalised in Chapters 4–7. Table 2.1 sets the prevailing practice against the choice made here on each of the five design dimensions.

Table 2.1: Analytical positioning of this thesis across the five design dimensions that define the analytical instrument, set against the prevailing practice in the European typological literature.

Design dimension	Prevailing practice in the literature	This thesis
Input space	Objective health and resource indicators alone	Objective and subjective dimensions on equal footing
Role of subjective measures	External validators of an objective partition	Constituent inputs to the partition
Comparative design	Multi-country descriptive, with latent classes pooled across countries	Two-country comparison on a single welfare-regime axis
Interpretive anchor	Country treated as a residual control	Welfare-regime institutions, with contrasts read as institutional signatures
Cross-country alignment	Share differences across pooled latent classes	Explicit structural matching of within-country typologies

Note. The five design dimensions correspond to the three explicit choices set out in §2.7, together with the multi-dimensional input and the structural-matching procedure. The "prevailing practice" column summarises the dominant approach in the European typological literature on SHARE and comparable cohorts (for example, Netuveli and Blane, 2008, Wahrendorf and Siegrist, 2010, and Lestari et al., 2022).

Chapter 3

Methodology

Data, variable selection, preprocessing, factor structure, clustering pipeline

3.1 Empirical strategy

The empirical strategy of this thesis rests on three commitments. The first is methodological: the multidimensional structure of late-life experience is recovered from a data-driven segmentation on a harmonised analytical set, not from a manual partition of cross-tabulations. The second is comparative: two segmentations are estimated independently in Italy and Sweden, the leading Mediterranean-familistic and Nordic-universalistic cases, and compared on structurally meaningful axes. The third is conceptual: the analytical set includes both objective dimensions (health, economic resources, digital and social integration, cognitive function) and subjective dimensions (CASP-12, loneliness, hope for the future, interest, expected survival), with the latter entering as constituent inputs rather than external validation outcomes. These commitments are operationalised through the pipeline summarised in Figure 3.1.

This chapter sets out the operational implementation of that pipeline.

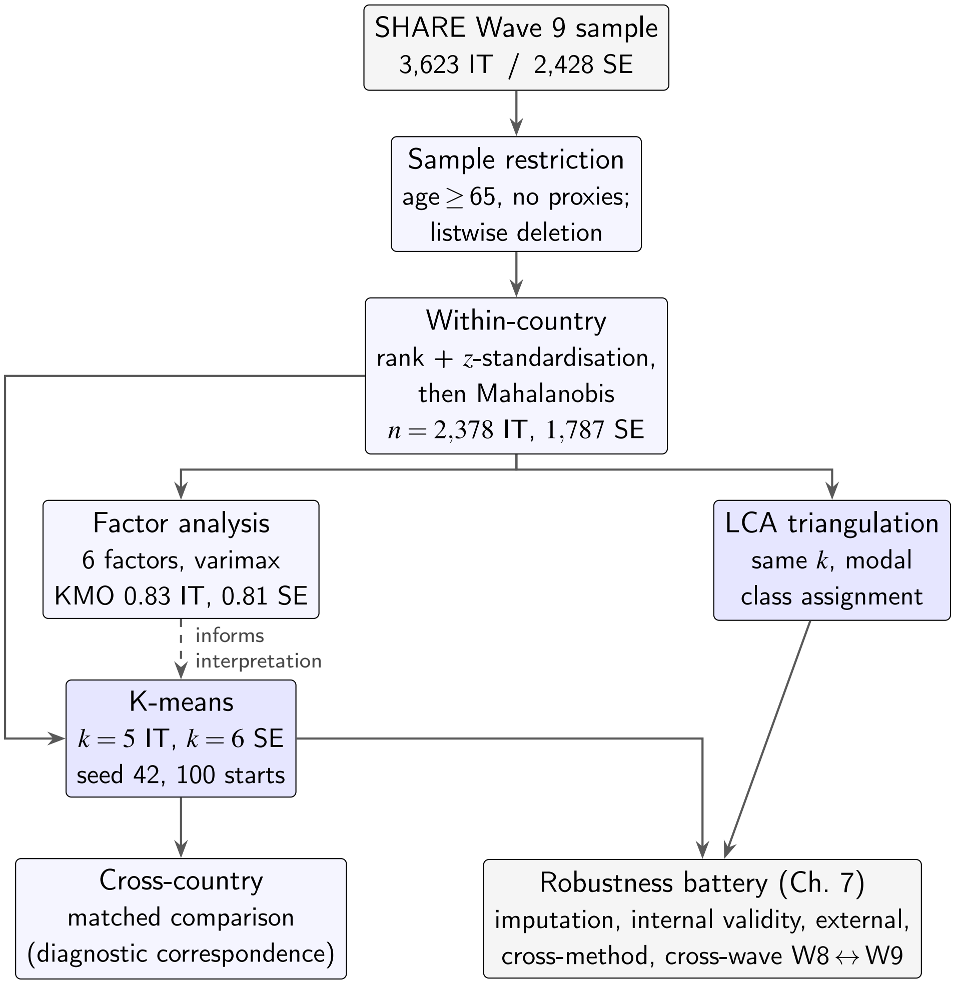


Figure 3.1: Analytical pipeline. The SHARE Wave 9 sample passes through restriction, listwise deletion, within-country rank-and-z standardisation, and Mahalanobis screening on the standardised matrix; the harmonised matrix then feeds the K-means estimation directly. Factor analysis is run on the same matrix as a pre-clustering diagnostic that informs the interpretation of the solution (dashed link) rather than supplying its input; the triangulation path estimates a Latent Class Analysis on a discretised representation of the same variables (§3.7) for cross-method validation. K-means supplies the centroids for the cross-country matched comparison of Chapter 6, and both K-means and LCA feed the robustness battery of Chapter 7.

3.2 Data and analytical sample

The Survey of Health, Ageing and Retirement in Europe (SHARE) is a multidisciplinary, cross-national panel of individuals aged 50 and older resident in 28 European countries, harmonised ex ante on questionnaire content and ex post on variable definitions.[1] Wave 9 was fielded between June 2021 and August 2022, covering the period that follows the most acute phase of the COVID-19 pandemic. The wave is the most recent fully released SHARE wave at the time

[1]This thesis uses data from SHARE Wave 9 (DOI: 10.6103/SHARE.w9.900) (SHARE-ERIC, 2024b) and, for the cross-wave analysis of §7.6, SHARE Wave 8 (DOI: 10.6103/SHARE.w8.900) (SHARE-ERIC, 2024a), release version 9.0.0; see Börsch-Supan et al. (2013) for methodological details. The full funding acknowledgement required by the SHARE-ERIC Conditions of Use is reproduced in Appendix C.

of writing. The pre-pandemic Wave 8 (2019–2020) is used in §7.6 as the comparator for the cross-wave robustness analysis.

Italy and Sweden are selected on theoretical grounds. They occupy opposite ends of the welfare-regime typology of Esping-Andersen (1990). Italy is the leading Mediterranean-familistic case (high reliance on intra-family care, a fragmented social-protection system, a predominance of cash transfers in elderly care); Sweden is the prototypical Nordic-universalistic case (extensive public long-term care, dense community-based social services, largely de-familised elderly care). The two also differ markedly in digital adoption among older adults: Eurostat (2022) reports 51% internet use in the past three months for Italian residents aged 65–74, against 96% in Sweden.

The analytical sample is restricted to respondents aged 65 or older at the date of interview, with proxy interviews excluded. The SHARE multiple-imputation procedure supplies the health, economic and cognitive variables, and the analytical sample uses the first imputation (`implicat` = 1). Table 3.1 reports the sample derivation. The Italian first-imputation sample (all ages) contains 3,623 individuals. Restriction to the over-65 population yields 2,804; listwise deletion on the 31 analytical variables retains 2,431 (a 13.3% loss); and Mahalanobis-distance outlier screening at $\alpha = 0.001$ removes a further 53 (2.2%), for a final analytical sample of 2,378. The Swedish sequence is parallel: 2,428 individuals, 2,045 over-65, 1,831 after listwise deletion (a 10.5% loss), and 1,787 after Mahalanobis screening (44 removed, 2.4%).

In both countries the listwise deletion is the larger source of attrition and is concentrated in the subjective and social block (self-rated survival, CASP-12, the social-integration index, and the cultural-participation items), which carries the highest item-level missingness; the Mahalanobis outlier exclusion is modest and nearly symmetric across the two countries (2.2% Italy, 2.4% Sweden), as visualised in §3.4 below (Figure 3.2).

These counts are taken on the SHARE first-imputation generated-variables sample (`implicat` = 1), the subset of the Wave 9 release carrying the imputed health, economic and cognitive variables. The broader Wave 9 country release is larger, but it is not the operative base for the imputed inputs the segmentation requires.

All data used here are de-identified SHARE microdata, released under the survey's research-use conditions and collected under its harmonised ethics and informed-consent governance (Börsch-Supan et al., 2013). No individual is identifiable in the analytical sample or in any reported result.

Table 3.1: Analytical sample, by country and stage

Stage	Italy	Sweden
Wave 9 sample, all ages (first imputation)	3,623	2,428
Aged 65 or older at first imputation	2,804	2,045
After listwise deletion on 31 variables	2,431	1,831
After Mahalanobis screening ($\alpha = 0.001$)	2,378	1,787
Final analytical sample	**2,378**	**1,787**
Active analytical variables	29	31
Number of clusters retained (k)	5	6

Note. The number of clusters is selected by joint reading of the Duda–Hart pseudo-T^2 statistic, the within-cluster sum of squares scree, and substantive interpretability (§3.6).

3.3 Variable inventory

Thirty-one analytical variables span five conceptual dimensions, following the SHARE conceptual framework on the multidimensional measurement of late-life experience (Wahrendorf & Siegrist, 2010). The activities-of-daily-living (ADL) and instrumental-activities-of-daily-living (IADL) scales are among the items carried for functional health. Where multiple SHARE items operationalise the same construct (for example, mobility limitations, ADL and IADL for functional health), all candidate items are retained and dimensionality reduction is delegated to the preprocessing and to the clustering algorithm itself; the full inventory is listed in Table 3.2. The health, economic and cognitive variables are imputed via the SHARE multiple-imputation procedure. Two cultural-engagement binary indicators (`ac035d4`, `ac035d7`) have zero variance in the Italian analytical sample and are dropped at the variance-screening stage, accounting for the 29-versus-31 active-variable asymmetry across countries (§3.4). The cognitive items are near-complete in both samples; the listwise loss falls mainly on the subjective and social-engagement block, as reported in §3.2.

Table 3.2: Analytical variable inventory by conceptual dimension

Dimension	Variables (SHARE codename)
Physical and functional health (8)	sphus, chronic, adl, iadl, mobility, bmi, phinact, eurod
Economic resources (5)	log_thinc, log_hnetw, ypen1, home_own, fdistress
Digital and social integration (10)	internet, sn_size_w9, social_integration, ac035d1, ac035d5, ac035d8, sp002_, sp008_, plus two further cultural-engagement items (ac035d4, ac035d7)
Cognitive function (3)	fluency, memory, orienti
Subjective well-being (5)	casp, loneliness, hope_future, interest, expect_alive

Note. Variable codes follow the SHARE Wave 9 release naming convention. Italy retains 29 active variables (ac035d4, ac035d7 dropped); Sweden retains the full 31 items.

3.4 Preprocessing pipeline

The preprocessing pipeline operates in five stages: sample selection, variable selection with variance screening, listwise deletion, scale harmonisation by rank transformation followed by z-standardisation, and Mahalanobis-distance screening of multivariate outliers on the standardised matrix. The harmonised matrix feeds the K-means estimation reported in §3.6. Each stage is implemented as a numbered R script with fixed seed 42.

Listwise deletion removes a respondent if any of the 31 analytical variables is missing. SHARE imputes the health, economic and cognitive variables ex ante (Christelis, 2011). Multiple-imputation pooling across replicates is not used at the segmentation stage: the K-means objective is not invariant to imputation noise, and Rubin-style consensus algorithms for high-dimensional clustering are not yet well established. The analytical sample therefore uses the first imputation (§3.2), and the diagnostics of §7.2 quantify the residual sensitivity to the imputation draw.

Continuous and ordinal variables are then rank-transformed within each country, mapping the original distribution onto a uniform rank distribution; binary variables pass through unchanged. The rank-transformed and binary variables are z-standardised within each country, so each enters K-means with mean zero and unit variance. Standardisation is performed at the country level rather than on the pooled sample, to preserve the within-country interpretation of the centroids. A respondent classified “Connected Active” in Italy is so classified relative to the Italian distribution, and is then compared in Chapter 6 with the analogous “Connected Wealthy” archetype defined relative to the Swedish distribution. The partition’s sensitivity to the weighting of the binary indicators, including a Gower-dissimilarity alternative and an explicit binary down-weighting, is examined in §A.2.

After standardisation, the sample is screened for multivariate outliers using Mahalanobis distance computed on the standardised matrix, with respondents above the chi-squared critical value at $\alpha = 0.001$ excluded; the two country densities are closely aligned and the threshold removes only the thin right tail in each (53 respondents, 2.2%, in Italy and 44, 2.4%, in Sweden; Figure 3.2, with the losses by country also reported in §3.2). The sensitivity of the partition to the $\alpha = 0.001$ threshold is quantified for Sweden at the profile level in §7.3.1, with the full robustness battery documented in §A.1.

The SHARE design and post-stratification weights are not applied at the segmentation stage: the segmentation operates unweighted. Weighting would distort the within-cluster geometry, and the choice of k in §3.6 depends on that geometry. The substantive findings are accordingly stated as claims about cluster centroids and structural relations rather than probabilistic generalisations to the national over-65 populations. A post-hoc projection using the

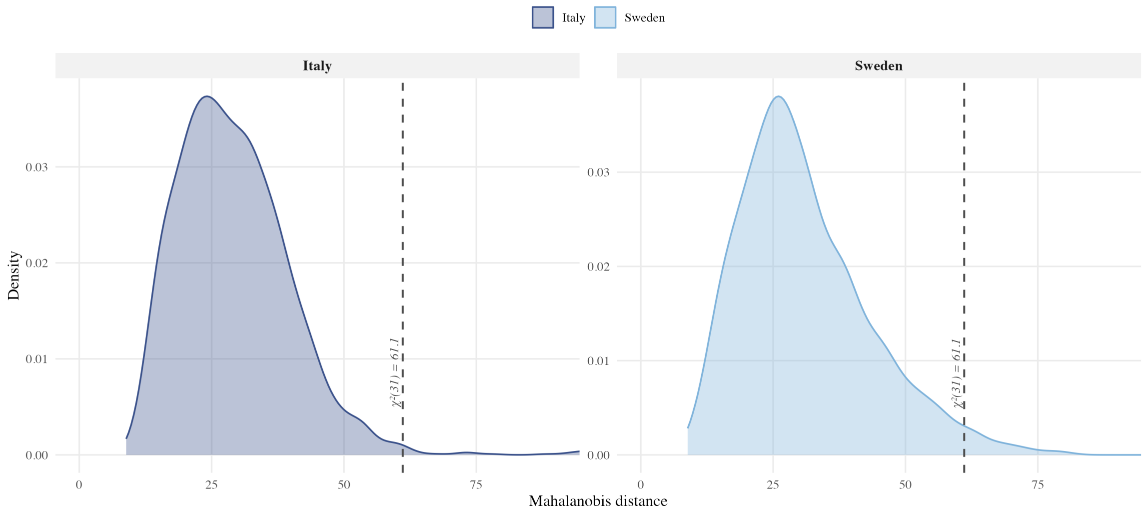


Figure 3.2: Mahalanobis distance distribution, Italy and Sweden. Kernel density on the country-specific standardised analytical matrix; dashed line marks the chi-squared critical value at $\alpha = 0.001$ used for outlier exclusion. Italy loses 53 (2.2%); Sweden loses 44 (2.4%) in the right tail.

SHARE Wave 9 design weight (`cciw_w9`) confirms that the descriptive cluster shares are stable under re-weighting, with differences uniformly within 3.2 percentage points across all eleven profiles (Appendix B). The national-population projection in Figure 4.2 applies these weighted post-hoc shares to the 2024 Istat over-65 estimate.

3.5 Factor structure of the harmonised input space

Before estimating the K-means partition, the preprocessed input space is examined through factor analysis. The purpose is twofold: a sanity check on whether the variables share sufficient common structure to be analysed jointly, and a low-dimensional representation that supports the substantive interpretation of the cluster solution.

The common-factor model assumes that the p-dimensional vector of standardised observed variables $\mathbf{X}$ admits the decomposition

$$\mathbf{X} = \Lambda \mathbf{F} + \varepsilon, \qquad \mathrm{Cov}(\mathbf{X}) = \Lambda\Lambda^\top + \Psi, \tag{3.1}$$

where Λ is the $p \times m$ matrix of factor loadings on m latent common factors $\mathbf{F}$, Ψ is the diagonal matrix of unique variances, and the standard identifying assumptions hold ($\mathrm{E}[\mathbf{F}] = \mathbf{0}$, $\mathrm{Cov}(\mathbf{F}) = \mathbf{I}$, $\mathrm{Cov}(\varepsilon) = \Psi$, $\mathrm{Cov}(\mathbf{F}, \varepsilon) = \mathbf{0}$). Principal Axis factoring estimates Λ iteratively by spectral decomposition of the reduced correlation matrix $\mathbf{R} - \Psi$, replacing the unit diagonal of $\mathbf{R}$ with the current estimate of communalities until convergence. The Italian solution uses this principal-axis estimator; the Swedish solution uses maximum-likelihood extraction, which estimates Λ and Ψ by maximising the Gaussian likelihood of the observed correlation matrix and is adopted because it resolves a Heywood case that principal-axis extraction produces on the social-integration index (§A.3).

The Kaiser–Meyer–Olkin (KMO) measure is 0.832 in Italy and 0.808 in Sweden, both above the 0.70 threshold conventionally taken as acceptable adequacy (Kaiser, 1974); the Bartlett test of sphericity (Bartlett, 1950) rejects the null in both countries ($p < 0.001$). The number of factors is selected by joint reading of three criteria: parallel analysis (Horn, 1965), BIC for m between 2 and 8, and substantive interpretability of the rotated loadings. The country-level BIC decreases across the tested range, reaching its within-range minimum at $m = 8$ (Italy, principal-axis) and $m = 6$–7 (Sweden, maximum-likelihood); the parallel-analysis criterion suggests $m = 6$ (Italy) and $m = 8$ (Sweden); substantive interpretability is highest at $m = 6$ in both countries. The six-factor solution is therefore retained on cross-country-comparability and interpretability grounds; it is conservative relative to the BIC minimum in Italy ($m = 8$) and at the lower edge of the BIC range in Sweden ($m = 6$–7). The RMSEA at $m = 6$ is 0.054 in Italy and 0.033 in Sweden, against 0.040 and 0.029 at the respective within-range BIC minimum. Figure 3.3 reports the four fit indices for m between 4 and 8.

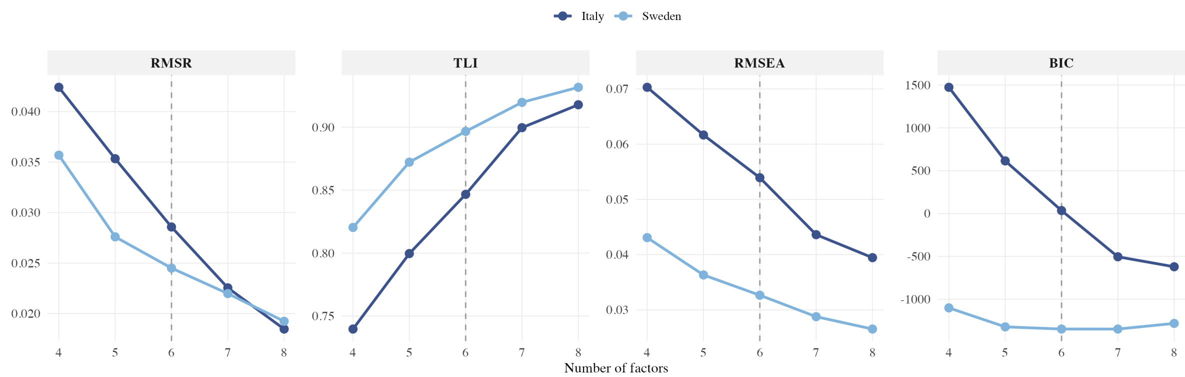


Figure 3.3: Factor-analysis fit indices across number of factors. Four indices (RMSR, TLI, RMSEA, BIC) reported for $m = 4$ to 8 factors, separately for Italy (principal-axis) and Sweden (maximum-likelihood). Dashed reference line marks the $m = 6$ design choice.

Loadings are rotated by varimax (Kaiser, 1958). Figure 3.4 reports the Italian and Swedish solutions side by side; the six factors carry the substantive labels Factor 1 *physical and functional fragility*, Factor 2 *subjective burden*, Factor 3 *quality of life and engagement*, Factor 4 *social integration*, Factor 5 *household income*, Factor 6 *household wealth*.[2] The Swedish solution recovers the same six-factor structure with broadly similar loadings on the health, social and economic blocks; cross-country differences in the subjective and digital factors are

[2]The R `psych` package outputs the same factors in variance-explained order as PA1, PA5, PA2, PA3, PA4, PA6 respectively. The substantive renumbering used throughout the thesis preserves the conceptual progression from objective fragility to economic resources via subjective and social dimensions.

developed in Chapter 6. The six factors jointly explain 38.4% of the variance in the 29-variable Italian active set.

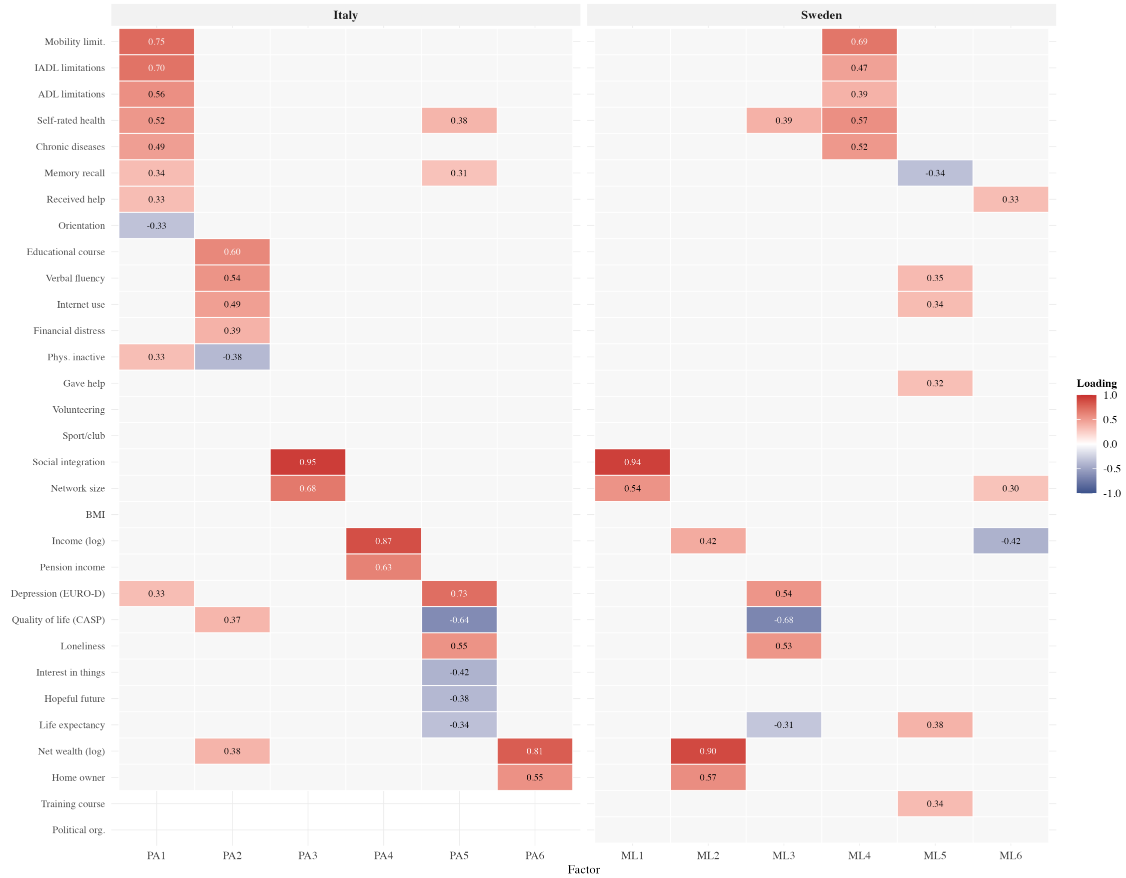


Figure 3.4: Factor loadings matrix, Italy and Sweden. Six factors with varimax rotation, extracted by principal-axis factoring (Italy) and maximum-likelihood factoring (Sweden); cell entries are loadings ≥ 0.30 (smaller values suppressed). Variable ordering follows the dominant Italian factor. KMO = 0.832 (IT), 0.808 (SE). Columns are the extraction factors (PA, principal-axis, Italy; ML, maximum-likelihood, Sweden); the substantive Factor 1–6 of §3.5 correspond to the Italian columns in the order PA1, PA5, PA2, PA3, PA4, PA6.

K-means estimation operates on the standardised analytical matrix directly, not on the EFA-extracted factor scores.[3] The factor analysis is therefore an interpretative companion to the segmentation, not an intermediate dimensionality-reduction step.

[3]The Swedish factor analysis uses maximum-likelihood extraction. An earlier principal-axis specification produced a Heywood case on the social-integration index (loading 1.11, communality 1.38), signalling that the variable was empirically over-determined within the social factor under PA factoring. The ML specification resolves the case cleanly while preserving comparable fit (RMSEA 0.033 and TLI 0.897 under both extractions; KMO 0.808). The principal-axis alternative and a 30-variable specification that drops social-integration are documented in §A.3.

3.6 K-means clustering: estimation and choice of k

K-means (MacQueen, 1967) with squared-Euclidean distance is the segmentation algorithm, implemented with the Hartigan–Wong update rule (Hartigan & Wong, 1979) on the country-specific standardised matrix. Given a sample $\{\mathbf{x}_1, \ldots, \mathbf{x}_n\}$ and a target number of clusters k, K-means seeks the partition into mutually exclusive sets $\mathscr{C}_1, \ldots, \mathscr{C}_k$ that minimises the within-cluster sum of squares,

$$\arg\min_{\mathscr{C}_1,\ldots,\mathscr{C}_k} \mathscr{J}(\mathscr{C}_1,\ldots,\mathscr{C}_k) = \sum_{i=1}^{k} \sum_{\mathbf{x}\in\mathscr{C}_i} \left\|\mathbf{x}-\mu_i\right\|^2, \qquad \mu_i = \frac{1}{|\mathscr{C}_i|} \sum_{\mathbf{x}\in\mathscr{C}_i} \mathbf{x}, \tag{3.2}$$

where μ_i is the centroid of cluster i and $\|\cdot\|$ is the Euclidean norm. The objective is non-convex; the estimation is therefore run with 100 random initial-centroid configurations and a fixed seed (42), and the configuration with the lowest $\mathscr{J}$ is retained. K-means is preferred to agglomerative alternatives (Ward, complete, average), which returned lower within-cluster R^2 and silhouette values in exploratory runs; the internal validity of the retained partition is reported in §7.3.

The Italian K-means is estimated on $p_{\mathrm{IT}} = 29$ active analytical variables and the Swedish on $p_{\mathrm{SE}} = 31$ (Table 3.1). The two-variable asymmetry reflects the variance-screening step of §3.4. Two cultural-engagement binary indicators (`ac035d4`, participation in a training course, and `ac035d7`, participation in a political or community organisation) have zero variance in the Italian sample and are dropped: under squared-Euclidean K-means a zero-variance variable contributes zero to every inter-cluster distance and adds no information. Both indicators retain non-trivial variance in Sweden and are kept there. The cross-country alignment of Chapter 6 operates on a common set of 11 analytical dimensions in which the two excluded indicators do not figure (§6.3). The sensitivity of the comparison to the 29-versus-31 variable asymmetry is probed in Appendix A.4, where perturbed re-inclusion of the two items on the Italian side leaves the partition essentially unchanged (ARI 0.977).

The number of clusters k is selected on three criteria. The within-cluster sum of squares scree (k between 2 and 10; Figure 3.5) places the elbow at $k = 5$ (Italy) and $k = 6$ (Sweden). The Duda–Hart pseudo-T^2 statistic on the Ward dendrogram (Duda et al., 2001) peaks independently at the same values. Substantive interpretability is decisive at the boundary cases. In Italy, at $k = 4$ the fragile stratum collapses into a single archetype that obscures a substantively important contrast between "Resigned" and "Depressed" fragile sub-types (§4.3); at $k = 6$ the middle stratum splits into two unstable profiles. In Sweden, at $k = 5$ the welfare-attached and digitally active strata collapse together, obscuring a contrast that the cross-country mapping of Chapter 6 relies on; at $k = 7$ the wealthy stratum fragments into unstable sub-profiles. Table 3.3

reports the internal validity at the chosen k.

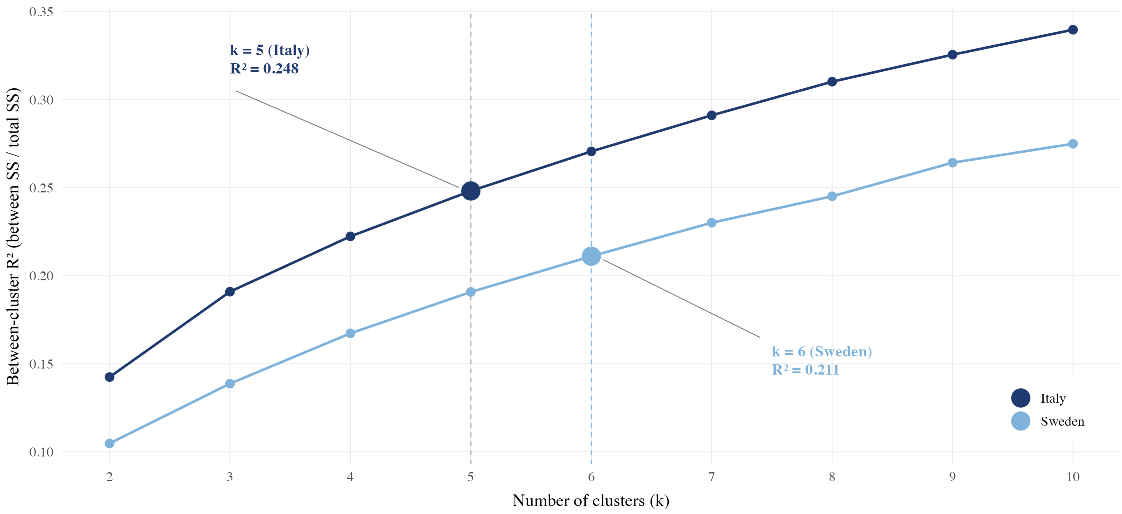


Figure 3.5: K-selection scree diagnostic, Italy and Sweden. Between-cluster R^2 (between/total SS) for $k = 2 \ldots 10$ on the standardised analytical matrix. The Italian scree elbows at $k = 5$ and the Swedish at $k = 6$, in agreement with the Duda–Hart pseudo-T^2 statistic and substantive interpretability (§3.6).

Table 3.3: Internal validity of the K-means partition at the chosen k

Index	**Italy** ($k = 5$)	**Sweden** ($k = 6$)
Between-cluster R^2 (between/total SS)	0.248	0.211
Mean silhouette width	0.069	0.060
Duda–Hart pseudo-T^2 peak at	$k = 5$	$k = 6$
ANOVA F on life satisfaction (external)	124.6	44.3

Note. Between-cluster R^2 is between/total sum of squares in the standardised input matrix. Mean silhouette is computed on Euclidean distances. The Duda–Hart pseudo-T^2 is the peak value of the statistic on the Ward dendrogram. ANOVA F is computed on the SHARE Wave 9 life-satisfaction item (single-item, eleven-point scale) external to the analytical input set.

The mean silhouette values of 0.069 (Italy) and 0.060 (Sweden) are low in absolute terms, below the 0.250 threshold conventionally taken as indicative of a cohesive cluster structure (Kaufman & Rousseeuw, 1990). This is a routine consequence of clustering on a high-dimensional input matrix spanning the six factor subspaces of §3.5, where Euclidean compactness is intrinsically limited and within-cluster heterogeneity remains material along the dimensions on which the partition does not separate. The Duda–Hart pseudo-T^2 peak and the substantive interpretability of the profiles therefore carry decisive weight in the choice of k, with the silhouette diagnostics reported in Chapter 7. Given these values, the K-means solution is framed less as a discovery of natural clusters than as a structured segmentation of a continuous multidimensional space.

3.7 Latent Class Analysis triangulation

The profile labels used below are introduced substantively in Chapters 4 and 5; here they serve only as cluster identifiers. The K-means hard assignment is triangulated using Latent Class Analysis fitted on a discretised representation of the input variables: continuous and ordinal items are partitioned at country-level tertile cut-points into three-category ordinal variables, with items whose distribution is dominated by a single modal value ($\geq 50\%$ of the mass) instead split at the mode (modal versus above-modal categories) to avoid degenerate cells; binary items are passed through unchanged. LCA assumes that the joint distribution of the J observed categorical items $\mathbf{Y} = (Y_1, \ldots, Y_J)$ arises from a finite mixture of K latent classes:

$$\Pr(\mathbf{Y} = \mathbf{y}) = \sum_{k=1}^{K} \pi_k \prod_{j=1}^{J} \Pr(Y_j = y_j \mid C = k), \tag{3.3}$$

where π_k is the prior probability of class k ($\sum_k \pi_k = 1$) and $\Pr(Y_j = y_j \mid C = k)$ is the conditional response probability for item j given class membership, under the local-independence assumption. The model is fitted by the EM algorithm (Vermunt & Magidson, 2002) with five random starts; class membership is then assigned by the modal posterior probability $\arg\max_k \Pr(C = k \mid \mathbf{Y} = \mathbf{y})$. The number of latent classes K is held fixed at the K-means k for each country (so $K = 5$ for Italy and $K = 6$ for Sweden), making the two partitions directly comparable.

Figure 3.6 reports the agreement between the K-means hard assignment and the LCA modal-class assignment by profile. Italian profiles converge between 54.6% (Moderate Isolated) and 95.6% (Fragile Resigned); Swedish profiles between 49.3% (Wealthy Digital) and 86.9% (Connected Wealthy). Random assignment on $k = 5$ Italian classes would yield 20% per class (17% on the $k = 6$ Swedish classes); the 50% reference line is not a canonical statistical cutoff but a deliberately conservative bar, requiring that more than half of a cluster's K-means members fall in the same LCA modal class, far above those chance levels. The corner archetypes (Fragile Resigned and Connected Active in Italy; the corresponding Swedish poles, Fragile and Connected Wealthy) and the Italian Traditional Social profile (78.3%) converge well above this bar, while the remaining middle-stratum profiles sit closer to it, with only Wealthy Digital (49.3%) falling marginally below.

The convergence pattern itself carries methodological information. The corner archetypes occupy regions of the input space that are dense and well separated from the rest of the population: K-means draws their boundaries by Euclidean compactness around a centroid, LCA recovers them as high-probability latent classes under a categorical mixture model, and the two converge because the underlying configurations are unambiguous. The middle-stratum profiles, Moderate Isolated in Italy and Wealthy Digital in Sweden, sit instead in transitional regions of the latent space, where the cluster boundary is geometrically thin and the assignment of marginal respondents is intrinsically uncertain. The lower convergence (49.3–54.6%) on these profiles is therefore not a failure of either algorithm but a substantive finding: it identifies the segments for which classification is most sensitive to the modelling choice, and for which any policy or product application should anticipate non-trivial assignment uncertainty. The robustness diagnostics in §7.5 and §7.6 quantify this uncertainty using cross-method assignment stability and bootstrap resampling.

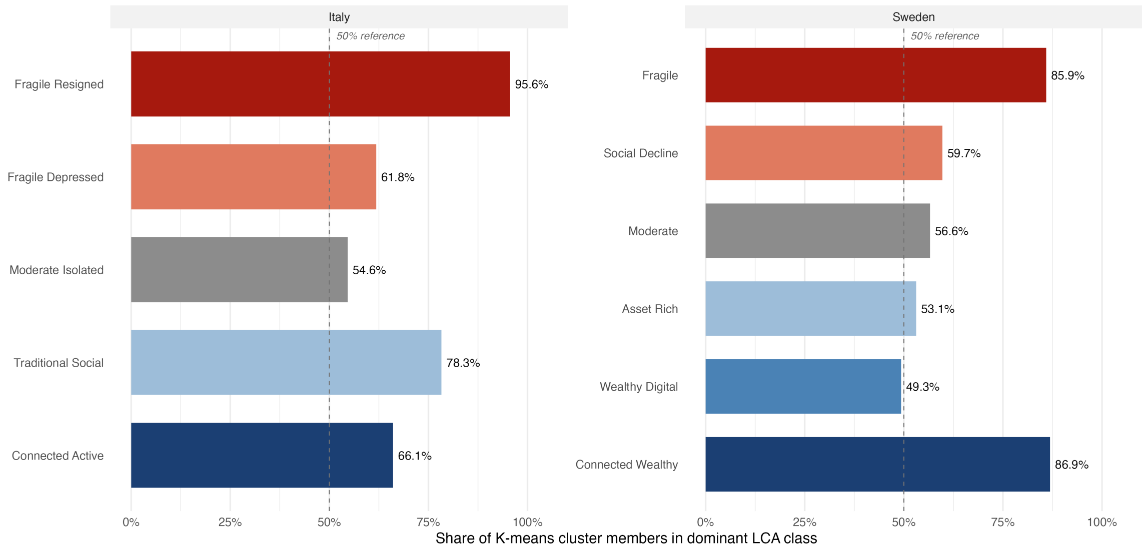


Figure 3.6: Cross-method convergence between K-means and LCA, by profile and country. Bars report the share of K-means cluster members whose LCA modal class coincides with the K-means assignment. The dashed 50% line is a deliberately conservative reference (a majority of a cluster's members in the same LCA class), not a canonical cutoff, and lies far above the 20% (Italy) and 17% (Sweden) levels expected under random assignment.

3.8 Methodological commitment: subjective dimensions as constituent inputs

A central methodological choice of this thesis is that the five subjective measures listed in §3.3 (CASP-12, loneliness, hope for the future, interest, and self-rated probability of survival) enter the K-means and LCA stages on equal footing with the objective indicators, rather than as external validators. The common architecture in older-adult segmentation studies clusters on objective indicators alone and treats subjective dimensions as downstream outcomes (Wahrendorf & Siegrist, 2010); the architecture adopted here treats them instead as constituent components of late-life experience, with empirical content partially independent of objective state. The position rests on three foundations: the gerontological tradition that frames late life as a phase shaped jointly by objective resources and subjective interpretation (Lalive d'Epinay et al., 2003); the evidence that self-rated health predicts mortality independently of clinically observed health (Idler & Benyamini, 1997); and the design of the instruments themselves, CASP-12 (Hyde et al., 2003) ($\alpha \approx 0.84$) and UCLA-3 loneliness (Hughes et al., 2004) ($\alpha \approx 0.72$), which were constructed to capture substantive components of older-age experience rather than outcomes of objective state.

The empirical justification for the commitment is a 4D-versus-5D sensitivity test that re-runs the pipeline with the subjective block removed, holding all other parameters constant. External discriminative power is measured by the F-statistic of a one-way ANOVA of life satisfaction (SHARE item AC012, eleven-point scale) on cluster membership; life satisfaction is external to both specifications. The Adjusted Rand Index between the 5D and 4D partitions measures the magnitude of individual reassignment; both diagnostics are reported in Table 3.4.

Table 3.4: Sensitivity to the subjective block: 4D versus 5D specifications

Country / Specification	**Silhouette**	**Between R^2**	**ANOVA F**	**ARI vs 5D**
Italy—5D (with subjective)	0.069	0.248	124.6	—
Italy—4D (no subjective)	0.089	0.264	73.8	0.41
Italy—Δ (4D − 5D)	+0.020	+0.016	−50.8 (−41%)	
Sweden—5D (with subjective)	0.060	0.211	44.3	—
Sweden—4D (no subjective)	0.079	0.229	34.6	0.52
Sweden—Δ (4D − 5D)	+0.019	+0.018	−9.7 (−22%)	

Note. The 4D specification omits the five subjective measures from the K-means input set; k and all other parameters are held constant. ANOVA F is computed on life satisfaction (SHARE item AC012, eleven-point scale), external to both specifications. ARI vs 5D is the Adjusted Rand Index between the 4D and 5D cluster assignments on the same respondents.

Dropping the subjective block raises internal-validity indices modestly (the routine consequence of clustering on a smaller space) but reduces external discriminative power on life satisfaction by 41% in Italy and 22% in Sweden (Figure 3.7). The 4D–5D ARI is 0.41 in Italy and 0.52 in Sweden, indicating only moderate agreement between the two partitions, well below the 0.80 level conventionally taken as substantive equivalence; the subjective block therefore materially reshapes the partition. The substantive cost in Italy is the collapse of the Fragile Resigned vs Fragile Depressed contrast (§4.3) into a single fragile cluster, eliminating an affective distinction the 5D solution preserves. The asymmetry (−41% vs −22%) is consistent with the cross-country finding that subjective well-being gaps are wider than objective ones (Chapter 6).

The circularity objection, that the typology cannot be validly evaluated against subjective outcomes because subjective measures enter as inputs, rests on a predictive framing the analysis does not adopt. The strategy is descriptive: the typology partitions multidimensional configurations of late-life experience with all dimensions on equal footing, rather than pre-

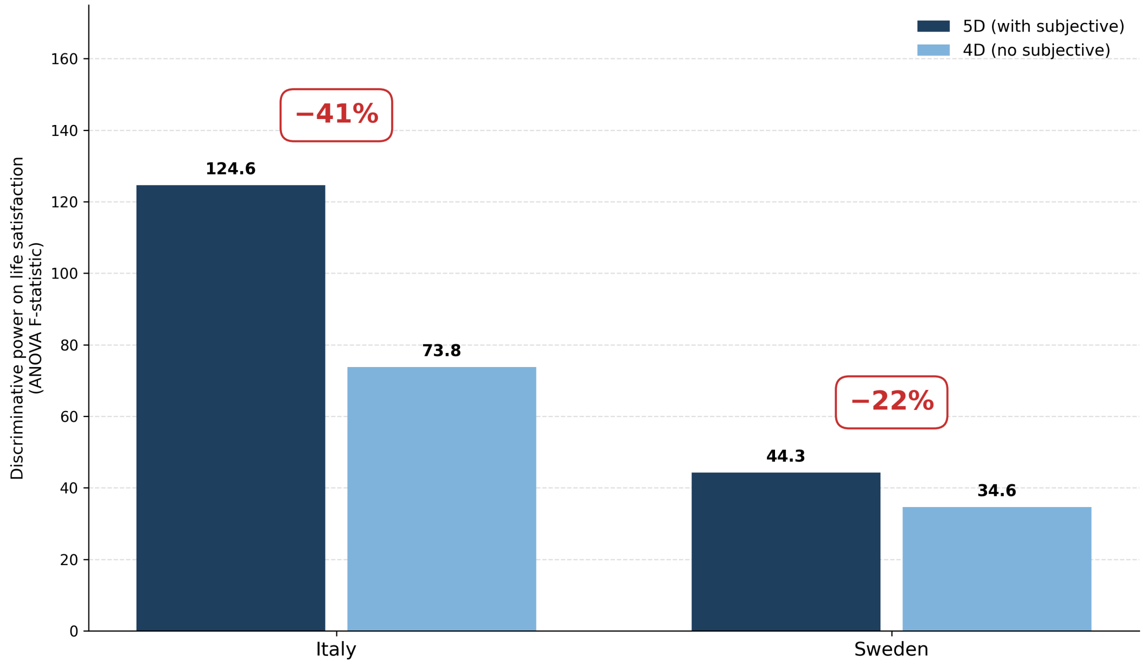


Figure 3.7: Sensitivity of the segmentation to the inclusion of the subjective block. Bar chart of the ANOVA F-statistic on life satisfaction (external outcome) under the 5D specification (with subjective inputs) and the 4D specification (objective inputs only). The 41% reduction in Italy and 22% reduction in Sweden quantify the discriminative gain attributable to the subjective dimensions.

dicting subjective well-being from objective state. The 4D–5D test against life satisfaction (an outcome correlated with but not identical to the five subjective inputs, administered in a separate questionnaire block) is therefore closer to construct validity than to a tautology. Two features sharpen the distinction. Life satisfaction is a single global evaluation, whereas the subjective block enters as five specific facets (CASP-12, loneliness, hope, interest, expected survival). The 41% (Italy) and 22% (Sweden) drop in its discrimination when the block is removed therefore reflects shared late-life content rather than mechanical overlap. The healthcare-utilisation outcomes, moreover, share no measurement channel with the subjective inputs at all, so the relevant validity argument is the convergent reading across a measurement-adjacent outcome (life satisfaction) and measurement-independent outcomes (healthcare utilisation), not either alone. Genuinely external validation against life satisfaction (§7.4) and healthcare utilisation (§7.4) uses outcomes entirely separate from the clustering inputs; further external validations are listed in §9.6.

3.9 Limitations of the methodological approach

Two classes of limitations bound the substantive scope. The first is the sample design: SHARE Wave 9 is treated cross-sectionally, precluding causal identification (cross-wave W8–W9 structure is exploited in §7.6 only for partition stability); the single multiple-imputation replicate (§3.4) introduces residual sensitivity quantified in §7.2; the exclusion of proxy interviews biases the typology toward respondents capable of self-reporting on the subjective block; and the two-regime comparison cannot be generalised to corporatist, liberal or post-socialist regimes without separate investigation. The second class concerns modelling. Low silhouettes (0.069 IT, 0.060 SE; §3.6) reflect the multi-factor input matrix, and the conservative k choice favours cross-country comparability over single-index optimality. The 4D/5D ARI of 0.41 IT and 0.52 SE (Table 3.4) indicates only moderate partition agreement, so the subjective block materially reshapes the partition rather than perturbing it at the margins. The robustness battery of Chapter 7 quantifies these sensitivities and bounds the substantive claims to the structural shape of the partition.

3.10 Reproducibility protocol

The pipeline is implemented in twenty numbered R scripts and two helper modules in a versioned git repository, building on the analytical scaffold of Piccarreta and Trentini (2025). Each script is parameterised by a single seed (42); all intermediate outputs are written to a versioned directory and indexed by the generating script; and the full pipeline runs in approximately forty minutes on a standard laptop with R-package versions pinned via renv. A companion web application mirroring the analytical findings is documented as an operational deliverable in Chapter 8.

Data and code availability. The microdata analysed here are SHARE Wave 9 (SHARE-ERIC, 2024b) and SHARE Wave 8 (SHARE-ERIC, 2024a) for the cross-wave robustness check of §7.6, release 9.0.0; they are available to registered researchers from SHARE-ERIC under its Conditions of Use, are governed by a data-use agreement, and are not redistributed with this thesis. The full analysis code—the numbered R pipeline and the companion web application of Chapter 8—is openly available in the project repository on GitHub.

Chapter 4

The Italian Typology

Five archetypes of late-life experience in Italy

4.1 Overview

The K-means segmentation of Chapter 3 recovers five archetypes spanning the Italian over-65 population, organised along a fragility-and-resources gradient: two fragile profiles at one pole, two well-resourced profiles at the other, and a country-specific middle cluster, *Moderate Isolated* (26.9% of the analytical sample). This middle cluster is marked by social and digital thinness without health or economic adversity. This profile has no structural counterpart in the Swedish typology of Chapter 5 and is the empirical foundation for the *Isolation Paradox* developed in §4.7.

All centroid coordinates are reported as z-scores within Italy on the analytical sample ($n = 2{,}378$). Cluster shares are reported as percentages of the analytical sample; see §3.4 for the descriptive/predictive interpretation of the partition.

4.2 The five archetypes

The K-means partition ($k = 5$) groups the 2,378 Italian respondents into five clusters. Figure 4.1 reports cluster mean values on eight diagnostic variables, with profiles ordered from most fragile (left) to most connected (right). Table 4.1 lists the archetypes with their headline signature and sample shares.

Two raw-scale observations from Figure 4.1 anchor the standardised description that follows. First, the age gradient: the Fragile Resigned cluster has a mean age of 81.5 years, against 77.3 (Fragile Depressed), 72.8 (Moderate Isolated), 73.8 (Traditional Social) and 73.9 (Connected Active). The Fragile Resigned are on average approximately seven years older than the four other archetypes, identifying the oldest-old segment of the population. Second, the digital divide: 8.6% in Fragile Resigned versus 74.8% in Connected Active, a near nine-fold

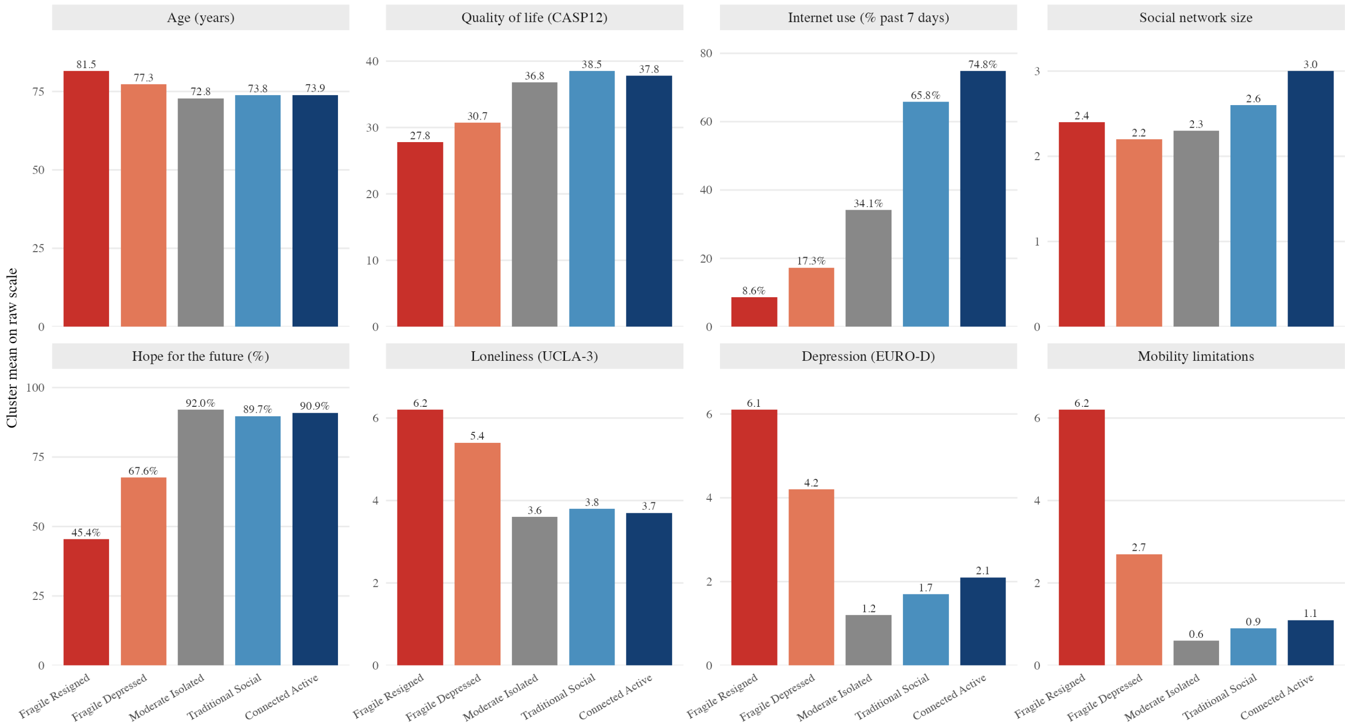


Figure 4.1: Profile characteristics, Italy. Cluster means on eight key dimensions: age, CASP-12, internet use (%), social network size, hope for the future (%), loneliness (UCLA-3), EURO-D, mobility limitations. Profiles ordered from most fragile to most connected. $n = 2{,}378$.

Table 4.1: Headline signature of the five Italian archetypes

Archetype	Share	Headline signature
Fragile Resigned	13.2%	Worst health, oldest age (mean 81.5 years), lowest internet use (8.6%) and CASP (27.8). Marked depressive load (EURO-D 6.1, well above the four-point clinical threshold). Withdrawn evaluative orientation (hope 45.4%).
Fragile Depressed	26.7%	Persistent depressive affect (EURO-D 4.2, loneliness 5.4) with intermediate functional health and below-average resources. The largest fragile cluster.
Moderate Isolated	26.9%	Average-to-good health (mobility 0.6, EURO-D 1.2) and resources, but restricted social engagement (network 2.3, internet 34.1%). The largest single archetype and the country-specific Italian configuration.
Traditional Social	7.7%	Good health and resources, dense cultural participation (hope 89.7%), moderate internet adoption (65.8%). Highest CASP (38.5). A small cluster in W9 (see §4.3).
Connected Active	25.4%	Highest digital adoption (74.8%), largest network (3.0), best resources, intact health. CASP 37.8. The silver-economy frontier of the Italian over-65 population.

Note. Headline values are raw means of the diagnostic variables of Figure 4.1. Standardised centroid coordinates are reported in §4.3. Shares may not sum to 100% due to rounding.

gap on a single behavioural indicator. From the silver-economy perspective, this gap defines a boundary. On one side are segments served by digital products; on the other, those for whom traditional welfare and family-based provision remain the only operational channel.

4.3 Profile descriptions

The five archetypes' within-sample shares and the corresponding weighted projections to the national over-65 population are reported in Figure 4.2. The five archetypes are described below in the same order as Table 4.1, from most fragile to most connected. Standardised centroid values are reported in z-units within Italy ($\bar{x} = 0$, $\sigma = 1$); raw means are given in parentheses where useful for interpretation.

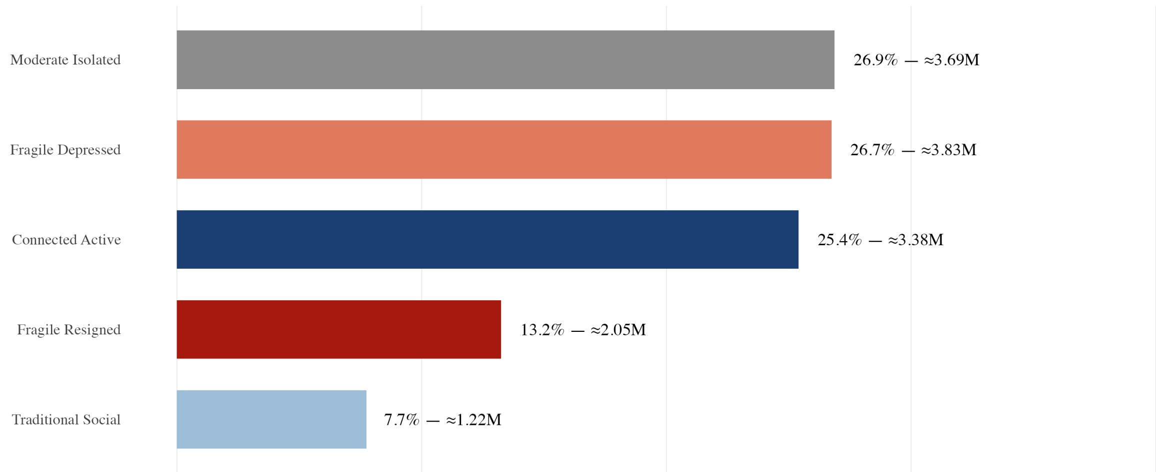


Figure 4.2: Sample shares and population projections of the five Italian archetypes. Bars report within-sample shares of the K-means partition on the SHARE Wave 9 Italian analytical sample ($n = 2{,}378$); each bar label gives the within-sample share and the weighted national projection (in millions). Projections to the national over-65 population apply post-hoc re-weighting with the SHARE design weight (`cciw_w9`; see Table B.1) and the 2024 Istat over-65 estimate (14.18 million). See §4.3 for the substantive interpretation of the small Traditional Social cluster.

Fragile Resigned (13.2%; mean age 81.5) occupies the most adverse position in the typology on every health and functional dimension (IADL +1.97, ADL +1.74, mobility +1.38, EURO-D +1.04). Economic resources are only modestly below the mean; the principal axis of fragility is functional autonomy, not income. The subjective signature is the lowest in the typology on every evaluative dimension: CASP-12 −1.02 (raw 27.8), loneliness +0.88, only 45.4% reporting hopes for the future against roughly 90% in the non-fragile profiles, with parallel deficits on interest, expected survival and cognition, supporting the label "Resigned". The cluster is the principal target of the Italian long-term-care welfare branch and the lower bound of the silver-economy market: digital and behavioural channels are essentially closed, and service consumption is mediated by family caregivers and institutional providers (Saraceno, 2010).

Fragile Depressed (26.7%; mean age 77.3), the second-largest archetype, occupies an intermediate position on functional and self-rated health (sphus +0.47, mobility +0.36) coupled with elevated depressive affect (EURO-D +0.54, raw 4.2, just above the four-point clinical threshold). Economic resources are below the Italian mean across all indicators. The subjective signature is the second-lowest in the typology but qualitatively distinct from Fragile Resigned: CASP −0.67, loneliness +0.59 (raw 5.4), hope −0.28 (67.6% reporting hopes, above 45.4% but below the non-fragile profiles). This is elevated depressive affect and loneliness without the same withdrawal from hope and interest, an active depressive episode rather than a settled resignation. The contrast with Fragile Resigned is one of the principal arguments for retaining $k = 5$ over $k = 4$ (§3.6); the cluster is over-represented among Italian women (31% of women vs 22% of men; §4.6). With internet use at 17.3%, behavioural and digital channels remain partially open, but the affective barrier is the binding constraint on adoption.

Moderate Isolated (26.9%; mean age 72.8) is the largest archetype and the youngest. It carries no objective adversity: health, functional and economic indicators sit at or near the Italian mean (EURO-D −0.68, raw 1.2 below the clinical threshold; mobility −0.61; income and wealth within 0.33 SD). What defines it is the social, cultural and digital block, where the centroid lies systematically below the mean: network size −0.22 (raw 2.3 confidants), internet use −0.11 (raw 34.1%), three cultural-engagement indicators between −0.15 and −0.37. No single variable is extreme; the cluster is moderately but consistently thinner across the whole social-information block, and its content rests on that consistency rather than on any one gap. It does not report itself as lonely despite the thin network (loneliness raw 3.6), in line with the finding that late-life loneliness reflects the gap between desired and actual ties more than absolute network size (Hawkley & Cacioppo, 2010). It is the most distinctive feature of the Italian typology and has no counterpart in Sweden (Chapter 5). Its reading as the signature of the Mediterranean familistic regime is developed in Chapter 6, and its healthcare consequences in §4.7 as the Isolation Paradox.

Traditional Social (7.7%; mean age 73.8), the smallest archetype, sits in the upper region of resource and health distributions, broadly comparable to Connected Active. The diagnostic feature is cultural and social engagement, where the centroid reaches its most extreme position in the typology (+3.29 on the cultural-engagement indicator ac035d5, with parallel positive loadings on educational participation, help to grandchildren, and social integration +0.62). Internet use is +0.53 (raw 65.8%), above the Italian mean but below Connected Active (74.8%): a dense traditional pattern of cultural attendance and intergenerational support with moderate digital engagement that complements rather than substitutes offline activities. CASP-12 is the highest in the typology (raw 38.5), loneliness low, hope positive (89.7%). The small size in

SHARE Wave 9 ($n = 184$) is itself a substantive finding. Pre-pandemic European segmentation studies consistently recovered a substantial "traditional" cluster combining dense offline networks with limited digital engagement; in the W9 Italian sample (2021–2022), by contrast, the densely connected over-65 are also the most digitally integrated and largely fall into Connected Active. The residual cluster retains the historical signature (high cultural participation + moderate-only internet) and represents a numerically narrow but distinct market for hybrid intermediated designs layering a digital interface onto an existing offline trust relationship.

Connected Active (25.4%; mean age 73.9), the largest well-resourced profile, sits at the high-resource end of the typology on every economic dimension (log_thinc $+0.72$, log_hnetw $+0.64$) and every health dimension (EURO-D -0.25, mobility -0.30). The diagnostic feature is digital and informational engagement: internet use $+0.72$ (raw 74.8%, the highest), educational participation $+0.74$, verbal fluency $+0.63$ (the highest cognitive score), network size $+0.42$ (raw 3.0, the largest). The configuration matches the "successful ageing" pattern (Rowe & Kahn, 1997) on a contemporary Italian sample (CASP-12 raw 37.8, 90.9% reporting hopes), and maps onto the Swedish Connected Wealthy archetype at a smaller share (25.4% vs 30.0%; Chapter 6). It is the immediate ready-to-serve segment for direct-to-consumer digital products targeted at the Italian over-65 population. Together with the smaller Traditional Social cluster, the high-resource pole accounts for 33.1% of the analytical sample; the addressability and design-logic implications of this two-fold split are developed in Chapter 6.

4.4 Integrative view: the typology fingerprint

The five-archetype centroid configuration is visualised in two complementary forms. Figure 4.3 reports the profile fingerprint: the standardised mean of each archetype across the 29 active variables, plotted as one line per archetype, with variables grouped by dominant factor loading (§3.5). Figure 4.4 plots all Italian respondents on the Factor 1 × Factor 2 plane (Factor 1: physical and functional fragility; Factor 2: subjective well-being, plotted inverted so high values correspond to elevated EURO-D and loneliness), with diamonds marking cluster centroids and ellipses marking 50% confidence regions per profile.

The two figures provide complementary readings. The fingerprint (Figure 4.3) confirms the three-fold structure already described profile by profile in §4.3: the two fragile profiles, the country-specific middle-stratum split between Moderate Isolated and Traditional Social, and the high-resource pole. The Factor 1 × Factor 2 scatter (Figure 4.4) confirms that the five profiles occupy distinct regions of the latent space rather than a single severity gradient, with Connected Active at the low-fragility pole and Fragile Resigned at the opposite corner on both axes, the visual counterpart of the argument of §4.3.

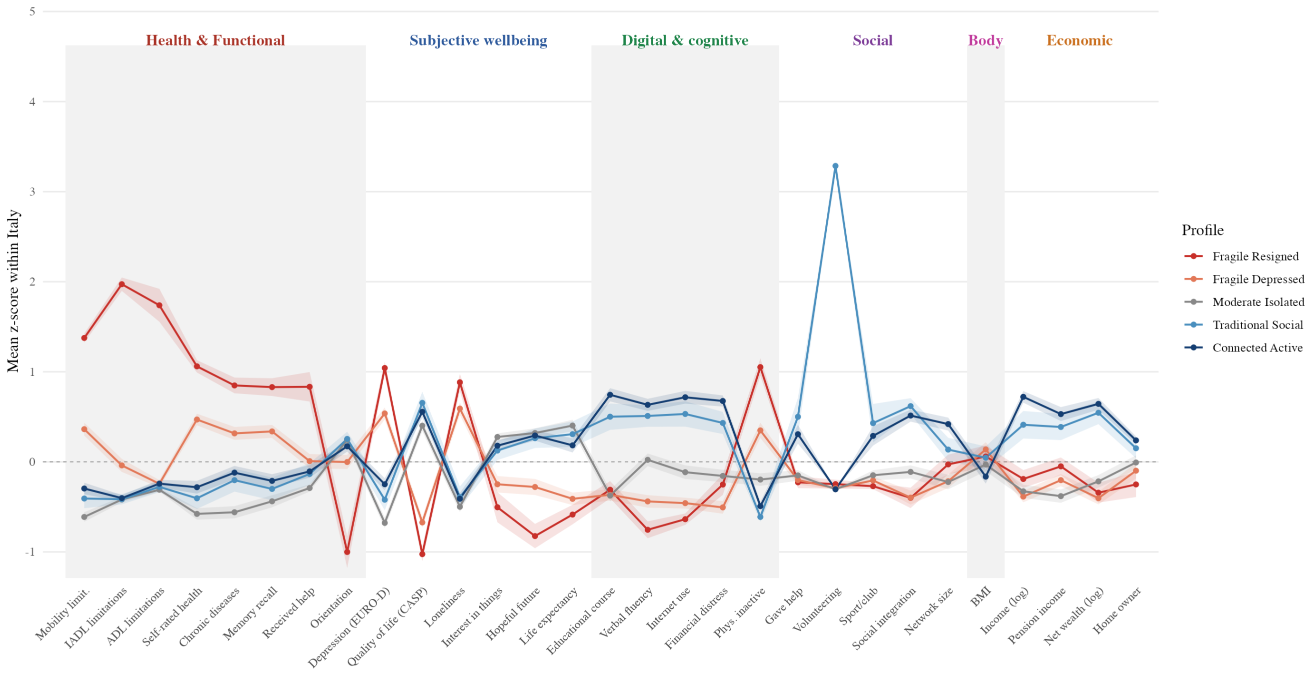


Figure 4.3: Profile fingerprint, Italy. Mean z-score per profile across the 29 active analytical variables, with 95% bootstrap confidence ribbons ($n = 2{,}378$). Variables grouped by dominant factor loading and ordered within factor; x-axis labels colour-coded by dimension.

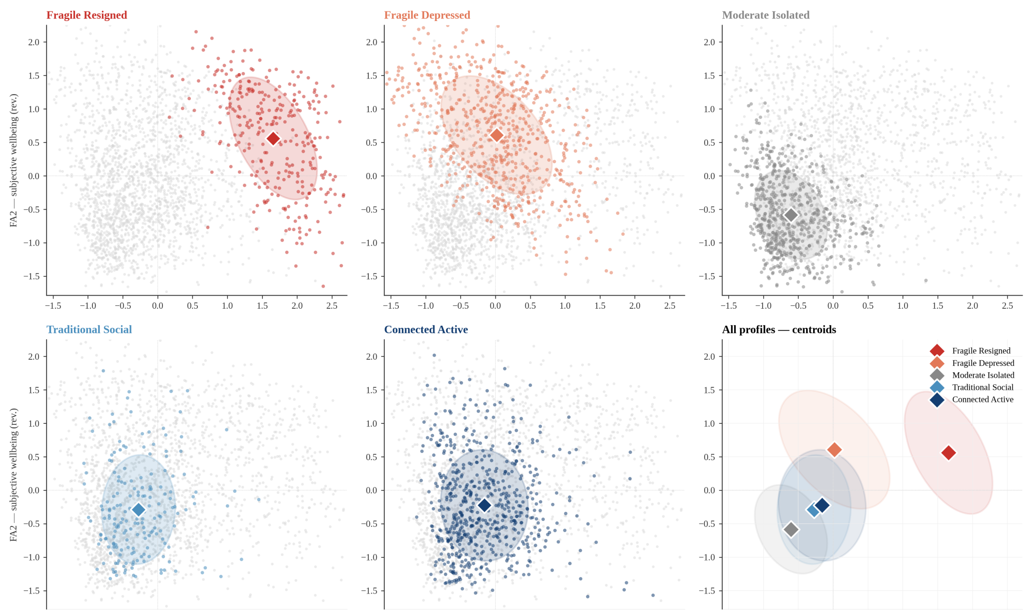


Figure 4.4: Profile scatter on the Factor 1 $\times$ Factor 2 plane, Italy. Small-multiple layout: one panel per archetype highlighting that profile against the rest of the sample, plus a summary panel showing the five cluster centroids on the common plane. Each point is one respondent on the two dominant factors (Factor 1: physical and functional fragility; Factor 2: subjective well-being, plotted inverted so high values correspond to elevated EURO-D and loneliness). Diamonds mark cluster centroids; ellipses mark 50% confidence regions per profile.

4.5 External validation: life satisfaction

Life satisfaction (SHARE Wave 9 item AC012, eleven-point scale), administered in a separate questionnaire block from the analytical input set, provides an external validation outcome (§3.8). Figure 4.5 reports the distribution across the five Italian archetypes; the F-statistic of a one-way ANOVA is reported in the caption.

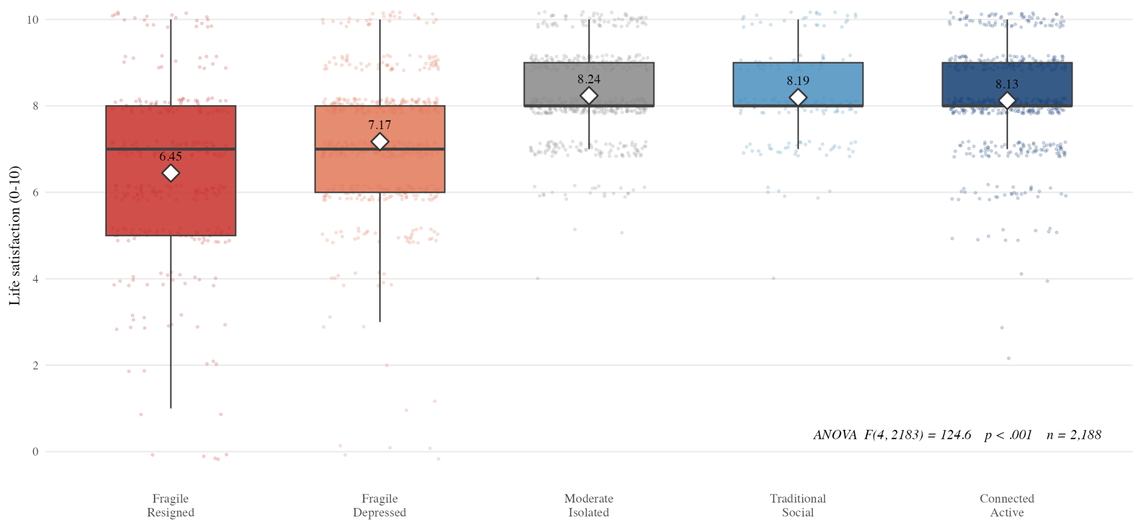
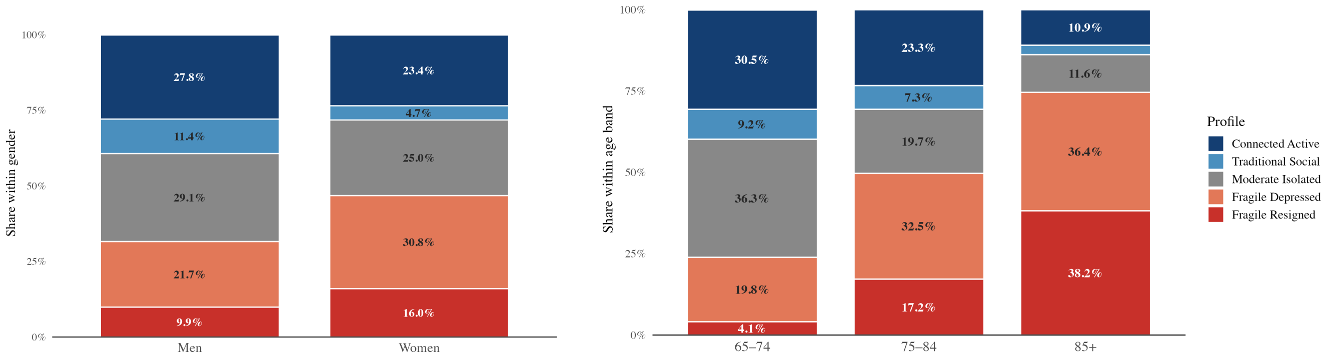


Figure 4.5: Life satisfaction by profile, Italy. Boxplots of SHARE Wave 9 item AC012 (eleven-point scale, 0–10) by cluster membership; diamonds report cluster means. One-way ANOVA: $F(4,2183) = 124.6$, $p < .001$. The contrast is principally between fragile (6.5–7.2) and non-fragile profiles (8.1–8.2). Life satisfaction is external to the K-means input set. $n = 2{,}188$.

Mean life satisfaction is 6.5 in Fragile Resigned, 7.2 in Fragile Depressed, and clusters around 8.1–8.2 in the three non-fragile profiles, which are close in mean and not separated in the boxplot distributions. The F-statistic of 124.6 on (4, 2183) degrees of freedom ($p < 0.001$) confirms substantial discriminative power of the typology on an outcome external to the input set, mainly along the fragile/non-fragile divide. The life-satisfaction outcome detects principally the binary contrast between fragile and non-fragile profiles (a gap of between one and one-and-a-half points on the eleven-point scale); the within-non-fragile contrasts are minor, consistent with the observation that life satisfaction in older age is dominated by the absence of major health and affective burdens. This supports including the affective and evaluative subjective block as a constituent input: it preserves the within-fragile contrast (Fragile Resigned vs Fragile Depressed) that life satisfaction alone would fail to detect.

Life satisfaction is external in measurement terms (administered in a separate questionnaire block from the analytical input set) but is conceptually adjacent to the subjective block constituents (CASP-12, loneliness, and hope for the future), all of which enter the K-means partition as inputs. The validation reported here is therefore best read as a construct-validity



(a) By gender

(b) By age band (65–74, 75–84, 85+)

Figure 4.6: Profile composition by gender and age band, Italy. Each column shows the percentage of the demographic group falling in each archetype; columns sum to 100%. Panel (b) reports the age gradient discussed in §4.6. Legend on panel (b) applies to both panels.

test on a partially overlapping latent dimension rather than a fully independent confirmation; complementary external validation against indicators with no measurement adjacency to the input set (healthcare-utilisation patterns at the country-aggregate and matched-pair level) is developed in Chapter 6 (§6.5) and consolidated in Chapter 7 (§7.4).

4.6 Demographic distributions across profiles

The five archetypes are not uniformly distributed across gender and age. Figure 4.6 reports profile composition by gender (panel a) and by age band 65–74, 75–84, 85+ (panel b), each as a 100% stacked bar.

Two patterns emerge. First, the feminisation of fragility: 47% of Italian over-65 women fall in one of the two fragile profiles (16% Fragile Resigned + 31% Fragile Depressed), against 32% of men (10% + 22%). Symmetrically, 39% of men are in the high-resource profiles (28% Connected Active + 11% Traditional Social), against 28% of women. The pattern is consistent with the broader epidemiological literature on the differential ageing of Italian women: longer life expectancy paired with higher widowhood rates, lower lifetime labour-market attachment, and lower individual pension income.

Second, the age gradient is strong. Comparing the 65–74 and 85+ bands, the Fragile Resigned share rises from 4% to 38%, the Fragile Depressed share from 20% to 36% (with a monotonic increase across the three bands: 20% $\rightarrow$ 33% $\rightarrow$ 36%), the Moderate Isolated share declines from 36% to 12%, the Traditional Social share from 9% to 3%, and the Connected Active share from 31% to 11%. Both well-resourced profiles thus contract sharply with age while both fragile profiles expand, a pattern consistent with both age-related shifts out of the autonomous clusters and cohort-level differences in engagement that the cross-sectional design cannot disentangle. The most vulnerable group (women, oldest old, fragile) is also the segment for whom standard digital channels of the silver economy are essentially inaccessible without family or institutional intermediation, a finding with direct implications for the design of long-term-care services and digital products targeting the Italian over-65 population.

Beyond gender and age, two further demographic dimensions vary systematically across the typology. The educational gradient is monotone: mean years of schooling rise from 6.8 (Fragile Resigned) to 10.9 (Connected Active), and the share with at most lower-secondary schooling (ISCED 0–2) contracts from 82% to 52% (72% in Moderate Isolated). Living arrangement shows a more selective pattern: the two fragile profiles report 40–45% of respondents living alone, against 15–19% in all three non-fragile profiles. Notably, the Moderate Isolated archetype (the cluster developed as the empirical signature of the Isolation Paradox in §4.7) reports only 18% of respondents living alone, close to Connected Active (15%). Its social thinness therefore persists *within* intact coupled households and is not residentially reducible to widowhood or solitary living. This pattern is consistent with the loneliness–network configuration discussed in §4.3 and reinforces the substantive claim of §4.7: the diagnostic feature of Moderate Isolated is a thin informational and social ecosystem, not the structural absence of a co-resident partner.

4.6.1 Demographic predictors of cluster membership

To quantify how much of the typological structure recovered in §4.3 is reducible to standard demographic stratifiers, a multinomial logistic regression of cluster membership on age, gender, years of education, living-alone status and homeownership is estimated on the $n = 2{,}188$ Italian respondents with complete predictor coverage (the same complete-case subsample as §4.5). The per-cluster counts in Table 4.2 are therefore the complete-predictor subset and lie below the full cluster sizes of §4.2 (for example, Moderate Isolated $n = 548$ against a full-cluster $n = 639$). Moderate Isolated is set as the reference category to mirror the focus of §4.7 on the Isolation Paradox. Table 4.2 reports the resulting odds ratios with 95% confidence intervals.

The likelihood-ratio test rejects the null of demographic non-association at a very large margin: $\chi^2(20) = 763.7$, $p < .001$. The fit measures, however, locate the bulk of the typological variability outside the demographic input set. Nagelkerke pseudo-$R^2 = 0.309$ indicates that approximately one third of the cluster distribution is captured by age, gender, education, living arrangement and homeownership combined. The remaining two thirds reflect the conceptual structure of the segmentation: factor-analytic dimensions of physical health, subjective burden, digital and social engagement (§3.5, §4.3). The typology is therefore demographically structured but not demographically reducible.

Table 4.2: Multinomial logistic regression of cluster membership on demographic predictors, Italy ($n =$ 2,188)

	Odds ratio vs Moderate Isolated [95% CI]			
Predictor	Fragile Resigned	Fragile Depressed	Traditional Social	Connected Active
Age (per year from 75)	1.20*** [1.17, 1.23]	1.09*** [1.07, 1.12]	1.02 [0.99, 1.05]	1.04*** [1.02, 1.06]
Female (vs male)	1.86*** [1.33, 2.60]	1.42** [1.09, 1.84]	0.54** [0.37, 0.80]	1.25 [0.97, 1.62]
Years of education (per year from 8)	0.94** [0.90, 0.98]	0.93*** [0.90, 0.96]	1.10*** [1.06, 1.15]	1.13*** [1.10, 1.17]
Living alone (vs partnered)	1.89*** [1.32, 2.71]	2.24*** [1.67, 3.01]	1.32 [0.83, 2.09]	0.72 [0.51, 1.01]
Homeowner (vs renter)	0.47** [0.30, 0.75]	0.80 [0.54, 1.18]	1.71 [0.84, 3.49]	3.55*** [1.95, 6.46]
n in cluster	313	607	170	550

Note. Reference category is Moderate Isolated ($n = 548$). Cells report the odds ratio of cluster membership relative to Moderate Isolated for a one-unit change in the predictor, with 95% confidence intervals in brackets. Age is centred at 75 years (sample median) and years of education at 8 years (sample mean rounded). Female, Living alone and Homeowner are 0/1 binary indicators. Nagelkerke pseudo-$R^2 = 0.309$ (Cox–Snell 0.295, McFadden 0.114); likelihood-ratio $\chi^2(20) = 763.7$, $p < .001$. Significance: $^{***}p < .001$, $^{**}p < .01$, $^{*}p < .05$.

The pattern of odds ratios is consistent with the substantive descriptions of §4.3: education separates every profile from Moderate Isolated (lower for the two fragile profiles, higher for Traditional Social and Connected Active), older age and living alone distinguish the fragile profiles (living-alone odds of 1.89 and 2.24), and homeownership sharply marks the high-resource Connected Active profile (odds ratio 3.55 relative to Moderate Isolated).

The regression is descriptive rather than causal: it bounds the demographic explainability of the partition from above, and the residual variability ($1 - R^2_N \approx 0.69$) is the portion of the cluster signal that originates in the analytical input variables themselves and that the typology of §4.3 is designed to make legible.

4.7 The Isolation Paradox

The most distinctive finding of the Italian typology, and one of its central contributions, is the Isolation Paradox. A substantial Italian segment is functionally and economically autonomous yet socially and digitally disconnected, and it under-utilises the healthcare system relative to peers with comparable health, net of resource differences, but denser social engagement. That segment is the Moderate Isolated archetype (§4.3), 26.9% of the Italian analytical sample.

The empirical signature is documented descriptively in §4.7.1 and supported by the regression-adjusted analysis of §4.7.2, Table 4.3 (OLS and linear-probability models on the joint Moderate Isolated + Traditional Social sub-sample, controlling for age, chronic conditions, self-rated health, log household income, gender and depressive symptoms). The country-specific character of the configuration is anticipated in §4.7.3 and developed formally through the matched-pair analysis of Chapter 6 (§6.5.3); the Swedish typology that confirms the absence of an analogous configuration is presented in Chapter 5.

4.7.1 The empirical signature

The Moderate Isolated cluster shares two key features with Traditional Social: comparable health (mobility 0.6 vs 0.9 raw count, EURO-D 1.2 vs 1.7) and similar subjective evaluation of life (CASP 36.8 vs 38.5, both above the Italian mean). On economic resources, Traditional Social is moderately higher than Moderate Isolated (a gap of approximately 0.7 SD on income and wealth), which the regression-adjusted analysis of §4.7.2 controls for in isolating the social-engagement effect. The two clusters diverge sharply on social, cultural and digital engagement: Moderate Isolated sits at $z = -0.22$ on network size against $+0.14$ for Traditional Social, at $z = -0.11$ on internet use against $+0.53$, and at z between -0.15 and -0.37 on cultural-engagement indicators against $+0.43$ to $+3.29$. In raw figures: 2.3 confidants vs 2.6, 34.1% internet use vs 65.8%, half the cultural-event participation rate. Despite these differences, the two clusters report near-identical life satisfaction (8.2 vs 8.2), differing only on the small CASP-12 gap noted above: the social isolation is not strongly internalised as a deficit on subjective-well-being indicators.

4.7.2 The healthcare under-utilisation pattern

The Isolation Paradox is substantiated through two complementary analyses. The first reports the descriptive cluster means on the six healthcare-utilisation indicators considered. Relative to Traditional Social, Moderate Isolated reports 1.59 fewer doctor visits per year (raw means 4.61 vs 6.20 on the twelve-month reference period), comparable general-practitioner contacts (4.19 vs 4.37), 0.94 fewer specialist visits (1.08 vs 2.02), and a 26.8 percentage-point lower preventive dental-care attendance rate (raw 21.0% vs 47.8%). It also reports a 0.4 percentage-point higher rate of forgone care for cost (respondents who report having needed but not received medical attention; 5.8% vs 5.4%) and, on a small hospitalised sub-sample, 2.0 fewer nights in hospital per year (6.64 vs 8.67). Moderate Isolated lies below Traditional Social on five of the six indicators (the exception being forgone care, where the small gap is not informative against the country baseline of 10.1%). The descriptive signature is consistent with a configuration of systematic disengagement from the formal healthcare system rather than with delayed care-seeking.

The second analysis tests whether the descriptive gaps are robust to the joint adjustment for the principal confounders (age, chronic-condition count, self-rated health, log household income, gender, and EURO-D depressive-symptoms score) through ordinary-least-squares regression on the union of the two cluster sub-samples. Table 4.3 reports the cluster coefficient

(Moderate Isolated relative to Traditional Social as reference) for the five outpatient and preventive indicators with adequate sub-sample coverage (specialist contacts, preventive dental attendance, doctor visits, general-practitioner contacts, and forgone care for cost), excluding the inpatient nights measure, whose hospitalised sub-sample is too small to support the controlled regression.

Table 4.3: Regression-adjusted cluster gap, Moderate Isolated vs Traditional Social

Outcome	**Coef.**	**SE**	**95% CI**	*p*	*n*	**Sig.**
Specialist contacts (12m count)	−0.669	0.173	[−1.01, −0.33]	<0.001	699	***
Dentist 12m attendance (%)	−27.9	3.7	[−35.2, −20.6]	<0.001	823	***
Doctor visits (12m count)	−0.864	0.612	[−2.06, +0.34]	0.159	818	n.s.
GP contacts (12m count)	+0.264	0.348	[−0.42, +0.95]	0.448	700	n.s.
Forgone care for cost (%)	+0.5	2.0	[−3.4, +4.4]	0.788	823	n.s.

Note. OLS regression on the joint Moderate Isolated + Traditional Social sub-sample. Reference category: Traditional Social. Controls: age, chronic conditions, self-rated health, log household income, gender, EURO-D depressive-symptoms score. Coefficients on binary outcomes (dentist attendance, forgone care) are linear-probability-model estimates expressed in percentage points. Significance: *** $p < 0.001$, ** $p < 0.01$, * $p < 0.05$, n.s. $p \geq 0.05$.

The regression returns two robust gaps and three gaps that do not survive adjustment. The Moderate Isolated cluster reports 0.67 fewer specialist contacts per year and a 27.9 percentage-point lower dental-care attendance rate than the Traditional Social cluster, both highly significant ($p < 0.001$) and large relative to the Traditional Social baseline: a 0.67-contact reduction against a Traditional Social specialist mean of 2.02 per year, and a 27.9-percentage-point reduction against a Traditional Social dental-attendance rate of 47.8%. The doctor-visit gap observed in raw means is absorbed by the joint controls, principally by the age and chronic-condition adjustments; GP contacts show no meaningful gap in either the descriptive or the adjusted analysis, and the forgone-care gap is not statistically distinguishable from zero in either.

The substantively important reading is that the Isolation Paradox is robust on the two indicators that capture engagement with the discretionary, navigation-intensive segments of the healthcare system: the specialist channel, which in the Italian regional health-service architecture requires explicit information-seeking and out-of-pocket co-payment, and the preventive dental channel, which is largely outside the universal health-service entitlement and requires private expenditure. The under-engagement on these two channels (0.67 specialist contacts per year and 28 percentage points on dental attendance, both controlling for the principal confounders) is the operational core of the Isolation Paradox. The descriptive cluster means on doctor visits, specialist contacts, forgone care and hospital nights across all five archetypes are plotted in Figure 4.7. The aggregate doctor-visit and GP-contact behaviour is not informative once age and chronic load are controlled for, indicating that the descriptive volume of contact with the universal-coverage primary-care channel is approximately invariant to the social-and-informational thinness that distinguishes Moderate Isolated from Traditional Social.

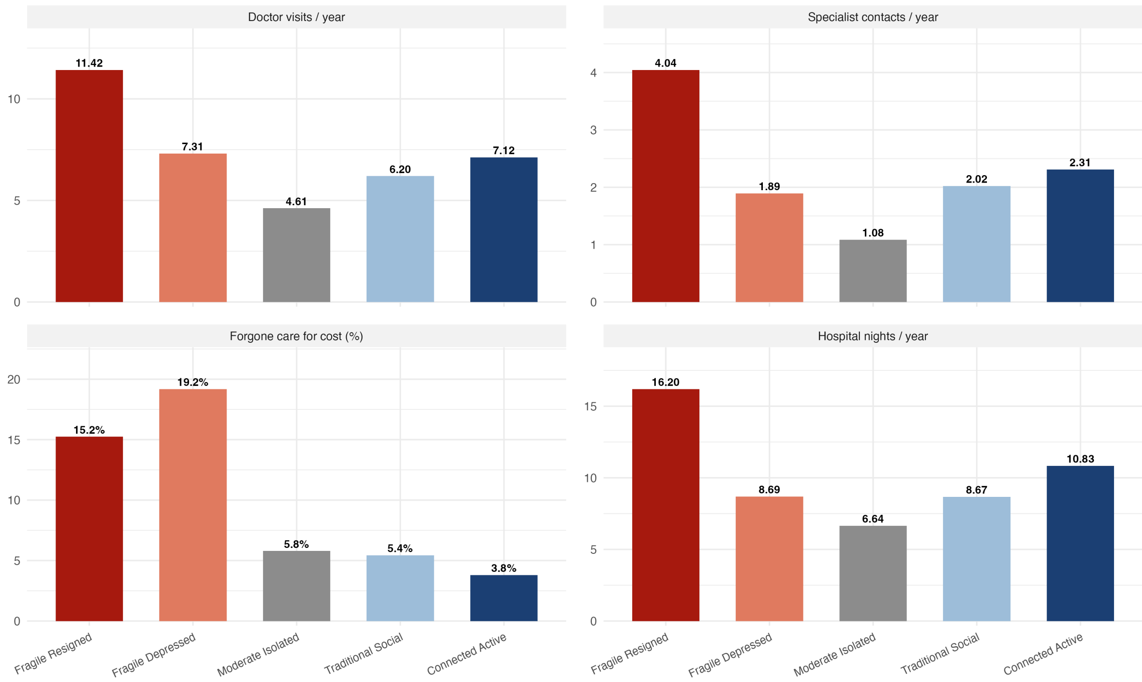


Figure 4.7: Healthcare utilisation across the five Italian archetypes. Cluster means on four indicators over the twelve months preceding interview: doctor visits per year, specialist contacts per year, forgone care for cost (%), and hospital nights per year. The GP-contact and preventive (dental) channels are not displayed here; both are reported in the regression-adjusted analysis of §4.7.2 and Table 4.3.

The mechanism is consistent with three convergent literatures. The literature on social capital and healthcare access (Berkman & Kawachi, 2000) documents the role of dense social networks in transmitting symptom information and signposting individuals towards appropriate care. The gerontological literature on the gateway role of grandchildren and adult children in mediating digital and informational access of older adults (Quan-Haase et al., 2017) identifies a channel to which the Moderate Isolated cluster, with its below-average network density and limited cultural and social participation, has materially reduced access. The Italian welfare-state literature (Lo Scalzo et al., 2009) documents a regional fragmentation of public health services that makes informal social capital a more critical input for healthcare navigation than in regimes designed for universal access.

4.7.3 Country-specificity of the Paradox: the comparative question

The Isolation Paradox has no structural counterpart in the Swedish typology of Chapter 5, where the Moderate archetype is socially integrated at the Swedish population mean. Its Italian specificity is established formally through the matched-pair analysis of Chapter 6 (§6.5.3).

4.7.4 Policy and silver-economy implications

The Isolation Paradox identifies a first-order policy and market reality rather than a residual category: at 26.9% of the analytical sample it is the largest single cluster, yet, facing neither health nor economic adversity, the cluster is invisible to the existing long-term-care and income-support branches of the Italian welfare system. Its binding constraint is not the absence of a digital channel (internet use at 34.1% is non-trivial) but the absence of the social and informational mediation that would route that channel towards healthcare and service utilisation; the candidate solution combines a digital channel with a social-mediation layer (peer-network platforms, community-based digital intermediaries, family-network-augmented service interfaces). The strategic case for this product-design gap as an opportunity for the Italian silver economy is developed in Chapter 6.

4.8 Synthesis

The Italian typology partitions the over-65 population into five archetypes spanning a fragility-and-resources gradient, with an affective contrast (Fragile Resigned versus Fragile Depressed) within the fragile stratum and a country-specific middle cluster, Moderate Isolated, that lies off the principal gradient. External validation against life satisfaction confirms substantial discriminative power ($F(4,2183) = 124.6$, $p < 0.001$). The headline finding is the Isolation Paradox: the Moderate Isolated segment, the largest single cluster at 26.9%, is functionally and economically autonomous yet systematically disengages from the formal healthcare system, in a pattern consistent with the welfare-regime literature on Italian familism and the gerontological literature on social capital and healthcare access. Whether this Isolation Paradox is a feature of late life everywhere or the signature of one welfare regime is the question the Swedish typology of Chapter 5 is built to answer—not by mirroring the Italian inventory, but by establishing the structural baseline against which the country-specificity of the paradox is measured in Chapter 6.

Chapter 5

The Swedish Typology

The structure of late-life experience under universal welfare

5.1 Overview

The K-means segmentation of Chapter 3 recovers six archetypes spanning the Swedish over-65 analytical sample. The Swedish typology is not a Nordic counterpart of the Italian inventory of Chapter 4: the two partitions differ in structural shape on three axes, developed in §5.4 and read in Chapter 6 as the empirical signature of the Nordic-universalistic welfare regime. Because the comparative payoff of Chapter 6 turns on these three structural axes rather than on a profile-by-profile inventory, this chapter treats the six archetypes more concisely than Chapter 4 and organises the analysis around the features that the cross-country reading projects.

All centroid coordinates are reported as z-scores within Sweden on the analytical sample (n = 1,787). Cluster shares are reported as percentages of the analytical sample; see §3.4 for the descriptive/predictive interpretation of the partition.

5.2 The six archetypes

The K-means partition ($k = 6$), estimated on the 31 active analytical variables, groups the 1,787 Swedish respondents into six clusters. Figure 5.1 reports cluster mean values on eight diagnostic variables, with profiles ordered from most fragile (left) to most connected (right). Table 5.1 lists the archetypes with their headline signature and sample shares.

Two raw-scale observations from Figure 5.1 anchor the descriptions that follow. First, the Fragile cluster is the only profile with non-trivial functional burden (mobility 3.79, ADL 0.69, IADL 1.54); the remaining five profiles are all in the autonomous range (mobility $\leq$ 1.55, ADL $\leq$ 0.10). Second, only the Fragile cluster is below 75% internet use (63.6%); the other five profiles span 75.1% to 98.7%. Neither observation has a parallel in the Italian typology, and both are developed as substantive findings in §5.4.

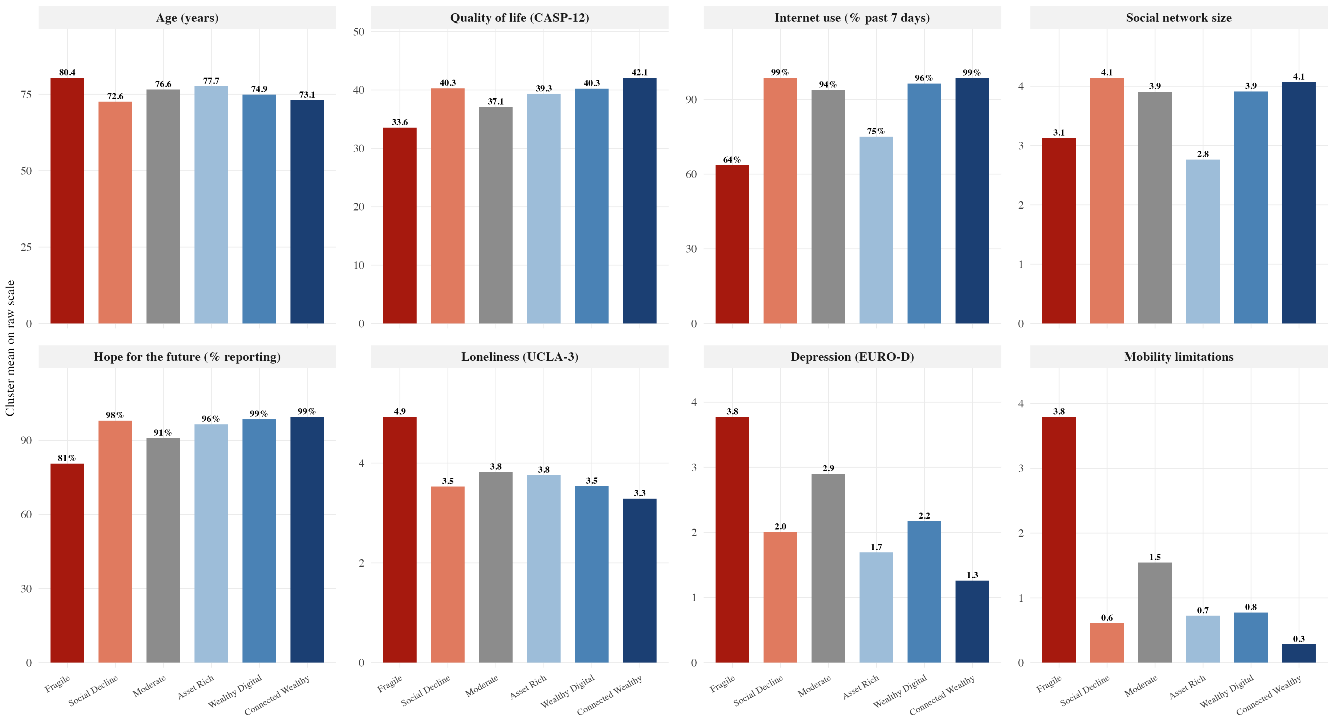


Figure 5.1: Profile characteristics, Sweden. Cluster means on eight key dimensions: age, CASP-12, internet use (%), social network size, hope for the future (%), loneliness (UCLA-3), EURO-D, mobility limitations. Profiles ordered from most fragile to most connected. $n = 1{,}787$.

Table 5.1: Headline signature of the six Swedish archetypes

Archetype	Share	Headline signature
Fragile	11.5%	Worst health (mobility 3.8, ADL 0.69, IADL 1.54), oldest age (80.4 yr), lowest internet use (63.6%), lowest CASP (33.6) and lowest hope (80.6%). The only true fragile configuration in Sweden.
Social Decline	8.3%	Despite the canonical label, the youngest archetype (72.6 yr) with full digital adoption (98.7%) and the largest network (4.14). A residual W9 cluster discussed in §5.3.
Moderate	23.3%	Average health and resources, internet 93.8%, network 3.91. The Swedish middle stratum.
Asset Rich	19.1%	Older (77.7 yr), wealth-anchored, intact health, internet 75.1%, smallest network (2.77).
Wealthy Digital	7.7%	Younger well-resourced cluster (74.9 yr), high digital adoption (96.4%), high CASP (40.3).
Connected Wealthy	30.0%	Near-universal digital adoption (98.5%), best resources, intact health, highest CASP (42.1) and lowest loneliness (3.3). The largest single archetype.

Note. Headline values are raw means of the diagnostic variables of Figure 5.1. Standardised centroid coordinates are reported in §5.3. Shares may not sum to 100% due to rounding.

5.3 Profile descriptions

The six archetypes are described below from most fragile to most connected, in the order of Table 5.1. Standardised values are z-units within Sweden ($\bar{x} = 0$, $\sigma = 1$); raw means in parentheses where useful.

Fragile (11.5%; mean age 80.4) is the only archetype with substantial burden on every health and functional dimension (mobility $+1.95$, IADL $+1.79$, ADL $+1.55$, EURO-D $+1.05$) and the lowest subjective signature in the typology (CASP -1.43, raw 33.6; loneliness $+1.07$; hope -1.45, 80.6% against $\geq 90\%$ in the non-fragile profiles). Its internet use, 63.6%, is materially below that of the other five Swedish profiles (all $\geq 75\%$). It is nonetheless several times higher than in the Italian fragile profiles (8.6% Fragile Resigned, 17.3% Fragile Depressed; §4.3). The cluster carries the entire age burden, its share rising from 6% in 65–74 to 37% in 85+ (§5.6); the contrast with the two qualitatively distinct Italian fragile profiles is the first distinctive feature (§5.4.1).

The remaining 88.5% partitions across five autonomous profiles that share intact functional health (mobility ≤ 1.55, ADL ≤ 0.10, IADL ≤ 0.23) and high subjective well-being (CASP ≥ 37, life satisfaction ≥ 8.0), separating on economic resources and engagement style. **Connected Wealthy** (30.0%; mean age 73.1) is the largest archetype and the high-resource, high-digital pole on every dimension (internet 98.5%, network 4.07, CASP 42.1, loneliness 3.3), the structural analogue of the Italian Connected Active with a larger share. **Asset Rich** (19.1%; mean age 77.7) is the older, wealth-anchored variant with intact health, lower digital engagement (internet 75.1%) and the smallest network (2.77). **Wealthy Digital** (7.7%; mean age 74.9) is a younger well-resourced cluster with high digital adoption (internet 96.4%, CASP 40.3) and a thin LCA boundary with Connected Wealthy (49.3% convergence). Asset Rich and Wealthy Digital are the two profiles with no Italian counterpart.

Moderate (23.3%; mean age 76.6) is the middle-stratum cluster centred on the Swedish mean (internet 93.8%, network 3.91), the counterpart of the autonomous middle stratum that in Italy splits into Moderate Isolated and Traditional Social. **Social Decline** (8.3%; mean age 72.6) is, despite the canonical label, the youngest and most residual cluster (internet 98.7%, network 4.14, CASP 40.3, hope 98.0%), capturing younger, fully digital, socially integrated individuals distinguished by moderate income rather than declining engagement; the label is retained for nomenclature continuity with the Wave 8 solution (Chapter 7).

5.4 Three distinctive features of the Swedish typology

Three substantive findings distinguish the Swedish partition from the Italian one and carry the cross-country comparison of Chapter 6: the binary structure of Swedish fragility (§5.4.1), the universal digital saturation across the autonomous profiles (§5.4.2), and the age-driven rotation of the high-resource pole (§5.4.3).

5.4.1 Binary fragility versus graduated fragility

The Swedish typology partitions the over-65 population along a binary fragility line: a single Fragile cluster (11.5%) carries the entire functional burden, against five autonomous profiles (88.5%) that share intact health and high subjective well-being. The Italian typology partitions the same population along a graduated fragility line: two qualitatively distinct fragile profiles (Fragile Resigned 13.2% and Fragile Depressed 26.7%, jointly 39.9%) span an active-depression-versus-settled-resignation contrast that drives separate policy responses (mental-health services in the first case, long-term-care provision in the second; §4.3).

The two countries place their principal fragile cluster in the same region of the joint functional-affective burden space, but Italy adds a second, qualitatively distinct cluster (Fragile Depressed) with high affective burden without proportionate functional burden, while Sweden produces no analogous second cluster (the cross-country centroid plane is shown for the matched-pair analysis in Figure 6.1). The regime reading is developed in §6.4.1.

5.4.2 Universal digital saturation

Internet adoption in the five autonomous Swedish profiles ranges from 75.1% (Asset Rich) to 98.7% (Social Decline), four of the five above 93%. The Italian profiles outside the principal fragility cluster span a far wider range, from 17.3% (Fragile Depressed, whose burden is affective rather than functional) and 34.1% (Moderate Isolated) to 65.8% (Traditional Social) and 74.8% (Connected Active; §4.3). Figure 5.2 plots the per-profile means against a 75% reference line: in Sweden only the Fragile cluster lies below it, in Italy no profile reaches it. The Swedish digital floor is the Italian digital ceiling.

The substantive point is not that Sweden is more digitally adopted on average, but that digital adoption cuts at the functional-fragility line in Sweden and at the resource line in Italy; the regime reading and product-design implications are developed in §6.4.2.

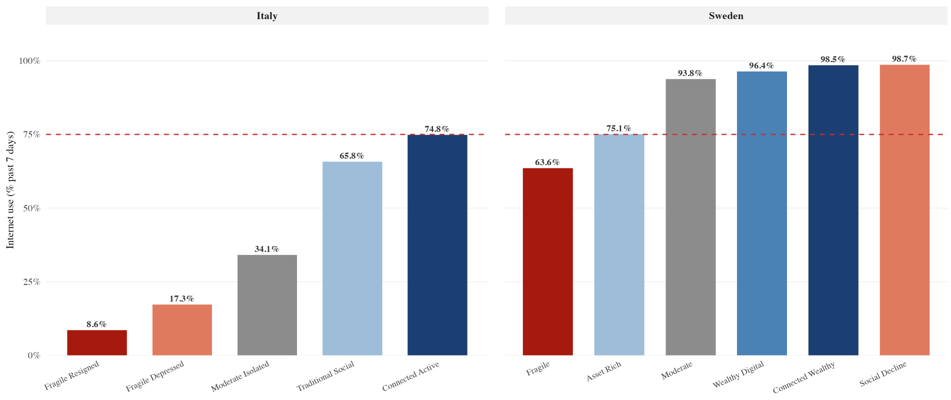


Figure 5.2: Internet use by profile, Italy and Sweden. Cluster means on the binary internet-use indicator (SHARE Wave 9). Dashed reference line at 75%. In Sweden only the Fragile cluster lies below 75%; in Italy no profile reaches 75%. The digital cut in Sweden tracks the functional-fragility line; in Italy it tracks the resource gradient.

5.4.3 The age-prosperity rotation

The Swedish high-resource pole is not occupied by a single archetype across the late-life trajectory. Figure 5.3 plots the within-age-band share of the two well-resourced profiles in each country across the 65–74, 75–84, 85+ bands.

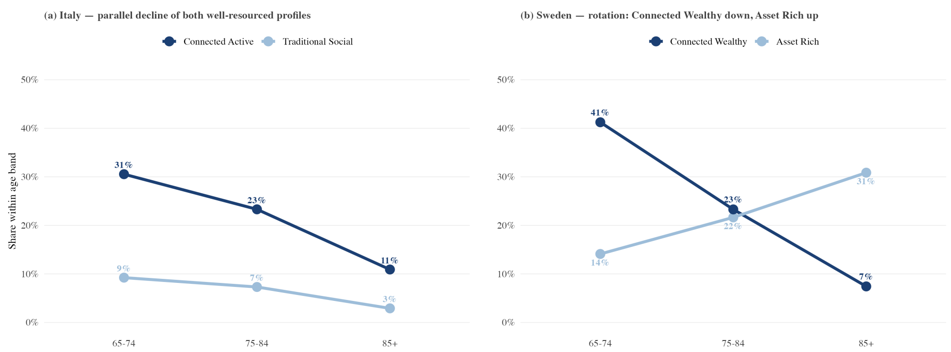


Figure 5.3: Share of well-resourced profiles within age band, Italy and Sweden. Italy (left) shows both Connected Active and Traditional Social declining in parallel with age. Sweden (right) shows Connected Wealthy contracting (41%→7%) while Asset Rich expands (14%→31%): the high-resource pole rotates from a digital-engagement anchor (young-old) to a wealth-accumulation anchor (oldest-old).

In Sweden the Connected Wealthy share declines from 41% to 7% across the three bands while the Asset Rich share rises from 14% to 31%, crossing over in the 75–84 band: the high-resource pole rotates from a digital-engagement anchor (Connected Wealthy, young-old) to a wealth-accumulation anchor (Asset Rich, oldest-old) as the cohort ages. The Italian typology has no analogue: both well-resourced profiles decline in parallel (Connected Active 31%→11%, Traditional Social 9%→3%). Because the design is cross-sectional, this age-band gradient is a between-cohort comparison rather than an observed within-cohort transition. One alternative to the ageing-trajectory reading is compositional: the wealth-anchored profile may simply survive into advanced age at a higher rate than the digitally anchored one. The two accounts cannot be separated without panel data (§9.5). The regime reading is developed in §6.4.3.

5.5 External validation: compressed variance under universal welfare

Life satisfaction (SHARE Wave 9 item AC012, eleven-point scale), administered in a separate questionnaire block from the analytical input set (§3.8), provides an external validation outcome. Figure 5.4 reports the distribution across the six Swedish archetypes; the F-statistic of a one-way ANOVA is reported in the caption.

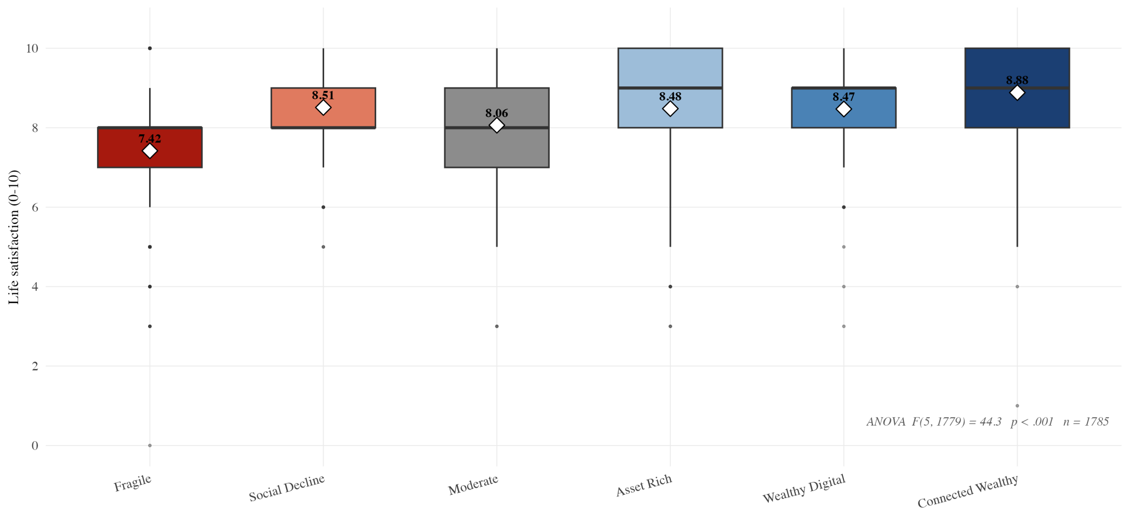


Figure 5.4: Life satisfaction by profile, Sweden. Boxplots of SHARE Wave 9 item AC012 (eleven-point scale, 0–10) by cluster membership; diamonds report cluster means. One-way ANOVA: $F(5,1779) = 44.3$, $p < .001$. Life satisfaction is external to the K-means input set. $n = 1{,}785$.

Mean life satisfaction is 7.42 in Fragile, against 8.06 (Moderate), 8.47 (Wealthy Digital), 8.48 (Asset Rich), 8.51 (Social Decline), and 8.88 (Connected Wealthy). The F-statistic of 44.3 on $(5,1779)$ degrees of freedom ($p < 0.001$) confirms significant discriminative power but at less than half the Italian magnitude ($F(4,2183) = 124.6$; §4.5). Because F is not comparable across partitions with different cluster counts, the effect size is reported as η^2: the typology accounts for approximately 19% of the life-satisfaction variance in Italy against 11% in Sweden. The non-fragile Swedish clusters span only 0.82 points (8.06–8.88), and even the most adverse Swedish segment (7.42) exceeds the means of both Italian fragile profiles (6.5 and 7.2). This pattern is consistent with the universal-welfare baseline compressing between-cluster variation

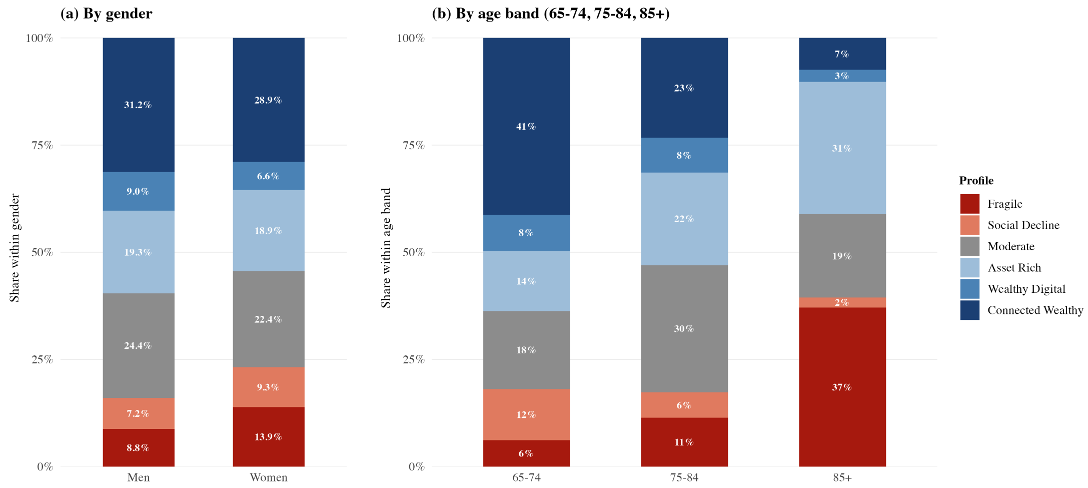


Figure 5.5: Profile composition by gender and age band, Sweden. Each column shows the percentage of the demographic group falling in each archetype; columns sum to 100%. Panel (b) shows the age-prosperity rotation discussed in §5.4.3. Legend on panel (b) applies to both panels.

in subjective well-being, in line with the welfare-regime literature on the de-commodification of late-life well-being under universal-access institutions (Esping-Andersen, 1990). As in Italy (§4.5), life satisfaction is external in measurement terms but conceptually adjacent to the subjective block (CASP-12, loneliness, hope) that enters the partition as input. It is therefore best read as a construct-validity check on a partially overlapping latent dimension. Measurement-independent validation is supplied by the healthcare-utilisation outcomes of §6.5 and §7.4.

5.6 Demographic distributions across profiles

Figure 5.5 reports profile composition by gender (panel a) and age band (panel b), each as a 100% stacked bar; the age-prosperity rotation visible in panel (b) is analysed in §5.4.3.

A gender gradient in late-life socio-economic position is present in Sweden but materially attenuated relative to Italy. The two profiles outside the high-resource and middle strata, Fragile (low functional health) and Social Decline (moderate income, not declining engagement), together account for 23% of Swedish women (13.9% Fragile + 9.3% Social Decline) against 16% of men (8.8% + 7.2%), a 7-percentage-point gap against the 15-percentage-point gap observed in Italy (§4.6). The high-resource profiles (Asset Rich, Wealthy Digital, Connected Wealthy) account for 60% of men against 54% of women, again a smaller asymmetry than in Italy. The reading is consistent with the welfare-regime literature on the de-familisation of elderly care in the Nordic countries (Anttonen & Sipilä, 1996): where institutional care provision is universal, the gender gap in late-life socio-economic outcomes is partially absorbed by public-service substitution rather than amplified by intra-household care obligations.

5.7 The absence of the Isolation Paradox

The Italian Isolation Paradox—the autonomous but socially and digitally disconnected Moderate Isolated configuration (§4.7)—has no Swedish counterpart: the Swedish Moderate cluster that occupies the corresponding position on the resource gradient is socially integrated at or above the population mean rather than disconnected. The matched-pair confirmation of the absent Swedish twin and the cross-country healthcare-utilisation quantification are developed in §6.5.3.

5.8 Synthesis

The Swedish typology partitions the over-65 population into six archetypes whose structural shape differs systematically from the Italian one. Three findings carry the chapter as the typological signature of the Nordic-universalistic welfare regime: the binary rather than graduated structure of Swedish fragility, the universal digital saturation that cuts at the functional-fragility line rather than the resource line, and the age-driven rotation of the high-resource pole from a digital-engagement to a wealth-accumulation anchor, with no analogous rotation in Italy. External validation against life satisfaction is significant ($F(5,1779) = 44.3$, $p < 0.001$) at less than half the Italian magnitude, and the Swedish typology contains no counterpart of the Italian Moderate Isolated configuration; the matched-pair cross-country comparison is developed in Chapter 6.

Chapter 6

Cross-Country Comparison

Welfare regime as the structuring institution of late-life experience

6.1 Overview

Chapters 4 and 5 have presented two parallel typologies of the over-65 population obtained through identical pipelines, differing only in the country of reference. The interpretation pursued throughout the empirical chapters has anticipated that the structural differences between the two partitions are not artefacts of sampling or modelling choices, but the typological imprint of two distinct welfare regimes operating on otherwise comparable European societies. This chapter develops that interpretation as the central comparative argument of the thesis.

All centroid coordinates are reported as z-scores within the country of origin to preserve the within-country structural reading of the typologies. Cross-country comparisons of raw indicators are reported on the original scales (percentages for binary indicators, count means for utilisation variables). Sample sizes are $n = 2{,}378$ for Italy and $n = 1{,}787$ for Sweden. Cluster shares are reported as percentages of the respective analytical sample; see §3.4 for the descriptive/predictive interpretation. The only inferential claims at the national level developed in this chapter are the scenario-based extrapolations of §6.6.3, which operate on the Eurostat 65–74 cohort time-series and not on the cluster assignments.

6.2 Welfare regime as the structuring institution

The two welfare regimes that surround the Italian and Swedish over-65 populations, and their documented effects on female labour-force participation, late-life poverty risk and the social distribution of healthcare access, are established in §2.3: the Nordic-universalistic regime defamilises care through an extensive public branch and a universal-access service infrastructure (Anttonen & Sipilä, 1996), the Mediterranean-familistic regime discharges it through intra-family responsibility over a residual and fragmented public infrastructure (Ferrera, 1996). This chapter extends the contrast from aggregate outcomes to the structural shape of the late-life partition that each regime produces.

The empirical content of this thesis adds a typological reading to that comparative literature: the two regimes do not only produce different aggregate outcomes for the over-65 population; they produce different *partitions* of it along the principal axes of late-life experience. The Italian over-65 population, treated as a unit by the welfare regime that surrounds it, organises into five clusters whose principal latent dimension is the joint resource-and-engagement gradient, and in which a country-specific middle-stratum profile (Moderate Isolated) emerges from intact autonomy combined with a thin informational environment. The Swedish over-65 population, treated as a unit by a structurally different regime, organises into six clusters whose principal latent dimension is a binary fragility line, and in which the high-resource pole rotates across age bands rather than contracting in aggregate. The thesis claim is that the two partitions are the empirical signature of the two regimes acting on the same demographic substrate. This claim is interpretive rather than causally identified: a two-country, cross-sectional design cannot isolate the welfare regime from the other respects in which Italy and Sweden differ, and the regime reading is advanced as the most parsimonious account of the structural contrast (§9.5).

The claim is developed through three arguments. The structural argument (§6.3): the two typologies share only four matched pairs out of seven cluster slots, and the three unmatched profiles carry most of the comparative content. The dimensional argument (§6.4): each of the three findings of Chapter 5 is the projection of the same regime contrast onto a different latent dimension. The policy argument (§6.5): the healthcare-utilisation gaps across the matched pairs and the unmatched Moderate Isolated profile identify a design space for the Italian silver economy that the Swedish data quantify.

6.3 Matched cross-country profile pairs

The two typologies are aligned by matching each Italian profile to the Swedish profile in the corresponding diagnostic position, using the matrix of pairwise centroid distances between Italian and Swedish profiles in the standardised analytical space to quantify the correspondence (Kaufman & Rousseeuw, 1990). The standardised distance is defined on the eleven analytical dimensions of the K-means partition (objective health: mobility, ADL, IADL, chronic conditions, self-rated health; affective and subjective dimensions: EURO-D, loneliness, hope, CASP; resources: log-income, log-wealth) with z-scoring performed within country to preserve the within-country structural reading. Each Italian profile is paired with at most one Swedish profile, and an indicative distance guide of $d \approx 2.0$ in standardised units marks the approximate upper range of structurally meaningful correspondence. The four matched pairs have centroid distances spanning roughly 0.6 to 2.1 standardised units—a continuous spread rather than a tight cluster well below the guide value. The matched/unmatched split is therefore anchored in the semantic correspondence of diagnostic positions (Table 6.1) and corroborated, but not mechanically determined, by these distances.

The three unmatched profiles are reported as unmatched on the grounds of qualitative diagnostic non-correspondence rather than of a large centroid distance: each in fact lies at a moderate distance (< 2.0) from its nearest cross-country neighbour, and what isolates it is the configuration of dimensions on which it is distinctive—social-and-informational and resource—rather than its overall proximity. The 2.0 figure is therefore a descriptive guide to the distance range, not a sharp cutoff separating two regimes.

The procedure returns four matched pairs and three unmatched profiles. Table 6.1 reports the matched pairs with their respective sample shares and a one-line characterisation of the structural correspondence. Figure 6.1 visualises the cluster centroids of the two typologies on a joint plane defined by physical/functional fragility (mean z-score on objective health and functional dimensions) and subjective burden (mean z-score on the subjective block). The unmatched profiles are: in Italy, the Moderate Isolated cluster (26.9%; §4.3); in Sweden, the Asset Rich cluster (19.1%; §5.3) and the Wealthy Digital cluster (7.7%; §5.3). The three unmatched profiles together cover 26.9% of the Italian analytical sample and 26.8% of the Swedish analytical sample, and they carry the bulk of the comparative content of the chapter. On the plane of Figure 6.1 the three unmatched profiles (red outlines) all fall in the low-fragility region, away from the matched fragility pair: the Italian Moderate Isolated and the two Swedish high-resource clusters (Asset Rich and Wealthy Digital) sit together in the autonomous, low-burden quadrant. What sets them apart from their cross-country neighbours is not their position on the two plotted axes but the dimensions on which this plane is silent—the social-and-informational block and resources.

Two observations on the matched-pair structure are substantively important. First, the matching is not a one-to-one identity: the four pairs differ materially on population share within the matched correspondence. The Connected pair is the largest in both countries (25.4% Italy, 30.0% Sweden), with the Swedish share approximately five percentage points above the Italian share, consistent with the deeper diffusion of the digitally integrated profile under universal welfare. The Socially-oriented pair is the most asymmetric: 7.7% in Italy against 23.3% in Sweden, reflecting the absorption into the Swedish Moderate cluster of the middle stratum that splits in Italy into Traditional Social and the unmatched Moderate Isolated. The Declining pair is asymmetric in the opposite direction (26.7% Italy against 8.3% Sweden), with the residual character of the Swedish Social Decline cluster (§5.3) accounting for the size mismatch. The Fragile pair is the only matched pair with comparable size in the two countries (13.2% Italy,

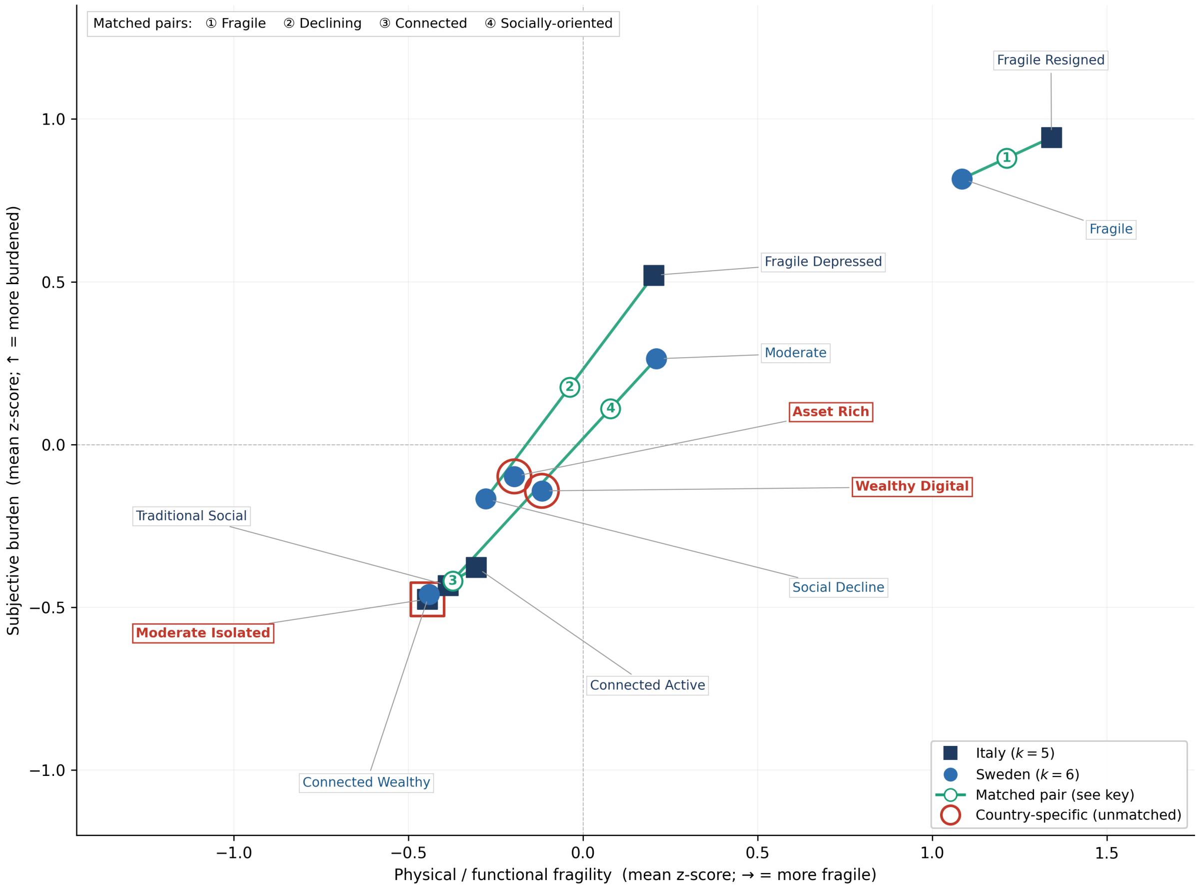


Figure 6.1: Italian and Swedish cluster centroids on the joint fragility-and-burden plane. Each marker is one within-country cluster centroid (squares for Italy, $k = 5$; circles for Sweden, $k = 6$); axes are the mean within-country z-score on the physical/functional block and on the subjective-burden block. Green lines connect the four matched pairs; red outlines mark the three country-specific unmatched profiles (Moderate Isolated in Italy; Asset Rich and Wealthy Digital in Sweden).

11.5% Sweden), confirming that the principal fragility profile is approximately identically distributed across the two welfare regimes despite the substantially different aggregate level of public elderly-care provision.

A caveat applies to the Declining pair specifically. The matching is corroborated by centroid proximity in the standardised space, a criterion that is agnostic to the substantive content of the matched profiles. As §5.3 documents, the Wave 9 Swedish Social Decline cluster is the residual cluster with the most ambiguous substantive content. Its pairing with the Italian Fragile Depressed cluster rests on broad structural correspondence in the joint analytical space rather than on a substantive declining-affect-and-engagement parallel. The Declining pair in §6.5.2 carries the largest dental gap and the largest forgone-care gap of the four matched pairs. These magnitudes should therefore be read as the upper bound of the within-pair Italy–Sweden differential under a structural matching that operates on the analytical centroid, and not as a regime-attributable gap on a substantively homogeneous pair.

Table 6.1: Matched profile pairs across Italy and Sweden

Pair	Italy	Sweden	Structural correspondence
Fragile	Fragile Resigned (13.2%)	Fragile (11.5%)	Oldest-old, high functional burden, low subjective evaluation. The Italian and Swedish principal fragility profiles.
Connected	Connected Active (25.4%)	Connected Wealthy (30.0%)	Best resources within country, with digital adoption at or near the top of each typology. The high-resource pole of the typology.
Socially-oriented	Traditional Social (7.7%)	Moderate (23.3%)	Dense social and cultural engagement with intact health; middle-stratum (Sweden) to upper-middle (Italy) resource position. Substantial size asymmetry.
Declining	Fragile Depressed (26.7%)	Social Decline (8.3%)	Above-mean affective burden on the Italian side; the Swedish counterpart matches on broad centroid correspondence only (see caveat in the text). Substantial size asymmetry.

Note. Pair names are introduced here for expository purposes. Sample shares as reported in Tables 4.1 and 5.1. The unmatched profiles are Moderate Isolated (Italy, 26.9%), Asset Rich (Sweden, 19.1%) and Wealthy Digital (Sweden, 7.7%).

Second, the unmatched profiles map directly onto the regime contrast. The single Italian unmatched profile (Moderate Isolated) is the configuration that the welfare regime literature predicts as the residual of a familistic regime that does not provide a substitute infrastructure for individuals not embedded in dense family networks: intact autonomy, thin social and digital engagement, and below-average healthcare utilisation. The two Swedish unmatched profiles (Asset Rich and Wealthy Digital) are the configurations that the welfare regime literature predicts as the residual of a universalistic regime that allows the high-resource pole to differentiate by age and by engagement style without contracting: a wealth-anchored profile of the oldest-old (Asset Rich, mean age 77.7) and a digitally anchored profile of the well-resourced young-old (Wealthy Digital, mean age 74.9). The three unmatched profiles together are not a residual category of the modelling exercise; they are the empirical content of the regime contrast.

6.4 Three structural contrasts as the regime signature

The three substantive findings developed in §5.4 are not three independent contrasts. They are three projections of the same regime distinction onto three different latent dimensions of late-life experience. This section reads them jointly.

6.4.1 Binary versus graduated fragility

Read as a regime signature, the binary-versus-graduated contrast (§5.4.1) is operational. In a regime that absorbs the functional burden of advanced old age into a public elderly-care branch with universal coverage (Anttonen & Sipilä, 1996), the infrastructure produces a single targetable segment, the Fragile cluster; affective deficits within the autonomous profiles do not separate into a second fragility cluster because they do not align with a functional or institutional cleavage, and the universal infrastructure absorbs them at the margin. In a regime that distributes the same burden across the household and the family (Ferrera, 1996), the infrastructure produces multiple distinguishable segments, because the burden is not absorbed institutionally and its operational characteristics (functional autonomy, affective trajectory, family proximity) co-vary with the household configuration in which the individual is embedded. The Italian graduated structure is what the regime produces; the Swedish binary structure is what the regime absorbs.

6.4.2 The digital cut: functional line versus resource line

The same internet-use indicator marks two different cleavages: the functional-fragility line in Sweden, the resource gradient in Italy (§5.4.2). In a regime whose public-service infrastructure operates through a universal-access digital channel that does not require informal mediation, the digital indicator measures whether the individual remains within the autonomous range of the typology: near-universal adoption across all profiles except the principal fragility cluster is the footprint of the universal channel. In a regime whose infrastructure operates through fragmented channels mediated by family and informal networks, the same indicator measures the cumulative socio-economic and informational endowment of the individual: the wide gap between lowest- and highest-adoption clusters is the footprint of the resource gradient. The same variable measures different things because the regimes produce different relations between the individual and the digital channel.

6.4.3 The age-prosperity rotation

The Italian high-resource pole declines in parallel with age while the Swedish pole rotates (§5.4.3). In a regime where the public elderly-care branch absorbs the functional burden of advanced old age, the asset-rich household can remain autonomous into the 85+ band without family-mediated care: Asset Rich is the late-life configuration that the universal infrastructure permits. In a familistic regime delivering elderly care through intra-household channels, the asset-rich older household has no analogous institutional substitute, and the trajectory contracts the high-resource pole into the fragile region rather than rotating it. The Italian parallel decline is late life under familistic absorption; the Swedish rotation is late life under universalistic absorption.

6.5 Healthcare utilisation across matched pairs

The cross-country comparison of healthcare utilisation across the four matched pairs and the unmatched Italian Moderate Isolated profile completes the comparative reading. The analysis focuses on the dental-care and forgone-care indicators of Figure 6.2, with internet use as contextual contrast; specialist contacts and CASP-12 are read at the aggregate level in §6.5.1. Cluster means are reported with Welch's t-test gaps (95% confidence intervals reported for the principal dental-care gaps) on the standardised analytical sample.

6.5.1 Country-aggregate gaps

At the country-aggregate level, healthcare utilisation shows a consistent pattern: Italian respondents report more doctor visits (approximately 7.0 per year against a Swedish mean of 5.5) and comparable specialist contacts (2.1 against 2.1), but materially higher rates of forgone care for cost reasons (10.1% against 2.7%) and substantially lower dental-care attendance (30.4% against 83.8%). These are country-aggregate figures; doctor-contact volume in particular is reported here as aggregate context and is not among the four matched-pair panels of Figure 6.2. The Italian system delivers higher generalist-contact volume at the aggregate but no comparable advantage on the navigation-intensive specialist channel, and it produces gaps in coverage at the population margin (Lo Scalzo et al., 2009). The Swedish system delivers comparable specialist volume with a substantially flatter access profile on the cost-sensitive and preventive channels, consistent with the universal-access infrastructure of the Nordic regime (Anttonen & Sipilä, 1996). The specialist parity is itself consistent with the Isolation Paradox reading: the specialist, the navigation-intensive channel, is precisely where Italian contact volume confers no advantage (§4.7).

6.5.2 Matched-pair gaps

The matched-pair analysis isolates the cross-country gaps within structurally comparable profiles. All gaps are in percentage points on the corresponding raw indicator, and the gap is Sweden minus Italy, so a negative forgone-care gap indicates *less* unmet need in Sweden; Table 6.2

reports the dental-care, forgone-care and internet-use gaps for the four matched pairs.

Table 6.2: Matched-pair gaps on dental attendance, forgone care and internet use, Italy versus Sweden

	Dental attendance			**Forgone care**			**Internet use**		
Pair (IT vs SE)	IT%	SE%	Gap pp	IT%	SE%	Gap pp	IT%	SE%	Gap pp
Fragile (FR vs F)	19.8	74.6	55 [48, 62]	15.2	7.8	−7	8.6	63.6	55
Declining (FD vs SD)	22.0	82.4	60	19.2	3.4	−16	–	–	–
Socially-or. (TS vs M)	47.8	86.6	39	5.4	1.0	−4	–	–	–
Connected (CA vs CW)	49.3	88.0	39 [34, 44]	3.8	0.9	−3	74.8	98.5	24

Note. Gap = Sweden − Italy, in percentage points on the corresponding raw indicator; a negative forgone-care gap indicates less unmet need in Sweden. Bootstrap 95% confidence intervals are given in brackets for the dental gaps reported with intervals in the text. "–" indicates not reported in this section.

The Declining pair (IT Fragile Depressed vs SE Social Decline) carries the largest gaps in the table, consistent with the residual character of the Swedish Social Decline cluster (§5.3).

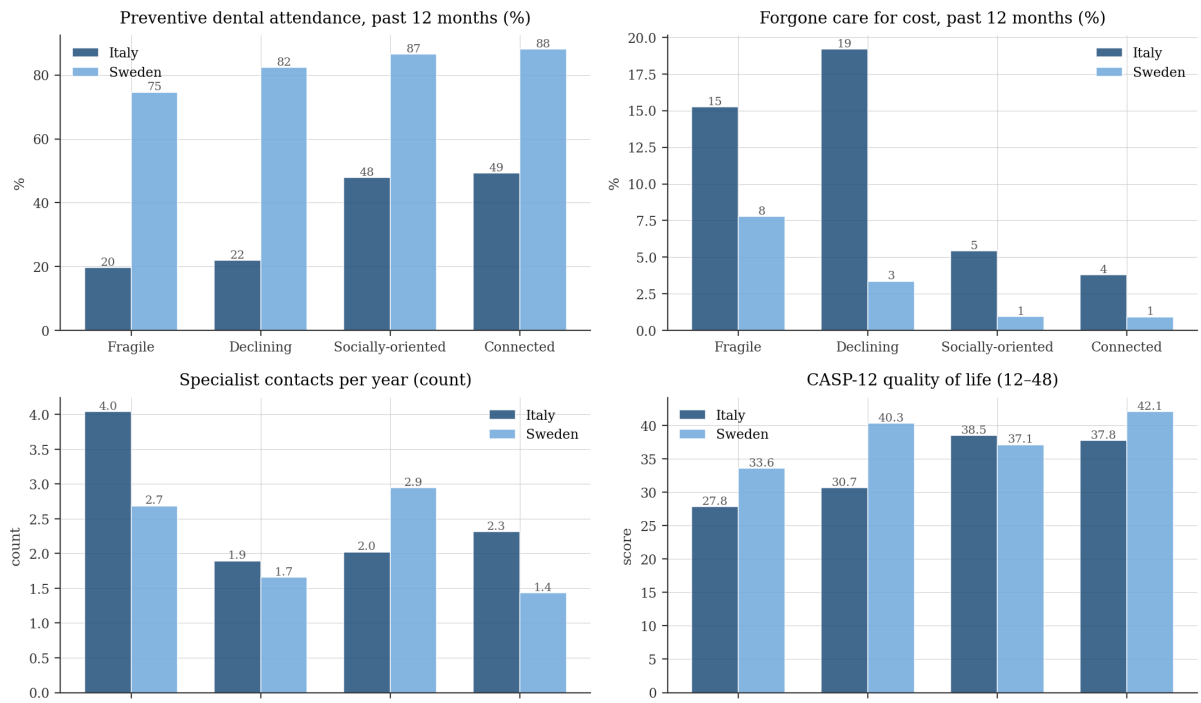


Figure 6.2: Cross-country healthcare utilisation across the four matched profile pairs. Italy (dark blue) and Sweden (light blue) cluster means on four indicators: preventive dental attendance and forgone care for cost (both over the past twelve months), specialist contacts per year, and CASP-12 quality of life. On the Socially-oriented pair the Italian profile is marginally higher on CASP-12 (38.5 against 37.1).

The gaps are systematic: across all four pairs Italian profiles report lower dental-care attendance and higher forgone-care prevalence than their Swedish twins, with magnitudes that scale with the substantive content of the pair, largest in the Declining and Fragile pairs (60 and 55 percentage points on dental attendance) and smaller in the Connected and Socially-oriented pairs (39 percentage points in both; Figure 6.2). The cross-country gap is thus not a uniform shift but a structurally graduated one that widens at the lower end of the resource-and-engagement gradient, consistent with a welfare-regime reading in which the Italian system's healthcare-access deficits are co-located with the fragile and declining segments of the population.

These cross-country gaps are descriptive differences between structurally matched profiles, reported without further covariate adjustment. The cross-country matching aligns the profiles on the eleven analytical dimensions, but residual compositional differences within a matched pair—for instance in age or chronic-condition load—are not partialled out; the gaps are therefore read as regime-consistent associations rather than as covariate-adjusted or causal estimates, and a fully adjusted cross-country specification is left to future work (§9.6). This is the cross-country counterpart of the within-country adjustment applied to the Italian Isolation Paradox in Table 4.3.

6.5.3 The Isolation Paradox in cross-country perspective

The Italian Moderate Isolated cluster (26.9% of the Italian analytical sample) is the unmatched profile in the typology and the central population segment of the Isolation Paradox developed in §4.7. What the cross-country comparison adds is that no Swedish profile occupies the corresponding diagnostic configuration, even though several Swedish clusters lie at moderate centroid distance from it. The comparative reading is that the Italian Moderate Isolated configuration is not a clinical or behavioural category that exists across welfare regimes; it is a regime-specific category that emerges where the public-service infrastructure does not provide a substitute baseline for individuals not embedded in dense informal networks (Lo Scalzo et al., 2009).

The substantive implication for healthcare-system design is direct. The Moderate Isolated configuration does not present in the existing Italian welfare branches: not in long-term care, where the cluster signature would not satisfy the eligibility criteria on functional autonomy; not in income support, where it would not satisfy the eligibility criteria on resources; not in mental health, where it does not present elevated affective scores. The segment is invisible to the operational categories of the regime that produces it. The Swedish counterpart of this configuration does not exist because the regime that produces the Swedish typology integrates the corresponding population segment into the universal-access infrastructure at the baseline. The Italian-only character of the configuration is the empirical signature of the regime contrast on the dimension of social and informational mediation.

6.6 Implications for silver-economy market structure

The cross-country comparison developed in §6.4 and §6.5 has direct implications for the structure of the silver-economy market in the two countries. The implications are not symmetric: the Swedish typology delivers a market that is structurally accessible to direct-to-consumer digital products across most of the Swedish analytical sample, while the Italian typology delivers a market that is structurally segmented into addressable and non-addressable cluster groups on the basis of dimensions that are themselves shaped by the regime.

6.6.1 Addressability under the two typologies

In Sweden, direct-to-consumer digital products can address the 88.5% of the Swedish analytical sample classified into the five autonomous profiles, with intermediated or alternative-channel designs required only for the 11.5% Fragile cluster. The digital floor across the autonomous profiles is high enough that internet-mediated services do not face a substantial penetration constraint within the autonomous range of the typology. The asset-rich households that occupy the high-resource pole in advanced old age (Asset Rich, 19.1%) provide a wealth-anchored autonomous segment whose product preferences differ from the digital-engagement-anchored Connected Wealthy segment but whose addressability through digital channels is preserved. The Swedish silver-economy market is structurally a single market with internal differentiation by engagement style and resource configuration.

In Italy, the same direct-to-consumer digital approach can address approximately one third of the analytical sample (Connected Active 25.4% + Traditional Social 7.7% = 33.1%), with the remaining two thirds requiring fundamentally different design logics. Within this two-thirds, the Fragile Resigned segment (13.2%) requires intermediated designs through family or institutional channels (8.6% internet use; §4.3), the Fragile Depressed segment (26.7%) requires affect-aware designs that address the binding behavioural constraint on adoption (17.3% internet use; §4.3), and the Moderate Isolated segment (26.9%) requires social-mediation designs that bridge the absence of dense informal channels in the configuration (34.1% internet use; §4.3). The Italian silver-economy market is structurally four markets with materially different design logics, of which only one is addressable through direct-to-consumer digital channels at scale.

6.6.2 The product-design gap

The substantive case for innovation is most direct on the Moderate Isolated segment. It combines intact autonomy with non-trivial digital adoption (34.1%), thin social and informational mediation, and systematic disengagement from the formal healthcare system (§4.7.2);

it is not addressable like the Connected Active segment, because the binding constraint is not the absence of a digital channel but the absence of a social-and-informational mediation layer that would route the available channel towards healthcare utilisation. The Swedish data quantify the magnitude of the corresponding shortfall, dental-attendance gaps of 39 to 60 percentage points across the four matched pairs (§6.5.2), though translating this descriptive cluster-centroid gap into a national market-size figure would require assignment-probability weighting, a conversion rate and a private-premium assumption that this thesis does not develop. The product-design candidate is therefore stated qualitatively: a digital channel combined with a social-mediation layer (peer-network platforms, community-based digital intermediaries, family-network-augmented service interfaces) that substitutes for the absent informal infrastructure. The Swedish typology, produced by a regime that does not require it, contains no precedent for this design.

6.6.3 Time to maturity of the digital channel

The Italian internet-use cohort 65–74 is reported at 65.6% in 2024 (Eurostat, 2024), against approximately 96% in Sweden on the same indicator (Eurostat, 2022). A linear extrapolation under three rate scenarios implies the roughly 30-percentage-point gap closes in approximately 30, 10 or 6 years respectively. The extrapolation is illustrative, not predictive: adoption curves decelerate near saturation, so the figure is a lower bound on convergence time. The reading is nonetheless robust. The digital divide between the typologies is not a transitory phenomenon that demographic replacement closes within a product cycle, and digital-first service designs face a materially different addressability profile in Italy than in Sweden over the medium term.

6.7 Synthesis

The two typologies partition populations whose welfare-regime literature predicts systematically different treatment of elderly care, household responsibility and service infrastructure. Only four of the seven cluster slots have a cross-country twin; the three unmatched profiles carry the contrast's empirical content. It projects onto three structural contrasts (binary versus graduated fragility, the digital cut at the functional versus resource line, the high-resource rotation versus parallel decline) and onto a healthcare-utilisation differential that scales the within-pair gaps, producing a country-only configuration the Italian welfare branches do not address. The silver-economy implication follows: Italy's market is structurally different, not a smaller Swedish one, and its dominant innovation is a social-mediation layer the regime does not provide.

Chapter 7

Robustness

The empirical battery behind the substantive findings

7.1 Overview

The comparative reading of Chapter 6—that the welfare regime structures the late-life partition, with the Italian Isolation Paradox as its sharpest signature—is only as strong as the partition it rests on. The substantive findings developed in Chapters 4–6 rest on a sequence of modelling choices: the imputation of missing data, the selection of k in K-means, the inclusion of the subjective block as input, and the pre-pandemic-versus-pandemic comparability of the wave. Each can in principle alter the partition that the pipeline returns. This chapter reports the empirical battery that quantifies the magnitude of those sensitivities and identifies the substantive claims that hold under stricter modelling alternatives. The chapter does not claim to eliminate the modelling sensitivity inherent to clustering a high-dimensional matrix; it documents where the partition is structurally stable and where the interpretation should be qualified.

All robustness diagnostics are computed on the analytical samples used throughout the empirical chapters ($n = 2{,}378$ for Italy, $n = 1{,}787$ for Sweden) and on the pre-pandemic Wave 8 sub-samples for the cross-wave analysis ($n = 1{,}277$ for Italy, $n = 1{,}814$ for Sweden), with the cross-wave panel intersection containing $n = 958$ Italian and $n = 1{,}320$ Swedish respondents observed in both waves.

7.2 Imputation sensitivity

The analytical sample is constructed from a single multiple-imputation replicate of the SHARE Wave 9 missingness pattern (predictive mean matching, missingness at random conditional on the observed covariates). The single-replicate strategy is adopted on parsimony grounds: K-means returns a deterministic partition for a given imputed dataset, and rerunning the pipeline across replicates would deliver a distribution of partitions with no closed-form summary. The diagnostic question is whether the reported partition is stable under alternative draws.

The proportion of analytical-block observations that are imputed is uniformly low in both countries: the principal subjective indicators (CASP-12, loneliness, hope, interest, expected survival) report item-level missingness below 6% in both samples, and the principal economic indicators (log-income, log-net-wealth) report missingness in the 4–9% range. The objective health and digital indicators have missingness below 3%. The combination of low item-level missingness and high pairwise correlation among the imputed variables (§3.5) implies that the imputation step should have limited leverage on the resulting partition. Alternative replicates should therefore differ from the deployed replicate on only a small subset of observations, concentrated in the lower tail of the resource distribution, where the imputation of log-net-wealth has the largest residual variance.

This hypothesis is tested directly in Appendix A (§A.5), which re-assembles both national samples under the second and third SHARE imputation replicates and re-runs the entire pipeline at the deployed k. The alternative replicates reproduce the deployed partition at an adjusted Rand index of 0.88–0.92 for Italy and 0.86–0.88 for Sweden—above the 0.80 robustness threshold—and at a cluster purity of 0.94–0.97, so between 94% and 97% of respondents retain their deployed profile irrespective of the draw. The residual disagreement falls, as the low-missingness reasoning predicts, at the boundary between the Fragile and Moderate strata, while assignments within the body of each cluster are essentially invariant. The reading is consistent with the cross-method convergence reported in §7.5, where the lowest convergence rates are observed precisely on the profiles whose centroid lies near a between-cluster boundary in the standardised analytical space.

7.3 Internal cluster validity

The internal-validity battery is reported through five indices: the mean silhouette coefficient (Kaufman & Rousseeuw, 1990), the between-cluster sum of squares as a fraction of total sum of squares, the Calinski–Harabasz index, the Davies–Bouldin index, and the Hopkins statistic of clustering tendency. Table 7.1 reports the indices for the retained partitions ($k = 5$ Italy, $k = 6$ Sweden); the trace of the indices across $k = 2, \ldots, 10$, used for the k-selection diagnostic, is summarised below.

The Hopkins statistic ($H = 0.75$ Italy, $H = 0.74$ Sweden) places Italy at the strong-tendency threshold and Sweden marginally below it, in both cases far from the $H \approx 0.5$ benchmark of a uniform distribution, and so supports the choice to estimate a partition rather than treat the over-65 population as a single homogeneous distribution. The mean silhouette coefficient is

Table 7.1: Internal validity diagnostics, Italy and Sweden

Index	**Italy** ($k = 5$)	**Sweden** ($k = 6$)
Mean silhouette	0.069	0.060
Between-cluster R^2 (%)	24.8	21.1
Calinski–Harabasz index	181.5	93.2
Davies–Bouldin index	2.94	2.97
Hopkins clustering tendency	0.75	0.74

Note. Silhouette and Calinski–Harabasz: higher is better; Davies–Bouldin: lower is better; Hopkins: $H \geq 0.75$ indicates strong clustering tendency, $H \approx 0.5$ indicates random/uniform distribution. Indices computed on the country-specific standardised analytical matrix with the deployed cluster assignments.

low in both countries (0.069 and 0.060, well below the 0.250 conventional cohesion threshold), but this reading is methodological rather than substantive. Silhouette coefficients are systematically depressed for K-means partitions of high-dimensional matrices spanning multiple factor subspaces, for the reasons set out in §3.6 (Steinley, 2006). The Calinski–Harabasz index (181.5 IT, 93.2 SE) and the Davies–Bouldin index (2.94 Italy, 2.97 Sweden) carry no absolute acceptance thresholds and are read comparatively (§3.6); the substantially higher Calinski–Harabasz of the Italian partition indicates stronger cluster separation, consistent with its higher external discriminative power on life satisfaction (§7.4).

The between-cluster sum of squares as a fraction of total sum of squares (the explained variance ratio) is approximately 24.8% in Italy at $k = 5$ and approximately 21.1% in Sweden at $k = 6$. Both values lie below the rule-of-thumb 50% threshold typically attained by partitions of low-dimensional input matrices but are within the range typical of high-dimensional segmentation studies on the SHARE platform. The trace of the silhouette, of the Calinski–Harabasz index, and of the Davies–Bouldin index across $k = 2, \ldots, 10$ does not produce a sharp elbow in either country: silhouette and Calinski–Harabasz both decay monotonically from their maximum at $k = 2$, and the Davies–Bouldin index oscillates within a narrow band (2.2–3.0) without a global minimum. The selection of $k = 5$ for Italy and $k = 6$ for Sweden is therefore conservative on cross-country-comparability and substantive-interpretability grounds rather than the optimum of any single-index optimisation. A smaller k would collapse the qualitative contrast between Fragile Resigned and Fragile Depressed (Italy) or the Connected Wealthy versus Asset Rich rotation (Sweden), while a larger k would split the principal clusters into sub-strata that the substantive interpretation of Chapters 4 and 5 does not require.

7.3.1 Robustness to the outlier-exclusion threshold

The Mahalanobis outlier-exclusion routine for Sweden (§3.4) removes 2.4% of the post-listwise sub-sample at the conventional $\alpha = 0.001$ threshold ($n = 44$ respondents, §3.2), a magnitude comparable to the Italian 2.2% (53 respondents). Although the exclusion is modest, its effect at the profile level warrants a check on whether the Swedish typology is robust to a stricter threshold. The deployed partition is re-derived under $\alpha = 0.0001$ (the χ^2 cutoff at the 99.99th percentile, which retains 30 additional respondents) and compared to the deployed partition over the intersection of the two samples. Table 7.2 reports the profile-level confusion of the deployed Swedish typology against the dominant $\alpha = 0.0001$ cluster for each of the six archetypes.

Table 7.2: Profile-level robustness of the Swedish typology to a stricter outlier-exclusion threshold ($\alpha = 0.0001$)

Deployed profile	n **deployed**	**Dominant cluster under** $\alpha = 0.0001$	**Concentration**
Wealthy Digital	138	cluster 2	138 / 138 (100.0%)
Fragile	206	clusters 4–5	206 / 206 (100.0%)
Connected Wealthy	536	cluster 3	478 / 536 (89.2%)
Asset Rich	341	cluster 1	293 / 341 (85.9%)
Moderate	417	cluster 6	354 / 417 (84.9%)
Social Decline	149	cluster 3	102 / 149 (68.5%)
Overall	1,787	ARI = 0.654; profile-purity = 0.822	

Note. Profile-level confusion of the deployed Sweden K-means partition ($\alpha = 0.001$, $n = 1{,}787$) against the partition re-derived under $\alpha = 0.0001$ ($n = 1{,}817$). The dominant cluster column reports the $\alpha = 0.0001$ cluster receiving the majority of the deployed profile's members. Concentration is the proportion of deployed-profile members assigned to the dominant cluster. The full sensitivity, including $\alpha = 0$ (no exclusion), is reported in §A.1.

Four of the six Swedish archetypes are stable under the stricter threshold at concentrations of 85% or above (with Moderate marginally below at 84.9%): the Wealthy Digital cluster preserves all 138 deployed members in a single $\alpha = 0.0001$ cluster, the Fragile cluster preserves all 206 members across two contiguous clusters (4 and 5) with no contamination into any non-fragile cluster, and the Connected Wealthy, Asset Rich and Moderate clusters retain dominant homes at 89.2%, 85.9% and 84.9% concentration respectively. The Social Decline cluster reports the lowest concentration at 68.5%, redistributing partly into the cluster also receiving the Connected Wealthy majority, a pattern consistent with the residual character of the Social Decline configuration (§5.3) and with its cross-method LCA convergence of 59.7% (§7.5). The $\alpha = 0.001$ specification is retained as the conventional χ^2 cutoff at the 99.9th percentile; the full sensitivity battery against $\alpha = 0.0001$ and $\alpha = 0$ (no exclusion) is documented in §A.1.

7.4 External validation: life satisfaction and healthcare utilisation

Two batteries of external outcomes, both administered in questionnaire blocks separate from the analytical input set, validate the typologies under the methodological commitment of §3.8.

Life satisfaction discriminates both typologies on an outcome external to the input set (§4.5, §5.5), with the typology accounting for approximately 19% of the life-satisfaction variance in Italy against 11% in Sweden (η^2); the larger Italian effect is consistent with the Swedish universal-welfare baseline compressing between-cluster variation rather than with a defect of the typology (Esping-Andersen, 1990). Healthcare utilisation discriminates the typologies in the same direction, including the Isolation Paradox signature on the navigation-intensive channels and the systematically higher Italian forgone-care prevalence (§4.7, §6.5.2).

7.5 Cross-method convergence: K-means versus LCA

Latent Class Analysis (LCA) was estimated as the cross-method probe of the K-means partition under the same input set and the same number of components ($k = 5$ for Italy, $k = 6$ for Sweden). LCA returns probabilistic assignments under a parametric mixture model rather than the geometric assignments returned by K-means, and convergence between the two assignments on the same respondents is read as a cross-method validation indicator (Vermunt & Magidson, 2002). The convergence is computed as the percentage of respondents in each K-means cluster who are assigned to the modal LCA class for that cluster, and is reported by profile in Table 7.3.

The convergence pattern of Table 7.3 is substantively informative. In Italy, the Fragile Resigned cluster reports the highest convergence (95.6%), consistent with its status as the most clinically distinct profile in the typology; the Moderate Isolated cluster reports the lowest convergence (54.6%), consistent with its position off the principal resource-and-engagement gradient and with the Isolation Paradox reading that places it as a country-specific configuration whose cluster boundary is structurally informative but geometrically thin. In Sweden, the Connected Wealthy and Fragile clusters report convergence above 85%, consistent with their status as the structurally extreme profiles of the binary Swedish typology; the Wealthy Digital cluster reports the lowest convergence (49.3%), consistent with §5.3. The modal-class column makes the same point structurally: Connected Wealthy, Social Decline and Wealthy Digital all converge on the same LCA class in Sweden, as do Traditional Social and Connected Active in Italy, so the latent-class solution merges the profile pairs whose K-means boundaries are geometrically thinnest rather than reproducing the partition one-to-one. The cross-method evidence supports retention of all eleven profiles but signals that the Moderate Isolated and Wealthy Dig-

Table 7.3: K-means/LCA convergence by profile and country

Country	Profile	n	LCA modal class	Convergence %
Italy	Fragile Resigned	315	4	95.6
Italy	Traditional Social	184	2	78.3
Italy	Connected Active	604	2	66.1
Italy	Fragile Depressed	636	3	61.8
Italy	Moderate Isolated	639	1	54.6
Sweden	Connected Wealthy	536	2	86.9
Sweden	Fragile	206	5	85.9
Sweden	Social Decline	149	2	59.7
Sweden	Moderate	417	4	56.6
Sweden	Asset Rich	341	6	53.1
Sweden	Wealthy Digital	138	2	49.3

Note. Convergence is the percentage of K-means cluster members assigned to the modal LCA class. LCA estimated on the same variables as K-means, discretised as described in §3.7, with the same k. The Italy mean is approximately 71% and the Sweden mean is approximately 65%; rates above 80% indicate strong cross-method confirmation, rates between 60% and 80% indicate substantial agreement, and rates below 60% indicate that the K-means cluster boundary is geometrically thin in the standardised analytical space.

ital cluster boundaries should be treated as informative of substantive content rather than as sharp geometric divides.

The complementary specification-sensitivity diagnostic, the 5D specification against the 4D specification that drops the subjective block, is documented in §3.8 (Table 3.4). The 4D specification raises the silhouette by approximately 0.020 but reduces the external-outcome F-statistic by 41% in Italy and 22% in Sweden, with ARI between specifications of 0.41 (Italy) and 0.52 (Sweden), and it collapses the Fragile Resigned/Fragile Depressed contrast that the 5D specification preserves (§4.7). The inclusion of the subjective block is thus empirically supported by the gain in external discriminative power and substantively required by the typological content.

7.6 Cross-wave panel stability: Wave 8 versus Wave 9

The K-means pipeline of Chapter 3 was applied unmodified to the pre-pandemic Wave 8 analytical samples (2019–2020) and the resulting partitions compared to the Wave 9 partitions on two indices: the panel-level Adjusted Rand Index (ARI) on the intersection of respondents observed in both waves, and the structural alignment of cluster centroids via the Hungarian assignment algorithm. The Italian panel intersection contains $n = 958$ respondents (ARI = 0.16, 95% CI [0.14, 0.20]); the Swedish intersection contains $n = 1{,}320$ respondents (ARI = 0.23, 95% CI [0.20, 0.27]). Read against interpretive bands defined for this analysis (ARI $\geq$ 0.40 substantial, $\geq$ 0.20 moderate)—deliberately lenient relative to the recovery thresholds of the cluster-validation literature, because the comparison spans a two-year window containing the pandemic shock—the Swedish value indicates moderate panel-level agreement and the Italian value falls below the moderate band: most respondents transition between clusters across the inter-wave window, but the reassignment is not random and the structural shape of each partition is preserved. The Swedish ARI is materially higher than the Italian one, with bootstrap intervals that meet only at the boundary value of 0.20, consistent with the welfare-regime literature on the absorbing role of universal-access institutions during exogenous shocks (Bambra, 2007).

Table 7.4 reports the structural alignment via Hungarian-matched centroid distances, visualised in Figure 7.1. The threshold $d \leq 2.0$, the indicative guide of §6.3, defines a structurally meaningful match.

Table 7.4: Wave 8 to Wave 9 Hungarian-matched cluster alignment, by country

Wave 8 cluster	**Wave 9 match**	**Distance**	**Label**
Italy			
Fragile Resigned	Fragile Resigned	0.47	preserved
Fragile Depressed	Fragile Depressed	0.77	preserved
Connected Active	Connected Active	1.10	preserved
Traditional Social	Moderate Isolated	1.32	swapped
Moderate Isolated	Traditional Social	5.71	swapped, $d > 2$
Sweden			
Connected Wealthy	Connected Wealthy	0.62	preserved
Wealthy Digital	Wealthy Digital	0.72	preserved
Moderate	Moderate	0.79	preserved
Social Decline	Asset Rich	0.83	swapped
Fragile	Fragile	1.05	preserved
Asset Rich	Social Decline	5.90	swapped, $d > 2$

Note. Centroid distance is computed in the country-specific standardised analytical space. The threshold $d \leq 2.0$ defines a structurally meaningful match. The single "swapped, $d > 2$" row in each country identifies the single cluster slot per country that undergoes substantive structural rearrangement between Wave 8 and Wave 9; the other "swapped" rows are within-threshold rearrangements that the Hungarian algorithm absorbs without loss of structural correspondence.

As Figure 7.1 shows, four of five Italian and five of six Swedish cluster slots match a cross-wave counterpart at $d \leq 2.0$. The single exception per country (Moderate Isolated in Italy, Asset Rich in Sweden) undergoes a parallel label exchange whose within-threshold half the Hungarian algorithm absorbs, while the $d > 2$ half marks a genuine structural rearrangement. What persists through the pandemic shock is therefore the centroid configuration—the location and identity of the cluster archetypes in the standardised space—rather than the assignment of individuals to them: the typology is structurally recurrent at the centroid level—more strongly in Sweden than in Italy, an asymmetry consistent with the regime contrast of Chapter 6—while individual memberships reshuffle substantially.

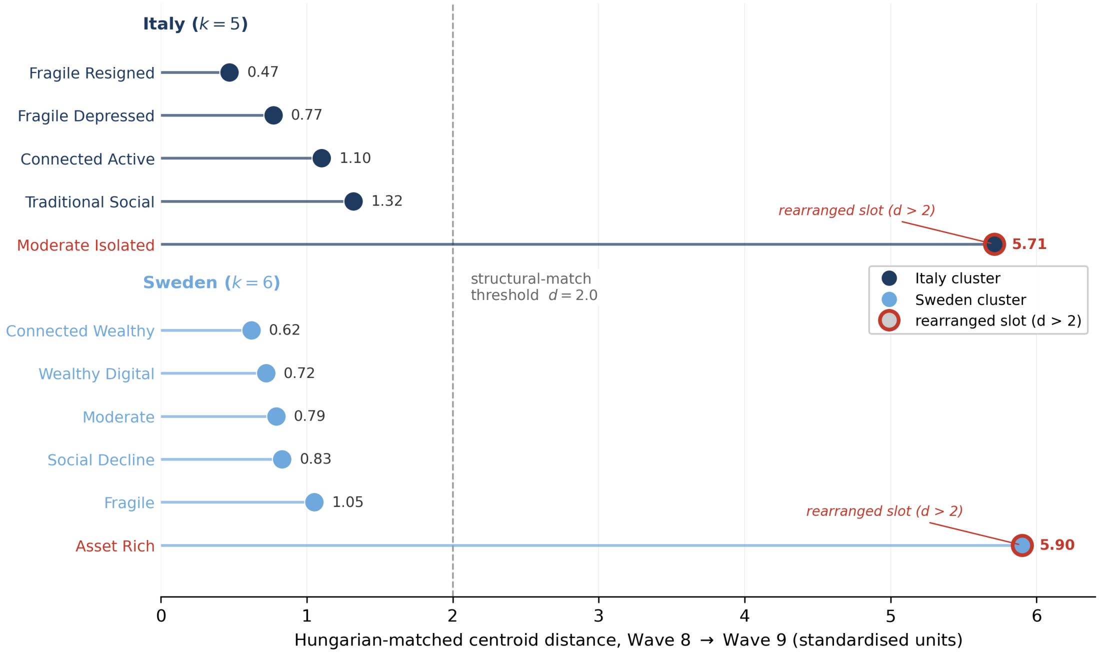


Figure 7.1: Cross-wave structural stability: Hungarian-matched centroid distance between each Wave 8 cluster and its Wave 9 counterpart, by country. Distances are reported in the country-specific standardised analytical space (the values of Table 7.4); the dashed line marks the $d \leq 2.0$ structural-match threshold. Four of five Italian and five of six Swedish clusters fall below it; exactly one slot per country (Moderate Isolated in Italy, Asset Rich in Sweden) rearranges beyond it under the parallel label exchange discussed in the text.

7.7 Synthesis

The robustness battery supports the substantive findings of Chapters 4–6 under stricter modelling alternatives: K-means/LCA convergence ranges from thin-boundary to strong (49.3–95.6%), the inclusion of the subjective block delivers a material gain in external discriminative power over the 4D specification, and cross-wave panel agreement is low in Italy (ARI 0.16) and moderate in Sweden (ARI 0.23) while the Hungarian-matched centroid distances keep all but one cluster slot per country within the structural threshold. The substantive findings should be read as claims about cluster centroids and structural relations rather than about marginal individual assignments. With the centroid structure established as recurrent, the typology is ready to leave the analytical sample: Chapter 8 transports it into an operational instrument, the Italian Silver Atlas, built around the very segment this battery singles out as the country-specific signal—the Moderate Isolated.

Chapter 8

The Italian Silver Atlas

An operational instrument for the social-mediation layer

8.1 Overview

The substantive analysis of Chapters 4–7 identifies a single dominant innovation case for the Italian silver economy: the Moderate Isolated segment (26.9% of the analytical sample) is autonomous and non-trivially digital yet disengaged from the formal healthcare system, and its binding constraint is the absence of a social-and-informational mediation layer rather than the absence of a digital channel (§6.6.2). This chapter documents the operational counterpart of that argument: the *Italian Silver Atlas* (invisible-profiles.vercel.app), a companion platform built as a working prototype of the mediation interface that the Italian welfare regime does not currently provide.

The platform is presented here as an operational artefact and design proposition, not as a tested intervention; empirical validation through user-comprehension studies and B2B field-trials is set out as a future-research direction in §9.6. The chapter first describes what the platform delivers and how an organisation uses it (§8.2–§8.4), and then summarises the statistical machinery that lets the typology be transported to an organisational cohort under explicit uncertainty (§8.5).

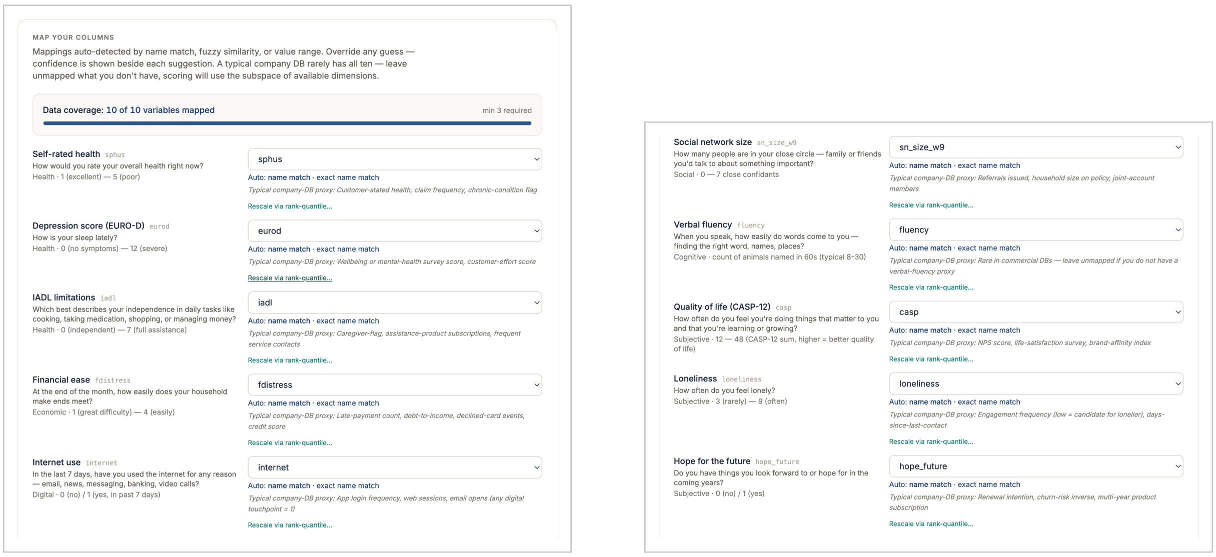
8.2 What the platform delivers

The Atlas has two faces. The public face makes the academic findings navigable, with a per-profile passport page for each archetype and a methodology section mirroring Chapters 3 and 7; the operational contribution, however, lies elsewhere.

The operational core is a business-to-business cohort-segmentation interface for organisations that hold a senior-population dataset (the platform names healthcare-practice managers, pharmacy-chain analysts and silver-economy product managers as its intended users). The interface assigns each record in the organisation's cohort to one of the Italian (or Swedish) archetypes, reports the cohort's segment composition against the national SHARE baseline, and shows the share of the cohort falling into the country-specific configurations identified in the thesis. The instrument occupies a position that population-level segmentation tools (Eurostat, Eurobarometer), commercial silver-economy reports segmenting by age and income alone (Vuik et al., 2016), and clinical risk scores do not: a cohort-screening instrument that operationalises a multidimensional academic typology of late-life experience.

8.3 The B2B cohort workflow

The user uploads a CSV with one row per individual and maps the available columns to the ten profiler variables through a guided wizard (Figure 8.1). Mappings are auto-detected by name match, fuzzy similarity and value range, with a confidence label beside each suggestion; each profiler variable is presented as a plain-language question with its SHARE coding range and a *typical company-database proxy* hint (for example, internet use proxied by application-login frequency, financial distress by late-payment count or credit score, quality of life by an NPS or brand-affinity index). A coverage meter reports how many of the ten variables are mapped; the scoring requires a minimum of three, and unmapped dimensions are simply left out of the subspace rather than defaulted.



(a) Coverage meter and the first five variables.

(b) The remaining five variables.

Figure 8.1: Column-mapping wizard, shown across its two scroll regions: panel (a) gives the coverage meter and the first five variables, panel (b) the remaining five. Each of the ten profiler variables is presented as a plain-language question with its SHARE coding range, an auto-detected mapping with a confidence label, a typical company-database proxy, and an optional rank-quantile rescaling control. The coverage meter (here ten of ten) governs how many dimensions enter the scoring subspace.

Before scoring, two safeguards are exposed. A missing-value policy lets the user skip incomplete rows (the conservative default) or impute with the cohort's own per-variable observed mean or median. A *sample-skew check* runs a per-variable two-sample test of the cohort against the SHARE Wave 9 reference and flags, in a non-blocking amber panel, any variable whose distribution departs materially from it, making the boundary of validity visible rather than enforcing it silently (§8.5).

Scoring produces the cohort dashboard (Figure 8.2). Four headline cards report the cohort size, the *Isolation Paradox flag* (the share assigned to the Moderate Isolated configuration of §4.7), the digitally addressable share, and the intermediated-need share, the latter three each reported against the national SHARE baseline. A statistical-fit line reports a Pearson χ^2 test of the cohort distribution against the national baseline, and a cluster-distribution chart plots the cohort share against the SHARE Wave 9 baseline for every archetype.

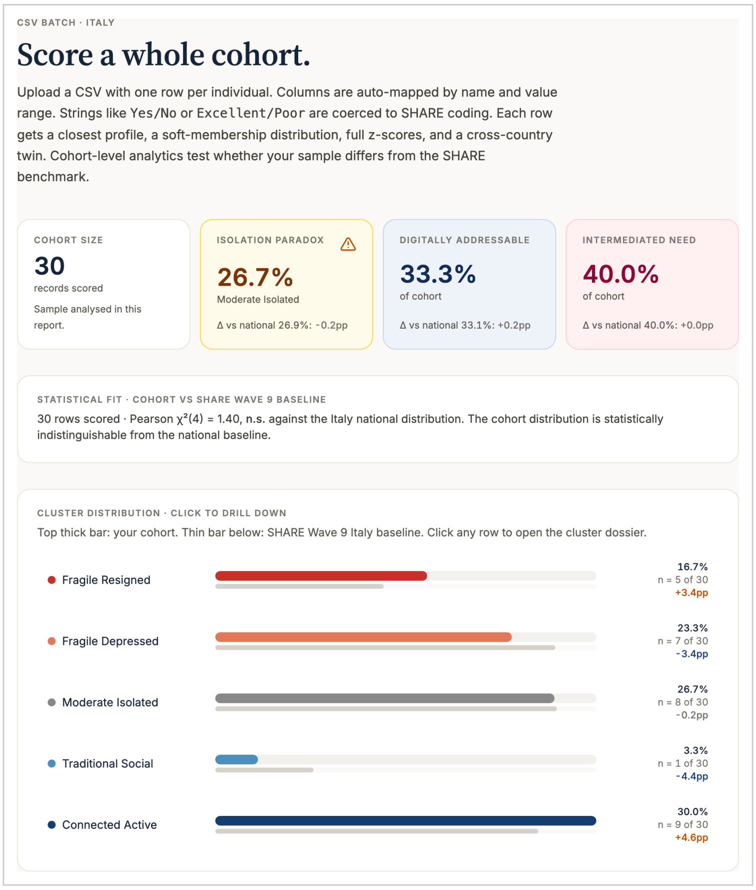


Figure 8.2: Cohort dashboard for a thirty-record test cohort. The four headline cards report cohort size, the Isolation Paradox flag, the digitally addressable share and the intermediated-need share, the latter three each reported against the national SHARE baseline. The statistical-fit line reports a Pearson χ^2 test against the national distribution, and the cluster-distribution chart plots cohort shares (thick bars) against the SHARE Wave 9 baseline (thin bars).

8.4 From cohort to action

Clicking any archetype opens a cluster dossier (Figure 8.3) that turns the segment share into an operational brief: a short "what this cluster makes visible" narrative, a per-record confidence summary, the cohort-versus-SHARE signature across all ten variables, the cohort's demographic composition, and the cluster's national-level healthcare-engagement gaps. A complementary per-record table (Figure 8.4) reports, for each individual, the assigned profile, an evidence label, the top-two profile probabilities, the distance percentile, and the cross-country twin.

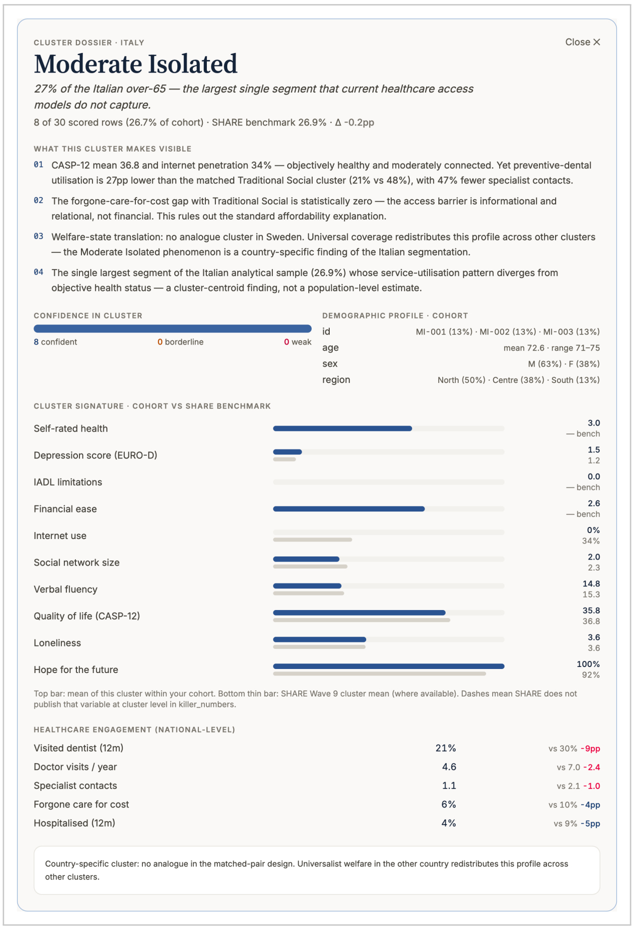


Figure 8.3: Cluster dossier for the Moderate Isolated segment of the test cohort. The dossier pairs the per-record confidence summary (the evidence ladder of §8.5) with the cohort-versus-SHARE cluster signature and the segment's national-level healthcare-engagement gaps, translating a population share into an operational brief.

PER-ROW PREDICTION

Show all 30

#	Profile	Evidence	Top-1	Runner-up	Top-2	Dist.	Pctile	Cross-country twin
1	Fragile Resigned	strong	99.8%	Fragile Depressed	0.2%	1.34	33	Fragile
2	Fragile Resigned	strong	100.0%	Fragile Depressed	0.0%	1.76	57	Fragile
3	Fragile Resigned	strong	99.8%	Fragile Depressed	0.2%	1.38	37	Fragile
4	Fragile Resigned	strong	100.0%	Fragile Depressed	0.0%	2.37	97	Fragile
5	Fragile Depressed	strong	99.0%	Fragile Resigned	0.7%	1.03	3	Fragile
6	Fragile Depressed	strong	99.5%	Moderate Isolated	0.3%	2.02	87	Fragile
7	Fragile Depressed	strong	87.9%	Fragile Resigned	12.0%	1.47	47	Fragile
8	Fragile Depressed	strong	85.1%	Moderate Isolated	13.9%	1.42	43	Fragile
9	Fragile Depressed	strong	87.4%	Fragile Resigned	12.6%	1.77	60	Fragile
10	Fragile Depressed	strong	85.6%	Moderate Isolated	13.7%	1.40	40	Fragile

Evidence = Strong (score ≥ 7), Moderate (4.5–7), Weak (< 4.5), where score combines coverage and imputation top-1/top-2 share on a 0–10 scale.
Pctile = within-cohort percentile of the row's distance to its centroid (high = atypical). Download to get full z-scores per row.

Figure 8.4: Per-record prediction table. Each individual is assigned a closest profile, an evidence label (Strong/Moderate/Weak), the top-two profile probabilities from the imputation distribution, a within-cohort distance percentile, and a cross-country twin. The cross-country twin is the nearest profile in the other country computed per record in the standardised analytical space, and need not coincide with the profile-level matched pairs of §6.3 (for example, a Fragile Depressed record may have the Swedish Fragile profile as its nearest twin rather than the Declining-pair counterpart Social Decline).

The three intended user roles map onto three uses of this output. A healthcare-practice manager reads the Isolation Paradox flag and the dossier's healthcare-engagement gaps to identify the autonomous-but-disengaged patients who slip through standard triage, and targets them with outreach rather than clinical escalation. A pharmacy-chain analyst reads the digitally addressable split to decide which segments a direct-to-consumer digital service can reach and which require an intermediated channel. A silver-economy product manager reads the aggregate addressability metrics (§6.6.1) as an estimate of the size of the design problem the thesis identifies. In each case the deliverable is the same: a screening instrument that routes a heterogeneous senior cohort to the design logic its empirical signature implies.

8.5 How records are scored

The typology of Chapters 4 and 5 is estimated on a clean probability sample under a fixed set of standardised variables. Applying it to an organisational cohort is an out-of-sample assignment problem in which several of those estimation conditions no longer hold: the cohort is a selected, non-probability sample; it observes only a subset of the analytical variables, often through proxies on arbitrary scales; and its marginal distributions need not match the SHARE reference on which the centroids and the dependence structure were estimated. The scoring layer is the set of devices that carry the partition across these departures and that report, rather than conceal, the uncertainty each one introduces.

Scoring operates on the ten profiler variables in the within-country standardised (z) space in which the K-means centroids of Chapters 4 and 5 are defined; a record with complete coverage is assigned to the nearest centroid in Euclidean distance. Distances are computed in the subspace spanned by the ten profiler variables, onto which the country centroids are projected; the remaining input dimensions do not enter the assignment. Four devices handle the ways an organisational cohort departs from the clean SHARE sample, and they share one design principle: each makes the uncertainty of a transported assignment visible rather than hiding it behind a point estimate.

Rank-quantile rescaling maps a proxy column on an arbitrary operational scale onto the SHARE coding through a monotone quantile transform, transporting the cohort's rank information without assuming a parametric proxy-to-target relationship. *Conditional-Gaussian imputation* handles partial coverage. The standardised vector is modelled as $z \sim \mathcal{N}(\mathbf{0}, \Sigma)$, with Σ the country's empirical correlation matrix from SHARE Wave 9. The missing coordinates M, given the observed coordinates O, then follow a Gaussian with conditional mean $\mu_{M|O} = \Sigma_{MO}\Sigma_{OO}^{-1}z_O$ and the corresponding (Schur-complement) covariance. Rather than collapse this to its mean, the platform draws 100 completions (by Cholesky factorisation of the conditional covariance), classifies each to its nearest centroid, and returns the modal cluster with the share of draws as its probability. A record near a boundary therefore returns a genuine distribution over profiles rather than a spurious point assignment.

This procedure rests on two assumptions the platform cannot verify from the cohort alone: that the analytical vector is approximately Gaussian in z-space, so that the conditional law takes the form above, and that the missingness is ignorable given the observed coordinates. Where either fails (markedly non-Gaussian dimensions, or coverage that is itself informative about the missing values), the imputation distribution is mis-calibrated, and the resulting profile probabilities are to be read together with the coverage-and-skew diagnostics below rather than on their own. A direct held-out calibration check—comparing the draw-based profile probabilities against known assignments on a labelled hold-out sample, where the Gaussian and ignorability assumptions are most exposed—is the natural first empirical test of this layer, and is left to the validation work of §9.6.

The *Kolmogorov–Smirnov pre-flight* (Figure 8.5) compares each mapped variable's cohort distribution to the SHARE reference, $D = \sup_x |F_{\text{cohort}}(x) - F_{\text{SHARE}}(x)|$, and flags material departures.

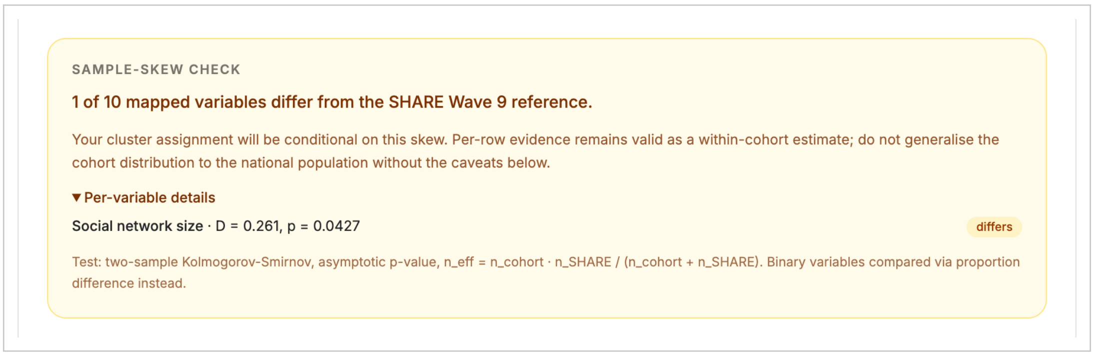


Figure 8.5: Sample-skew pre-flight. A non-blocking panel reports, for each mapped variable, a two-sample Kolmogorov–Smirnov comparison of the cohort against the SHARE Wave 9 reference, flagging the dimensions on which the transported assignment is least reliable.

The *evidence ladder* reports a per-record label on a 0–10 scale that adds a coverage component to a decisiveness component (the top-one probability and its margin over the runner-up), so a sharp peak resting on thin coverage is capped below the "Strong" band rather than over-trusted.

The platform is honest about what it does not establish: the typology is descriptive of the SHARE sample partition (§3.4), the proxy mappings transport rank rather than construct validity, and the cohort assignments are provisional screening signals rather than diagnoses. Because the interface processes individual-level records of a potentially vulnerable population, any operational deployment would also have to satisfy data-protection requirements (a GDPR lawful basis, data minimisation, and the right not to be subject to solely automated profiling) that lie outside the scope of this prototype, which is run here only on synthetic test data. The platform is currently supported analytically and untested in the field; the two validation routes, a qualitative comprehension study and a B2B field-trial, are developed in §9.6. So framed, the Atlas is the operational instrument through which the central finding of the thesis, that the Italian silver-economy gap is a missing social-and-informational mediation layer rather than a missing digital channel (§6.6.2), is made usable by the organisations positioned to close it.

Chapter 9

Conclusions

Welfare regimes, late-life partitions, and the silver-economy design gap

9.1 Overview

This thesis has developed a multidimensional segmentation of the over-65 population in Italy and Sweden using SHARE Wave 9 data, with a methodological commitment to including subjective dimensions of late-life experience as constituent inputs to the K-means partition rather than as external validation outcomes. The empirical core, comprising Chapters 4–7, produced two parallel typologies of five Italian and six Swedish archetypes, a formal matched cross-country comparison, and a robustness battery documenting the structural persistence of the centroid configuration.

9.2 Substantive findings

The empirical analysis has produced four findings that the thesis takes as its substantive contribution.

The first is the typological identification of the Isolation Paradox in the Italian over-65 population: the Moderate Isolated archetype (26.9% of the analytical sample; §4.3) combines intact functional health and economic resources with thin social and digital engagement, and significantly under-utilises the navigation-intensive and privately funded channels of the healthcare system (§4.7.1). The matched-pair analysis confirms the configuration is structurally Italian-only (§6.5.3): the empirical signature of a familistic regime that offers no substitute infrastructure for individuals outside dense informal networks.

The second is the binary-versus-graduated structure of late-life fragility: Italy splits the fragile stratum into two qualitatively distinct profiles (Fragile Resigned and Fragile Depressed, jointly 39.9%), spanning an active-depression-versus-settled-resignation contrast with separate policy responses, whereas Sweden partitions along a single binary fragility line, one Fragile cluster against five autonomous profiles (§6.4.1). The universalistic regime absorbs the burden into a single targetable institutional segment; the familistic regime distributes it across multiple distinguishable household-level configurations.

The third is universal digital saturation across the autonomous Swedish profiles against resource-graduated adoption in Italy (§6.4.2): the same diagnostic instrument cuts at the functional-fragility line in Sweden and at the resource-and-engagement gradient in Italy. The digital channel therefore substitutes for the family-and-community infrastructure of healthcare and service navigation under the familistic regime, and merely complements the universal-access infrastructure under the universalistic one.

The fourth is the age-driven rotation of the Swedish high-resource pole—from a digital-engagement anchor (Connected Wealthy) in the young-old to a wealth-accumulation anchor (Asset Rich) in the oldest-old—against the parallel decline of both well-resourced Italian profiles (§6.4.3). The rotation is the late-life trajectory permitted by a regime that absorbs the functional burden of advanced old age institutionally; the parallel decline is the trajectory where that burden falls on the family. That Asset Rich has no matched Italian counterpart (§6.3) is itself the structural expression of this rotation: an oldest-old wealth pole that the parallel-decline trajectory of the familistic regime never produces.

The four findings are not independent observations on a heterogeneous dataset. Each is the within-typology projection of the same regime contrast onto a different latent dimension of late-life experience. The thesis claim that the welfare regime is the structuring institution of the late-life partition is the joint reading of the four.

9.3 Theoretical contribution

The comparative welfare-regime literature reviewed in Chapter 2 establishes that the institutional treatment of social risk sorts the European welfare states into a small number of regime types with systematically different aggregate outcomes. The contribution of this thesis to that literature is to extend the regime reading from aggregate outcomes to the typological partition of the populations served by each regime: the Italian and Swedish over-65 populations differ not only on aggregate indicators of late-life experience but on the structural shape of the partition into archetypes that the same segmentation pipeline returns from each population.

The methodological contribution to the segmentation literature is the inclusion of the subjective block as constituent input to the K-means partition. The empirical justification (§3.8, §7.5) shows that dropping the subjective dimensions reduces external discriminative power on life satisfaction by 41% in Italy (22% in Sweden) relative to the full specification, and that the 4D specification collapses the Fragile Resigned versus Fragile Depressed contrast. The position is that the segmentation literature on older adults should treat the subjective dimensions as constituent rather than derivative.

9.4 Operational contribution to the silver-economy literature

The operational implications developed in §6.6 rest on three observations. First, addressability through direct-to-consumer digital products is structurally different across the two countries: in Sweden the autonomous profiles (88.5% of the analytical sample) reach internet adoption of 75–99% and are digitally addressable as a homogeneous block, against approximately 33% of the Italian sample (Connected Active 25.4% + Traditional Social 7.7%) at broadly comparable adoption (74.8% and 65.8% respectively). The Italian silver-economy market is structurally segmented into addressable and non-addressable cluster groups on dimensions that are themselves the signature of the regime.

Second, the unaddressed Italian Moderate Isolated segment (26.9%) is the most direct case for product-design innovation: the binding constraint is not the absence of a digital channel but the absence of the social-and-informational mediation layer that would route the available channel towards healthcare and service utilisation. The candidate design is a digital channel combined with the social-mediation designs of §6.6.2, a layer that the Italian regime does not currently provide and that the Swedish typology does not require.

Third, the cross-country gap is structural rather than transitory: the Italian 65–74 cohort closing the digital gap to the Swedish reference under three rate scenarios takes approximately 6 to 30 years (§6.6.3), and the cross-country dental-care gap is documented across the four matched pairs and concentrated, structurally, on the unmatched Moderate Isolated segment. The market-structure implication is therefore stated qualitatively: an addressable third of the Italian over-65 sample is served by direct-to-consumer and hybrid digital designs already targeted by the commercial silver-economy literature, against a non-addressable two thirds that would require the alternative design logics summarised in Table 9.1.

The operational counterpart of the Italian silver-economy design problem, a working prototype of the missing mediation layer built as a B2B cohort-segmentation instrument over the typology, is developed in Chapter 8.

Table 9.1: Italian over-65 typology by silver-economy addressability logic

Addressability	Profile	Share (%)	Operational design logic
Addressable through direct-to-consumer digital channel			
DTC digital	Connected Active	25.4	Direct-to-consumer digital products; high internet adoption (74.8%).
Hybrid intermediated	Traditional Social	7.7	Hybrid offline-digital designs; moderate internet (65.8%) layered onto offline trust networks.
Subtotal addressable		**33.1**	Approximately one third of the analytical sample.
Not addressable through direct-to-consumer digital channel			
Social-mediation	Moderate Isolated	26.9	Social-and-informational mediation layer over digital channel; non-trivial internet (34.1%) without dense informal networks.
Affect-aware	Fragile Depressed	26.7	Affect-aware designs addressing the binding behavioural constraint (depressive affect, EURO-D 4.2); 17.3% internet.
Family-mediated	Fragile Resigned	13.2	Family or institutional intermediation; 8.6% internet. Long-term-care welfare branch.
Subtotal non-addressable		**66.8**	Approximately two thirds of the analytical sample.
TOTAL		**99.9**	Italian over-65 analytical sample (§3.2).

Note. Profile shares are taken from Table 4.1 and refer to the SHARE Wave 9 Italian analytical sample. Subtotals add to 99.9% due to rounding. The table maps each typological segment to the operational design logic implied by its empirical signature on the digital and social-engagement dimensions (§4.3, §6.6.1); shares are descriptive of the sample partition (see §3.4).

9.5 Limitations

Three classes of limitations bound the substantive claims. The first is inferential and bears on the central claim: the regime reading rests on a cross-sectional, two-country comparison and cannot causally identify the welfare regime against the other respects in which Italy and Sweden differ (cohort composition, urbanisation, income level, cultural and historical context). The cross-wave evidence of §7.6 indicates persistence at the level of centroid structure—not of individual assignment—and not causal attribution; the regime interpretation is therefore the most parsimonious reading of the structural contrast rather than an identified effect, and generalisation beyond the two regimes requires the additional-country evidence set out in §9.6.

On the data side, the exclusion of proxy interviews biases the Fragile Resigned cluster downward in the prevalence of the most adverse configurations (severe cognitive impairment, institutional residence): the typology partitions the self-reporting over-65 cohort, not the full population. The single multiple-imputation replicate (§3.4) introduces residual sensitivity at the Fragile–Moderate boundary, although item-level missingness is materially low (below 6% on subjective indicators and below 9% on income and wealth; §7.2) and does not undermine the cluster centroids.

On the modelling side, silhouette values lie below the 0.250 threshold (§7.3), and K-means/LCA convergence ranges from thin-boundary to strong (49.3–95.6%), with the lowest rates on Wealthy Digital (49.3%), Asset Rich (53.1%) and Moderate Isolated (54.6%) flagging geometrically thin cluster boundaries. With 4D/5D ARI at 0.41 (Italy) and 0.52 (Sweden), the subjective block does not merely perturb the partition at the margins; it materially reshapes it. The Hopkins statistic ($H = 0.75$ IT, 0.74 SE) supports the existence of a real underlying structure, but the substantive findings of §9.2 should be read as claims about cluster centroids and structural relations, not about marginal individual assignments.

9.6 Future research

Five directions extend the analysis. First, applying the pipeline to additional SHARE countries (Germany, corporatist; Spain, conservative-Catholic; Poland and Czech Republic, post-socialist) would test whether the partition generalises across regimes. Second, a transition matrix on the cross-wave panel intersection would identify the drivers of late-life cluster trajectories and forecast segment-share evolution. Third, a field-trial pairing digital infrastructure with a community-based intermediary would test the social-mediation-layer proposition for the Moderate Isolated segment. Fourth, integrating the cluster assignments with within-Italy regional healthcare-access indices would yield a within-country counterpart. Fifth, the Italian Silver Atlas platform of Chapter 8 is supported analytically but remains to be tested in the field; the natural extensions are a comprehension study with Italian seniors across the five clusters and a B2B field-trial of the cohort dashboard.

9.7 Closing remarks

The same pipeline partitions the Italian and Swedish over-65 populations into structurally different typologies whose differences—centred on the Moderate Isolated segment—are the empirical signature of each welfare regime. For the European silver economy, that signature is not a statistical curiosity but a design constraint: the Italian market does not lack a digital channel so much as the social-and-informational mediation layer that would carry it to the Moderate Isolated. The Silver Atlas is an instrument built to make that segment visible to the organisations positioned to reach it.

Appendix A

Sensitivity Analyses

This appendix documents the five sensitivity analyses run on the deployed pipeline in response to the supervisor's methodological review. Each analysis perturbs one methodological choice of the Italian or Swedish branch of the pipeline, re-derives the resulting partition, and reports the adjusted Rand index against the deployed partition together with relevant fit diagnostics. The interpretive thresholds are stated upfront for each analysis; the verdicts in Table A.3 are read against those thresholds rather than as absolute statements of correctness. The quantitative detail of the distance-metric and factor-extraction analyses is tabulated in Tables A.1 and A.2; fuller per-analysis outputs are additionally archived as comma-separated value files under `v9/outputs/sensitivities/`.

A.1 Outlier exclusion threshold (Sweden)

The deployed Mahalanobis outlier-exclusion routine for Sweden (§3.4) operates at $\alpha = 0.001$, the conventional χ^2 cutoff at the 99.9th percentile. The procedure removes 44 respondents from the post-listwise Swedish sub-sample of 1,831, equivalent to 2.4% (Table 3.1, §3.2). The robustness check re-runs the full pipeline under two alternative thresholds, $\alpha = 0.0001$ (more conservative; outliers removed = 14) and $\alpha = 0$ (no exclusion). The K-means partition at $k = 6$ is recomputed under each variant on the 31-variable standardised input space with the same seed and initialisation count of the deployed routine, and the adjusted Rand index is computed against the deployed partition over the intersection of the two samples. The resulting ARI values are 0.654 at $\alpha = 0.0001$ (+30 rows retained) and 0.551 at $\alpha = 0$ (+44 rows retained). The values lie below the 0.80 reference threshold for substantive partition robustness.

A profile-level purity calculation qualifies the lower-than-threshold ARI. Each variant cluster is mapped to the dominant deployed profile, and purity is the proportion of observations in the majority-profile column. The purity is 0.822 at $\alpha = 0.0001$ and 0.727 at $\alpha = 0$. Four of the six macro-profiles retain identifiable variant-cluster homes under both alternatives, while the boundary observations between contiguous profiles (Moderate, Social Decline) migrate at the higher rate observed at $\alpha = 0$. The choice of $\alpha = 0.001$ is retained as the conventional threshold. The partition is read as robust at the macro-profile level under the stricter alternative and as moderately less robust under the no-exclusion alternative. The profile-level confusion against the stricter $\alpha = 0.0001$ alternative is reported in §7.3.1 (Table 7.2).

A.2 Distance metric: Gower versus Euclidean

The deployed K-means routine for both countries operates on the Euclidean distance over the standardised variable space. The robustness check substitutes the Gower dissimilarity computed via `cluster::daisy` on the original (non-standardised) data with all binary variables declared as asymmetric, and runs partitioning-around-medoids at the deployed k ($k = 5$ for Italy, $k = 6$ for Sweden). The adjusted Rand index between the Gower-PAM partition and the deployed K-means partition is 0.205 for Italy and 0.091 for Sweden. Both values are below the 0.70 reference threshold for partition convergence across distance choices.

The empirical disagreement between the two distance choices is read as substantively informative rather than as a failure of either algorithm. The Euclidean choice is retained on three grounds. First, the standardised input space carries an interpretation of cluster centroids as mean z-deviations from the population baseline, a quantity consumed directly by the ANOVA validation of §3.6 and the fingerprint visualisations of §4.4. The Gower dissimilarity, by normalising each variable to its own range, dissolves the cross-variable comparability that underpins the centroid reading. Second, the K-means and Euclidean combination is standard practice in the welfare-regime and typology literature on which the comparative reading of Chapter 6 builds. Third, the Gower and PAM combination produces a different partition because it weights the binary indicators differently. Its asymmetric-binary handling treats two zero-coded observations as not similar rather than as jointly absent, suppressing the role of co-absence in the digital and social blocks; the deployed reading, by contrast, treats co-absence as a substantively informative configuration of low engagement. The two partitions therefore represent two different definitions of what counts as similar, not two competing estimates of an underlying ground truth.

Profile-level analysis of the Gower partition confirms that the disagreement is pervasive rather than confined to a single stratum: the global profile-level purity is 0.52 in Italy and 0.38 in Sweden, and no Italian or Swedish profile concentrates more than 70% of its members on a single Gower-PAM cluster. The low purity is the profile-level signature of the same effect the adjusted Rand index quantifies, namely that the metric choice reorganises the partition broadly rather than isolating a robust subset of profiles. This is consistent with the cross-method evidence (§7.5) that several cluster boundaries are geometrically thin in the standardised space.

A complementary sensitivity holds the Euclidean K-means specification but down-weights the binary indicators, to probe the amplification that within-country z-standardisation introduces on low-prevalence binary variables. The 11 binary indicators in the Swedish active set (9 in the Italian active set, after the zero-variance exclusion of `ac035d4` and `ac035d7`) are re-scaled by a factor $w < 1$, leaving the 20 rank-transformed non-binary variables unchanged, and the K-means partition at the deployed k is re-derived under each weighting. The unit-weight case $w = 1$ reproduces the deployed partition exactly. At $w = 0.5$ the adjusted Rand index against the deployed partition is 0.370 for Italy and 0.335 for Sweden, and at $w = 1/\sqrt{2}$ it is 0.401 for Italy and 0.379 for Sweden. The binary indicators therefore carry substantial partition-determining weight: down-weighting them reassigns a non-trivial share of respondents. At the profile level the macro-typology is more stable than the adjusted Rand index alone implies, with the deployed profiles retaining dominant homes at a purity of 0.64–0.67 (Italy) and 0.60–0.64 (Sweden) across the two weightings, but the sensitivity is real and is not read as robustness.

The deployed unit-weighted specification is retained on two substantive grounds the down-weighting clarifies rather than refutes. The binary indicators of digital and social engagement are constitutive of the profiles the typology is built to recover (the Wealthy Digital and Connected configurations are defined precisely by these items), so down-weighting them dissolves distinctions the analysis is designed to capture. Any down-weight magnitude, moreover, is a free parameter with no measurement-theoretic anchor, whereas unit-weighted z-standardisation preserves the interpretation of cluster centroids as deviations on a single common scale across binary and non-binary indicators.

Taken together, the three components (Gower-PAM as an alternative metric, the profile-level breakdown of the disagreement, and binary down-weighting as a parametric alternative; Table A.1) converge on a single reading: the partition is sensitive to both the distance metric and the weighting of the binary indicators. That sensitivity reinforces, rather than undermines, the scope condition stated in §7.7, namely that the substantive findings are claims about the centroid configuration of the profiles and the relations among them, not about the assignment of marginal respondents, which is the level at which the metric and weighting choices exert their influence.

Table A.1: Distance-metric and binary-weighting sensitivity: adjusted Rand index and profile purity against the deployed partition

	ARI vs deployed		Profile purity	
Specification	**Italy**	**Sweden**	**Italy**	**Sweden**
Alternative distance metric				
Gower dissimilarity + PAM	0.205	0.091	0.52	0.38
Euclidean K-means, binary indicators down-weighted by w				
$w = 1$ (deployed)	1.000	1.000	—	—
$w = 1/\sqrt{2}$	0.401	0.379	0.64–0.67	0.60–0.64
$w = 0.5$	0.370	0.335	0.64–0.67	0.60–0.64

Note. ARI is computed against the deployed unit-weighted Euclidean K-means partition ($k = 5$ Italy, $k = 6$ Sweden). Reference thresholds: 0.70 for cross-metric convergence and 0.80 for substantive partition robustness. The Gower dissimilarity (`cluster::daisy`) declares all binary indicators asymmetric and is paired with partitioning-around-medoids (PAM). Profile purity for the two down-weightings is reported as the range across them (§A.2).

A.3 Alternative factor extraction methods (Sweden)

The deployed factor analysis for Sweden (§3.5) extracts six factors via maximum-likelihood factoring on the 31 active variables, with varimax rotation. An earlier principal-axis specification produced a Heywood case on `social_integration` (loading 1.11, communality $h^2 = 1.38$), which the maximum-likelihood specification resolves cleanly. Three alternative specifications are reported here to document the fit and clustering stability of the deployed choice. The first re-extracts the six factors via principal-axis factoring on the same 31 variables (the prior deployed specification); the second substitutes minimum-residual factoring; the third reverts to principal-axis factoring but on the 30 variables that exclude `social_integration`.

The principal-axis specification on the 31 variables, the prior deployed choice, retains the Heywood case on `social_integration`, with RMSEA = 0.033 and TLI = 0.897. The minimum-residual specification likewise retains the Heywood case, with RMSEA = 0.034 and TLI = 0.890, as expected from a principal-axis-type estimator. The deployed maximum-likelihood specification eliminates the Heywood case at effectively identical fit (RMSEA = 0.033, TLI = 0.897), indicating that the Heywood loading is specification-dependent rather than data-pathological. The 30-variable principal-axis specification, which drops `social_integration`, also eliminates the Heywood case and returns the best overall fit of the alternatives (RMSEA = 0.029, TLI = 0.908, KMO = 0.830 versus 0.808 on the 31-variable input). The factor structure is therefore not unique: the choice of extraction method and the inclusion of `social_integration` each shape whether the Heywood case appears, while every specification yields a defensible six-factor solution under standard fit criteria (Table A.2).

Table A.2: Alternative factor-extraction specifications for Sweden: fit and clustering stability

Specification	RMSEA	TLI	KMO	Heywood	ARI vs deployed
ML, 31 vars (deployed)	0.033	0.897	0.808	No	—
PA, 31 vars	0.033	0.897	0.808	Yes	1.000
MinRes, 31 vars	0.034	0.890	0.808	Yes	1.000
PA, 30 vars (drop social-integ.)	0.029	0.908	0.830	No	0.518

Note. All specifications extract six factors with varimax rotation on the Swedish active set. ML = maximum-likelihood, PA = principal-axis, MinRes = minimum-residual. KMO is a property of the input correlation matrix and is therefore identical across the three 31-variable specifications. ARI is computed between each specification's K-means partition and the deployed partition; it is unity for the 31-variable specifications because the clustering input is invariant to the factor extraction (§A.3).

The implication for the deployed clustering is independent of the factor choice for the three 31-variable specifications. The K-means routine operates on the standardised variable space and is invariant to the factor extraction by construction: the adjusted Rand index of the principal-axis and minimum-residual specifications against the deployed maximum-likelihood partition is unity, since the cluster input is identical. The 30-variable principal-axis specification does alter the K-means input space, and the resulting ARI is 0.518: the removal of `social_integration` from the cluster input induces a substantively different partition over the Swedish sample. The deployed specification (ML, 31 variables, `social_integration` retained) is therefore defensible on two grounds that the alternatives clarify rather than challenge. The factor analysis is descriptive and does not propagate into the clustering, which is invariant to the extraction method. The inclusion of `social_integration`, in turn, is required by the Isolation Paradox reading in Chapter 4, which the maximum-likelihood extraction now preserves without incurring the Heywood case of the principal-axis alternative.

A.4 Italian 29-versus-31 variable asymmetry

The Italian analytical sample exhibits zero variance on the two activity-indicator variables `ac035d4` and `ac035d7`, which are consequently dropped from the active variable set, yielding 29 active variables against the 31 active variables of the Swedish branch (§3.4). The 29-versus-31 asymmetry is documented in the main text; this section reports the preemptive robustness check that quantifies the impact of the asymmetry on the deployed Italian partition.

The two zero-variance variables are re-included with a small Gaussian perturbation ($\sigma = 0.05$ on the standardised scale) drawn under the deployed seed; the perturbation magnitude is calibrated to introduce non-degenerate variance without injecting structure. The full pipeline is re-derived on the 31-variable input space, with principal-axis factor analysis at six factors and varimax rotation, K-means clustering at $k = 5$, and identical seed and initialisation specifications. The factor analysis converges cleanly with no Heywood case (RMSEA = 0.049, RMSR = 0.027). The adjusted Rand index between the 31-variable variant partition and the deployed 29-variable partition is 0.977. The 29-versus-31 asymmetry is read as empirically benign: the deployed Italian partition is structurally identical, up to a negligible boundary perturbation, to the partition that would have been derived had the two zero-variance variables not been dropped.

A.5 Imputation-replicate sensitivity

The deployed analytical sample draws the imputed health, economic and cognitive variables from the first SHARE multiple-imputation replicate (`implicat == 1`; §7.2). This check quantifies the leverage of that single draw on the partition by re-running the entire pipeline under the second and third imputation replicates. For each of `implicat == 2` and `implicat == 3`, both national samples are re-assembled from the raw SHARE Wave 9 modules; only the `gv_imputations`-sourced variables change across replicates, while the non-imputed modules (social networks, activities, social participation, mental health, survival expectations) are held fixed. The full preprocessing (listwise deletion, rank transformation of the non-binary variables, z-standardisation, Mahalanobis outlier exclusion at $\alpha = 0.001$) and K-means clustering at the deployed k ($k = 5$ for Italy, $k = 6$ for Sweden) are then re-derived with the deployed seed and an initialisation count of 100. The adjusted Rand index and the cluster purity are computed between each replicate's partition and the deployed `implicat == 1` partition over the respondents common to both samples (matched on `mergeid`). The re-assembled `implicat == 1` pipeline reproduces the deployed Italian and Swedish partitions exactly (adjusted Rand index = 1.000 for both countries), which confirms that the comparison isolates the imputation draw rather than any incidental difference in the reproduction.

The partition is stable across the imputation replicates in both countries. For Italy the adjusted Rand index against the deployed partition is 0.917 under `implicat == 2` and 0.878 under `implicat == 3`; for Sweden it is 0.855 and 0.875 respectively. All four values lie above the 0.80 reference threshold for substantive partition robustness. The cluster purity—the share of respondents retaining their deployed cluster home under the alternative replicate—is 0.97 and 0.96 for Italy and 0.94 and 0.95 for Sweden, so between 94% and 97% of respondents are assigned to the same profile irrespective of the imputation draw. The residual disagreement is concentrated, as §7.2 anticipates, at the boundary between the Fragile and Moderate strata, where the imputation of log-net-wealth carries the largest residual variance; the macro-typology—the five Italian and six Swedish profiles and their centroid configuration—is invariant to the draw.

The single-replicate strategy of §7.2 is therefore empirically vindicated: the imputation draw exerts limited leverage on the partition, and the substantive readings of Chapters 4–6, which are claims about the profile centroids rather than about the assignment of marginal respondents, are not contingent on the choice of replicate. The per-replicate partitions and index values are archived in `v9/outputs/imputation_sensitivity.json`.

A.6 Summary

Table A.3 reports the five sensitivity analyses against the interpretive thresholds stated in the preceding sections. Two analyses close empirically: the 29-versus-31 variable asymmetry and the imputation-replicate check, each reproducing the deployed partition at an adjusted Rand index well above the 0.80 reference (§A.5). The outlier-threshold and the distance-metric and binary-weighting analyses are read as concerns that hold at the macro-profile level and are retained on the substantive grounds of §A.2 rather than as claims of invariance. The Heywood case is resolved in the deployed pipeline by maximum-likelihood extraction (§A.3), which removes it at unchanged fit and leaves the clustering invariant.

Table A.3: Sensitivity analyses: summary against interpretive thresholds.

Analysis	**Metric (vs. threshold)**	**Verdict**
Outlier exclusion threshold (Sweden, $\alpha = 0.0001$ / $\alpha = 0$)	ARI 0.654 / 0.551 vs. 0.80; purity 0.822 / 0.727 (Table 7.2)	Concern (macro-robust)
Distance metric and binary weighting, Italy / Sweden	ARI 0.09–0.40 across the metric and weighting variants, below the 0.70 / 0.80 thresholds (Table A.1)	Concern (defended)
Heywood case, Sweden factor extraction (PA $\rightarrow$ ML)	RMSEA 0.029–0.034, all six-factor solutions defensible; ML removes the Heywood case, clustering ARI 1.000 on the 31-variable input (Table A.2)	Resolved (ML deployed)
Italian 29-vs-31 variable asymmetry (perturbed re-inclusion)	ARI 0.977 vs. 0.80; no Heywood case	Pass
Imputation replicate (`implicat` 2/3), Italy / Sweden	ARI 0.88–0.92 (Italy) / 0.86–0.88 (Sweden) vs. 0.80; purity 0.94–0.97 (§A.5)	Pass

Appendix B

Unweighted versus Post-stratified Cluster Shares

The segmentation is estimated on the unweighted analytical sample (§3.4). To document the empirical magnitude of the SHARE survey design, a post-hoc projection re-weights the within-sample cluster fractions using the SHARE Wave 9 calibrated cross-sectional individual weight (`cciw_w9`), computing each profile's weighted share as $\sum_{i \in C} w_i / \sum_i w_i$, under the additional assumption that the proxy-interview exclusion of the analytical sample does not bias the weighted-share estimate. Table B.1 reports the Italian and Swedish cluster shares under the deployed unweighted specification and under this post-hoc re-weighting. The differences are uniformly within 3.2 percentage points across all eleven profiles, so the descriptive cluster shares of Chapters 4 and 5 are broadly stable under the weighting adjustment; the national-population projection of Figure 4.2 applies these weighted post-hoc shares to the 2024 Istat over-65 estimate.

Table B.1: Cluster shares under deployed unweighted specification and post-hoc weighted re-projection

Country	Profile	Unweighted (%)	Weighted (%)	Δ pp
Italy	Fragile Resigned	13.2	14.5	+1.2
Italy	Fragile Depressed	26.7	27.0	+0.3
Italy	Moderate Isolated	26.9	26.0	−0.9
Italy	Traditional Social	7.7	8.6	+0.9
Italy	Connected Active	25.4	23.9	−1.5
Sweden	Fragile	11.5	13.0	+1.5
Sweden	Social Decline	8.3	9.0	+0.7
Sweden	Moderate	23.3	20.1	−3.2
Sweden	Asset Rich	19.1	20.8	+1.8
Sweden	Wealthy Digital	7.7	8.0	+0.3
Sweden	Connected Wealthy	30.0	29.0	−1.0

Note. Unweighted shares are within-sample cluster fractions. Weighted shares are computed as $\sum_{i \in C} w_i / \sum_i w_i$ using the SHARE Wave 9 calibrated cross-sectional individual weight (`cciw_w9`). The population projection in Figure 4.2 applies the weighted shares to the Istat 2024 Italian over-65 population estimate (14.18 million). Differences computed on unrounded shares.

Appendix C

SHARE Data and Funding Acknowledgement

The empirical analysis uses data from SHARE Wave 9 (DOI: 10.6103/SHARE.w9.900) (SHARE-ERIC, 2024b) and, for the cross-wave analysis of §7.6, SHARE Wave 8 (DOI: 10.6103/SHARE.w8.900) (SHARE-ERIC, 2024a), release version 9.0.0; see Börsch-Supan et al. (2013) for methodological details. The funding acknowledgement below is reproduced in full as required by the SHARE-ERIC Conditions of Use.

The SHARE data collection has been funded by the European Commission, DG RTD through FP5 (QLK6-CT-2001-00360), FP6 (SHARE-I3: RII-CT-2006-062193, COMPARE: CIT5-CT-2005-028857, SHARELIFE: CIT4-CT-2006-028812), FP7 (SHARE-PREP: GA no. 211909, SHARE-LEAP: GA no. 227822, SHARE M4: GA no. 261982, DASISH: GA no. 283646) and Horizon 2020 (SHARE-DEV3: GA no. 676536, SHARE-COHESION: GA no. 870628, SERISS: GA no. 654221, SSHOC: GA no. 823782, SHARE-COVID19: GA no. 101015924) and by DG Employment, Social Affairs & Inclusion through VS 2015/0195, VS 2016/0135, VS 2018/0285, VS 2019/0332, VS 2020/0313, SHARE-EUCOV: GA no. 101052589 and EUCOVII: GA no. 101102412. Additional funding from the German Federal Ministry of Research, Technology and Space, the Max Planck Society for the Advancement of Science, the U.S. National Institute on Aging (U01_AG09740-13S2, P01_AG005842, P01_AG08291, P30_AG12815, R21_AG025169, Y1-AG-4553-01, IAG_BSR06-11, OGHA_04-064, BSR12-04, R01_AG052527-02, R01_AG056329-02, R01_AG063944, HHSN271201300071C, RAG052527A) and from various national funding sources is gratefully acknowledged (see https://www.share-eric.eu).

Bibliography

Anttonen, A., & Sipilä, J. (1996). European social care services: Is it possible to identify models? *Journal of European Social Policy*, *6*(2), 87–100.

Bambra, C. (2007). Going beyond The Three Worlds of Welfare Capitalism: Regime theory and public health research. *Journal of Epidemiology and Community Health*, *61*(12), 1098–1102.

Bartlett, M. S. (1950). Tests of significance in factor analysis. *British Journal of Statistical Psychology*, *3*(2), 77–85.

Berkman, L. F., & Kawachi, I. (2000). *Social epidemiology*. Oxford University Press.

Bettio, F., & Plantenga, J. (2004). Comparing care regimes in Europe. *Feminist Economics*, *10*(1), 85–113.

Börsch-Supan, A., Brandt, M., Hunkler, C., Kneip, T., Korbmacher, J., Malter, F., Schaan, B., Stuck, S., & Zuber, S. (2013). Data resource profile: The Survey of Health, Ageing and Retirement in Europe (SHARE). *International Journal of Epidemiology*, *42*(4), 992–1001.

Bowling, A. (2006). Lay perceptions of successful ageing: Findings from a national survey of middle aged and older adults in Britain. *European Journal of Ageing*, *3*(3), 123–136.

Campens, J., Vercruyssen, A., De Keyzer, W., & De Witte, N. (2024). Using Two-Step Cluster Analysis and Latent Class Analysis to construct a typology of internet users among community-dwelling older people. *Journal of Gerontological Social Work*, *67*(6).

Christelis, D. (2011). *Imputation of missing data in Waves 1 and 2 of SHARE* (SHARE Working Paper Series No. 01-2011). Munich Center for the Economics of Aging.

Courtin, E., & Knapp, M. (2017). Social isolation, loneliness and health in old age: A scoping review. *Health & Social Care in the Community*, *25*(3), 799–812.

Dannefer, D. (2003). Cumulative advantage/disadvantage and the life course. *The Journals of Gerontology Series B*, *58*(6), S327–S337.

Duda, R. O., Hart, P. E., & Stork, D. G. (2001). *Pattern classification* (2nd ed.). Wiley-Interscience.

Esping-Andersen, G. (1990). *The three worlds of welfare capitalism*. Princeton University Press.

Eurostat. (2022). Individuals — internet use — Digital Economy and Society statistics [Data extracted on internet use in the past three months by age group, accessed via Eurostat data browser.].

Eurostat. (2024). Individuals — internet use, by age group — Digital Economy and Society statistics [2024 release on internet use in the past three months among individuals aged 65–74, accessed via Eurostat data browser.].

Ferrera, M. (1996). The 'Southern model' of welfare in social Europe. *Journal of European Social Policy*, *6*(1), 17–37.

Hargittai, E., & Dobransky, K. (2017). Old dogs, new clicks: Digital inequality in skills and uses among older adults. *Canadian Journal of Communication*, *42*(2), 195–212.

Hartigan, J. A., & Wong, M. A. (1979). Algorithm AS 136: A K-means clustering algorithm. *Journal of the Royal Statistical Society. Series C (Applied Statistics)*, *28*(1), 100–108.

Hawkley, L. C., & Cacioppo, J. T. (2010). Loneliness matters: A theoretical and empirical review of consequences and mechanisms. *Annals of Behavioral Medicine*, *40*(2), 218–227.

Horn, J. L. (1965). A rationale and test for the number of factors in factor analysis. *Psychometrika*, *30*(2), 179–185.

Hughes, M. E., Waite, L. J., Hawkley, L. C., & Cacioppo, J. T. (2004). A short scale for measuring loneliness in large surveys. *Research on Aging*, *26*(6), 655–672.

Hyde, M., Wiggins, R. D., Higgs, P., & Blane, D. B. (2003). A measure of quality of life in early old age: The theory, development and properties of a needs satisfaction model (CASP-19). *Aging & Mental Health*, *7*(3), 186–194.

Idler, E. L., & Benyamini, Y. (1997). Self-rated health and mortality: A review of twenty-seven community studies. *Journal of Health and Social Behavior*, *38*(1), 21–37.

Istat. (2024). Resident population on 1 January by age, sex and marital status — Italy [Italian over-65 resident population estimate for 2024 accessed via the Istat statistical data warehouse.].

Kaiser, H. F. (1958). The varimax criterion for analytic rotation in factor analysis. *Psychometrika*, *23*(3), 187–200.

Kaiser, H. F. (1974). An index of factorial simplicity. *Psychometrika*, *39*(1), 31–36.

Kaufman, L., & Rousseeuw, P. J. (1990). *Finding groups in data: An introduction to cluster analysis*. Wiley.

Lalive d'Epinay, C., Cavalli, S., & Spini, D. (2003). The death of a loved one: Impact on health and relationships in very old age. *Omega: Journal of Death and Dying*, *47*(3), 265–284.

Lestari, S. K., et al. (2022). Frailty and types of social relationships among older adults in 17 European countries. *Archives of Gerontology and Geriatrics*, *101*, 104705.

Lo Scalzo, A., Donatini, A., Orzella, L., Cicchetti, A., Profili, S., & Maresso, A. (2009). Italy: Health system review. *Health Systems in Transition*, *11*(6), 1–216.

MacQueen, J. B. (1967). Some methods for classification and analysis of multivariate observations. *Proceedings of the 5th Berkeley Symposium on Mathematical Statistics and Probability, Volume 1: Statistics*, 281–297.

Naldini, M., & Saraceno, C. (2008). Social and family policies in Italy: Not totally frozen but far from structural reforms. *Social Policy & Administration*, *42*(7), 733–748.

Netuveli, G., & Blane, D. (2008). Quality of life in older ages. *British Medical Bulletin*, *85*(1), 113–126.

OECD. (2023). *Health at a Glance 2023: OECD Indicators*.

Piccarreta, R., & Trentini, F. (2025). Data analytics and visualization: Course materials.

Prince, M. J., Reischies, F., Beekman, A. T. F., Fuhrer, R., Jonker, C., Kivela, S.-L., Lawlor, B. A., Lobo, A., Magnusson, H., Fichter, M., van Oyen, H., Roelands, M., Skoog, I., Turrina, C., & Copeland, J. R. M. (1999). Development of the EURO-D scale: A European Union initiative to compare symptoms of depression in 14 European centres. *British Journal of Psychiatry*, *174*(4), 330–338.

Quan-Haase, A., Mo, G. Y., & Wellman, B. (2017). Connected seniors: How older adults in East York exchange social support online and offline. *Information, Communication & Society*, *20*(7), 967–983.

Rowe, J. W., & Kahn, R. L. (1997). Successful aging. *The Gerontologist*, *37*(4), 433–440.

Saraceno, C. (2010). Social inequalities in facing old-age dependency. *Journal of European Social Policy*, *20*(1), 32–44.

SHARE-ERIC. (2024a). Survey of health, ageing and retirement in Europe (SHARE) Wave 8 [Data set, release version 9.0.0. DOI: 10.6103/SHARE.w8.900].

SHARE-ERIC. (2024b). Survey of health, ageing and retirement in Europe (SHARE) Wave 9 [Data set, release version 9.0.0. DOI: 10.6103/SHARE.w9.900].

Statistics Sweden. (2024). Population by age and sex on 31 December — Sweden [Swedish over-65 resident population estimate for 2024 accessed via the Statistics Sweden statistical database.].

Steinley, D. (2006). K-means clustering: A half-century synthesis. *British Journal of Mathematical and Statistical Psychology*, *59*(1), 1–34.

Technopolis Group & Oxford Economics. (2018). *The silver economy: Final report* (tech. rep.) (Study prepared for the European Commission; Publications Office of the European Union. Values the European silver economy at EUR 3.7 trillion in 2015, projected to

EUR 5.7 trillion by 2025, with a forecast contribution of about 32% of EU GDP.). European Commission, Directorate-General for Communications Networks, Content and Technology. Luxembourg.

Vermunt, J. K., & Magidson, J. (2002). Latent class cluster analysis. In J. A. Hagenaars & A. L. McCutcheon (Eds.), *Applied latent class analysis* (pp. 89–106). Cambridge University Press.

Vuik, S. I., Mayer, E. K., & Darzi, A. (2016). Patient segmentation analysis offers significant benefits for integrated care. *Health Affairs*, *35*(5), 769–775.

Wahrendorf, M., & Siegrist, J. (2010). Are changes in productive activities of older people associated with changes in their well-being? Results of a longitudinal European study. *European Journal of Ageing*, *7*(2), 59–68.

Walker, A. (2005). A European perspective on quality of life in old age. *European Journal of Ageing*, *2*(1), 2–12.